

Asymptotics of higher rank $6j$ symbols and the tetrahedral Horn problem

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Abstract

For a compact reductive group G , Wigner $6j$ -symbols are maps between multiplicity spaces which encode the associativity constraint $(V_\alpha \otimes V_\beta) \otimes V_\delta \cong V_\alpha \otimes (V_\beta \otimes V_\delta)$ for unitary irreducible representations of G . For $G = SU(2)$, their asymptotic behavior for large weights was conjectured by Ponzano-Regge and established by Roberts using geometric quantization techniques.

In this paper, we address the problem of asymptotic behavior of $6j$ -symbols for $G = U(n)$. We show that their decay rate is polynomial or exponential depending on the existence of a tetrahedron in the space of Hermitian n by n matrices with edges determined by the 6 arguments of the $6j$ -symbol. Our proof is inspired by probabilistic techniques from quantum information theory.

As an application of our results, we obtain a new proof of the strongly subadditive entropy inequality for tetrahedra in the space of Hermitian matrices (the tetrahedral Horn problem).

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1 Introduction

Given a complex reductive group G with unitary irreducible representations (irreps) V_λ labelled by highest weights λ along with tensor-product multiplicity spaces, $M_{\alpha,\beta}^\gamma := \text{Hom}_G(V_\gamma, V_\alpha \otimes V_\beta)$, the $6j$ -symbols are maps between these multiplicity spaces which encode the associativity constraint $(V_\alpha \otimes V_\beta) \otimes V_\gamma \cong V_\alpha \otimes (V_\beta \otimes V_\gamma)$ between a triple of irreps, V_α, V_β , and V_δ of G . Specifically, a $6j$ -symbol depends on six irreps, $V_\alpha, V_\beta, V_\gamma, V_\delta, V_\epsilon$, and V_ϕ , and is a map between multiplicity spaces of the form

$$\left\{ \begin{matrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{matrix} \right\} : M_{\alpha,\beta}^\gamma \otimes M_{\delta,\gamma}^\epsilon \rightarrow M_{\alpha,\phi}^\epsilon \otimes M_{\beta,\delta}^\phi,$$

corresponding to the block-wise components of the canonical isomorphism from $\bigotimes_\gamma M_{\alpha,\beta}^\gamma \otimes M_{\delta,\gamma}^\epsilon$ to $\bigotimes_\phi M_{\alpha,\beta}^\gamma \otimes M_{\delta,\gamma}^\epsilon$. Whenever the group G is multiplicity-free, in the sense that the dimensions of all multiplicity spaces, $m_{\alpha,\beta}^\gamma := \dim M_{\alpha,\beta}^\gamma$, are either zero or one, the corresponding $6j$ -symbol is merely a scalar quantity.

Wigner’s $6j$ -symbol for $\text{SU}(2)$: The group $\text{SU}(2)$ is multiplicity-free and the $6j$ -symbols for $\text{SU}(2)$ are, depending on the normalization and/or phase convention, are also referred to as Wigner $6j$ -symbols or Wigner $6j$ -coefficients [Wig93] or the Racah W -coefficients [Rac42]. The $6j$ -symbols for $\text{SU}(2)$ play a foundational role in the quantum theory of angular momentum, modelling the recoupling theory of three angular momenta, and thus they are used frequently in calculations involving the scattering of multiple particles in nuclear, atomic and molecular physics [VMK88; Edm96; Aqu+12]. Beyond the direct application to the theory of angular momenta, the $6j$ -symbols for $\text{SU}(2)$ also constitute a primitive object in the theory of spin networks [Pen71; BS10], representing a minimal non-trivial closed spin network. General spin networks are networks of trivalent vertices and connections labelled by irreps (spins) of $\text{SU}(2)$ which provide a discrete/combinatorial and manifestly invariant model with applications to gauge theories on graphs or manifolds [Bae96], invariants of knots and manifolds [Res87; TV92], topological quantum field theory [DJN93], quantum computing [MR05], quantum change-point discrimination [AH11], and various aspects of proposed models for quantum gravity [Bae99; BC98; Bar03; Rov04].

Throughout this paper, by the *asymptotics* of $6j$ -symbols, we are referring to their behaviour in the regime wherein the six highest weights, $\lambda \in \{\alpha, \beta, \gamma, \delta, \epsilon, \phi\}$, are becoming larger and larger in a suitable sense. The asymptotics of the $6j$ -symbol for $\text{SU}(2)$, wherein the irreps of dimension $\ell + 1$ may be identified either by integers $\ell \in \{0, 1, \dots\}$ or half-integers $j = \frac{\ell}{2} \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$, were initially investigated by Wigner [Wig60, Chapter 27]. In the context of the quantum theory of angular momenta, the limit of large ℓ , or equivalently large spins j , may also be interpreted as the limit wherein Planck’s constant tends to zero, $\hbar \propto \frac{1}{k} \rightarrow 0$, which suppresses the effects of quantizing angular momentum; therefore, these asymptotic limits are also known as semiclassical limits. Given three particles with spins $\frac{a}{2}$, $\frac{b}{2}$ and $\frac{c}{2}$ and total spin $\frac{e}{2}$, the $6j$ -symbol, $\left\{ \begin{matrix} a & b & c \\ d & e & f \end{matrix} \right\}$, can be interpreted as the quantum mechanical amplitude to observe the first and second particles to have combined spin $\frac{e}{2}$ given the second and third have combined spin $\frac{f}{2}$ [Wig60; Rob99]. Wigner considered the probability for this event to occur in the asymptotic limit of large spins, i.e., the square of the $6j$ -symbol, and conjectured the estimate:

$$\left\{ \begin{matrix} a & b & c \\ d & e & f \end{matrix} \right\}^2 \approx \frac{1}{3\pi V}$$

where V is the volume of the Euclidean tetrahedron with side-lengths (a, b, c, d, e, f) . Building upon the work of Wigner, Ponzano and Regge conjectured a formula for the amplitude of the $6j$ -symbol itself in the

limit of large spins [PR68]. Later, using techniques from geometric quantization theory, Roberts rigorously proved an asymptotic formula which is equivalent to the conjecture of Ponzano and Regge (up to minor differences) [Rob99]:

Theorem 1 (Roberts). *If (a, b, c, d, e, f) are integer side-lengths of a Euclidean tetrahedron with volume V and corresponding exterior dihedral angles $(\theta_a, \theta_b, \theta_c, \theta_d, \theta_e, \theta_f)$, then*

$$\left\{ \begin{smallmatrix} ka & kb & kc \\ kd & ke & kf \end{smallmatrix} \right\} \stackrel{k \rightarrow \infty}{\sim} \sqrt{\frac{2}{3\pi V k^3}} \cos \left(\sum_{\ell} (k\ell + 1) \frac{\theta_{\ell}}{2} + \frac{\pi}{4} \right),$$

where the sum is over the six sides ℓ . Otherwise, if there does not exist a Euclidean tetrahedron with side-lengths (a, b, c, d, e, f) , then $\left\{ \begin{smallmatrix} ka & kb & kc \\ kd & ke & kf \end{smallmatrix} \right\}$ decays to zero at an exponential rate as $k \rightarrow \infty$.

It is worth emphasizing that Chen, Ismail, and Simeonov [CIS99] obtained a rigorous proof of the Ponzano and Regge asymptotics (with minor correction due to [Wat99]) slightly before Chen, Ismail, and Simeonov using estimates from the theory of hypergeometric functions. Additional proofs of this asymptotic correspondence between $6j$ -symbols for $SU(2)$ and Euclidean tetrahedra have emerged in recent years including the elementary proof by Gurau which applies Stirling's approximation directly to Racah's single-sum formula [Gur08] as well as group integral approach to evaluate so-called simple spin networks as Feynman graphs [FK00; FL03]. See also the paper by Bartlett and Nzaganya for an accessible presentation of these kinds of calculations [BN23]. Analogous asymptotic formulas have also been obtained for the $6j$ symbol for the quantum group $U_q(\mathfrak{sl}_2)$ for a k -dependent parameter q [TW06]. The asymptotics calculated by Taylor and Woodward encode (i) the existence of a spherical tetrahedron on the unit sphere with prescribed edgelengths whenever $q = \exp(i\pi/r(k))$ is an $r(k)$ th root of unity or (ii) the existence of a hyperbolic tetrahedron with prescribed edgelengths when $q = \exp(\pi/r(k))$ is positive real (here $r(k) = k(r - 2) + 2$ for $r > 2$ a fixed integer) [TW06].

More generally, that the asymptotics of representation-theoretic quantities (such as $6j$ -symbols) are controlled by the existence of some critical geometric object is explained by the theory of geometric quantization [Rob02]. Geometric quantization, broadly speaking, associates classical phase spaces modelled by symplectic manifolds to quantum phase spaces modelled by Hilbert spaces [Rob02; GS82; Kir04]. In particular, Kirillov's orbit principle [Kir04] proposes an association between (i) the coadjoint orbits of a compact Lie group G passing through an integral weight λ to (ii) the corresponding unitary irreps of G of highest weight λ . Furthermore, products of coadjoint orbits correspond to tensor products of irreps and moreover symplectic quotients of the resulting products of orbits are identified with invariant vectors in the resulting tensor products of irreps [Rob02; GS82]. For the aforementioned case of $SU(2)$, the coadjoint orbits can be identified the $SO(3)$ rotations of vectors in Euclidean $\mathbb{R}^3$ (i.e. spheres of fixed radius), and thus the asymptotics of representation-theoretic quantities for $SU(2)$ are controlled by the existence of corresponding geometric objects in Euclidean $\mathbb{R}^3$ with prescribed edge-lengths [Rob02].

$6j$ -symbols for $SU(n)$: The purpose of this paper is prove statements concerning the asymptotics of the $6j$ symbol for the group $U(n)$ (or equivalently for $SU(n)$) wherein the polynomial irreps V_{λ} are labelled by highest-weights λ which can also be identified with Young diagrams. One of the primary difficulties in proving statements about $6j$ symbols for $SU(n)$ for $n \geq 3$ is the presence of non-trivial multiplicity spaces, i.e. the dimensions of tensor-product multiplicity spaces for $SU(n)$, also known as Littlewood-Richardson coefficients, can be greater than one. Because of these difficulties, over the years, considerable attention has been placed on the study of $SU(n)$ (or $U_q(\mathfrak{sl}_n)$) $6j$ symbols which are multiplicity-free in the sense that the dimensions of each of the multiplicity spaces is either one or zero (in the latter case the $6j$ symbol defined to be zero). For instance, tensor products involving symmetric representations, where $\lambda = (k)$, or conjugates of symmetric representations, $\lambda = (k^{(n-1)})^1$, are necessarily multiplicity free; Nawata, Pichai, and Zodinmawia conjectured closed form expressions for the $U_q(\mathfrak{sl}_n)$ $6j$ symbols when a, b, d and e are integers and either $(\alpha, \beta, \delta, \epsilon) = ((a), (b^{(n-1)}), (d), (e^{(n-1)}))$ and $a + d = b + e$ (Type I) or $(\alpha, \beta, \delta, \epsilon) = ((a), (b), (d^{(n-1)}), (e^{(n-1)}))$ and $a + b = d + e$ (Type II) as well as the cases which can be related to these by known symmetries of the corresponding $6j$ symbol [NPZ13]. Building upon this work, Alekseev, Morozov, and Sleptsov studied both

¹Here $k^{(n-1)}$ is shorthand for a string $k, k, \dots, k$ consisting of $n - 1$ repetitions of k .

the asymptotics of and symmetries in each of the Type I and Type II cases [AMS20]. More recently, Alcock-Zeilinger et al. have derived formulae for the family of $SU(n)$ $6j$ symbols where at least one pair of non-adjacent edges, say (α, δ) correspond to a pair of fundamental/defining representations, i.e., $\alpha, \delta = (1)$ [Alc+23]. Shortly thereafter, Keppeler, Plätzer, and Sjö Dahl expanded upon these results to derive formulae for the family of $SU(n)$ $6j$ symbols where either (i) at least one pair of non-adjacent edges consists of fundamental or adjoint representations, i.e., $\alpha, \delta \in \{(1), (2, 1^{(n-2)})\}$, or (ii) at least one vertex, say (α, β, ϕ) , consists entirely of fundamental or adjoint representations, i.e., $\alpha, \beta, \phi \in \{(1), (2, 1^{(n-2)})\}$ [KPS24]. As stated by Coquereaux, “apart from very few special cases (low dimension of irreps and no multiplicity), ... we are not aware of any published general result.” [Coq23].

TC: Some residual research-notes concerning the $6j$ symbols:

- The graphical notion, specifically the use of an additional edge for multiplicity spaces can has been used for a long time, as early as [AB68; Mou84].
- Aquilanti et al. provides an excellent introduction to subject of classical $6j$ symbols, their history, diagrammatics, presentations and asymptotics [Aqu+12]
- There is a series of papers involving Alexey Sleptsov, starting with a computation of quantum $6j$ symbols for generic dimension n but with symmetric representations in terms of certain hypergeometric polynomials [MMS17], then with a discussion of the eigenvalue hypothesis and its potential connection to generalized Regge symmetries [MS18], then with a calculation of all multiplicity free quantum $6j$ symbols for arbitrary dimension [AMS20], and then finally with the discovery of novel “tug-the-hook” symmetries for quantum $6j$ symbols for all n [LS23].

Our contribution: Our approach bypasses the difficulties presented by multiplicities by focusing only on the *norm* (specifically the Hilbert-Schmidt, Frobenius or Schatten 2-norm) of the $6j$ symbol for $SU(n)$:

$$\left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_2 := \text{Tr} \left[\begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix}^\dagger \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right].$$

Following the procedure provided by geometric quantization, one expects the asymptotics of the $6j$ -symbol for $U(n)$ to be controlled by existence of some kind of tetrahedral object in space of coadjoint orbits for $U(n)$. In particular, the coadjoint orbits of $U(n)$ can be identified with the orbits of $n \times n$ Hermitian matrices, X , under the conjugation action, $X \mapsto U^{-1} X U$, by unitary $n \times n$ matrices $U \in U(n)$; in other words, coadjoint orbits are associated to sets of $n \times n$ Hermitian matrices sharing a common set of eigenvalues $\lambda' = (\lambda'_1 \geq \dots \geq \lambda'_n)$.

Theorem 2 (Informal). *If $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ are the eigenvalues of $n \times n$ of positive semidefinite matrices (A, B, C, D, E, F) satisfying*

$$A + B = C, \quad B + D = F, \quad C + D = E, \quad A + F = E, \tag{1}$$

then there exists a sequence of Young diagrams $(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k)$ satisfying

$$\frac{\alpha_k}{k} \rightarrow \alpha', \quad \frac{\beta_k}{k} \rightarrow \beta', \quad \frac{\gamma_k}{k} \rightarrow \gamma', \quad \frac{\delta_k}{k} \rightarrow \delta', \quad \frac{\epsilon_k}{k} \rightarrow \epsilon', \quad \frac{\phi_k}{k} \rightarrow \phi', \tag{2}$$

such that

$$\left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\|_2 \geq \frac{1}{\text{poly}(k)}. \tag{3}$$

Otherwise, if $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ are not the eigenvalues of any matrices satisfying Eq. (1), then for any sequence of Young diagrams satisfying Eq. (2), we have exponential decay in the value of $\left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\|_2$.

Before continuing, it is worth emphasizing the similarities and differences between our result and the asymptotic formula proven by Roberts above [Rob99]. First and foremost, our result goes beyond that of Roberts in the sense that it applies to the higher-rank Lie group $SU(n)$ for $n \geq 3$. Moreover, our result, even when restricted to the case for $SU(2)$, goes beyond the statement of proven by Roberts in the sense our

result is applicable to *irrational* eigenvalues/edge-lengths, whereas the former is only applicable to *integer*, or with a minor re-scaling, *rational* eigenvalues/edge-lengths. Nevertheless, one drawback of our result is that it merely promises the *existence* of some sequence of Young diagrams, $(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k)$, converging to the eigenvalues $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$, whereas the result of Roberts provides a concrete statement about the particular sequence of *scaled* Young diagrams $(k\alpha, k\beta, k\gamma, k\delta, k\epsilon, k\phi)$.

Another drawback of our result is that it merely provides a polynomial lower bound on the value of $\left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_2$ and does not provide an asymptotic formula which would serve as the a proper generalization of the Ponzano and Regge formula above [PR68]. The Ponzano and Regge formula for $SU(2)$ was traditionally obtained using the method of *stationary phases* which relies on both an integral formula for the $6j$ -symbol with exponential integrand along with a geometric understanding of the *isolated* critical points for the exponential phase [PR68; Gur08]. In the case of $SU(2)$ with generic edge-lengths, there is a formulation involving exactly *two* isolated critical points corresponding the $SO(3)$ -orbit $[\tau]$ of a Euclidean tetrahedron τ with given edge-lengths and (ii) the $SO(3)$ -orbit $[\tau']$ of the Euclidean tetrahedron τ' which is the mirror-image of the tetrahedron τ with the same fixed side lengths. Generalizing this type of criticality analysis to $SU(n)$ for $n \geq 3$ appears to be a significant challenge, and even if an analysis was performed, one would still need to compute a corresponding Hessian for each critical point and evaluate the total contribution [Rob02]. Perhaps another approach, such as the approach adopted by Taylor and Woodward (see also [CIS99]) to prove asymptotics for the $6j$ -symbol for $U_q(\mathfrak{sl}_2)$ using recursion relations [TW06] (see also the formulas for the $6j$ -symbols for $\mathfrak{gl}_n$ in terms of generalized hypergeometric functions due to [Art24] [Art23; Art24]).

Eigenvalue estimation: Our approach to proving the above theorem relies on the concept of *weak Schur sampling*, a probabilistic technique developed in the context of quantum information theory for estimating the eigenvalues of density matrices (trace one, positive semidefinite matrices), which model the state of quantum systems [KW01; Chr08; OW16; HM02; CHW07; Hay17]. **TODO: Finish the explanation of the eigenvalue-estimation idea once a suitable presentation of the all the “T” functions have been fixed.**

The tetrahedral Horn problem: The problem of determining the set of eigenvalues $(\alpha', \beta', \gamma')$ of Hermitian matrices (A, B, C) satisfying $A + B = C$ has been investigated for over a century [Wey12; Fan49; Lid53; Wie55; TF71; TF70] and has come to be known as *Horn’s problem* after Horn who, in 1962, conjectured a solution to the problem in the form of an inductively defined, finite set of homogeneous linear inequalities. In 1998, Klyachko provided a solution to Horn’s problem [Kly98] which, when combined with a proof of the so-called saturation conjecture, due to Knutson and Tao [KT99], constitutes a proof of Horn’s original conjecture [Ful98]. These linear inequalities, now known as the *Horn inequalities* describe the convex polytope of admissible values for $(\alpha', \beta', \gamma')$. Recent developments on Horn’s problem include the development of the geometric method by Sherman [She15] and Berline, Vergne, and Walter [BVW18] originally pioneered by the geometric proofs due to Belkale [Bel02; Bel05], as well the characterization and computation of the induced distribution over the eigenvalues γ of the sum $C = A + B$ if A and B are sampled uniformly and independently from the respective sets of Hermitian matrices with eigenvalues α' and β' by Coquereaux, McSwiggen, and Zuber in series of papers [CZ18; CZ18; CMZ19; Zub18]. For a pedagogical and historical introduction to Horn’s problem, we recommend the review articles of Bhatia [Bha01], Knutson [Knu00], and Fulton [Ful00] or alternatively [NP06, Appendix D]. See also the article by Daftuar and Hayden for an account of applications to topics in quantum information theory [DH05].

In parallel to Horn’s problem, numerous variations and extensions of Horn’s problem have been explored by various authors. For example, numerous authors have considered the admissible eigenvalues of unitary matrices and their product [AW97; Bel08; BK16], or the admissible singular values of matrices and their product [Spe05; TT71]. In 2000, it was shown by [Kly00] that *logarithms* of the admissible singular values of non-singular matrices and their product are identical to the admissible eigenvalues of Hermitian matrices and their sum, thus proving a conjecture due to Thompson [Kly00; AMW01].

In this paper, we consider a generalization of Horn’s problem investigated previously only briefly in [CŠW18]. While Horn’s problem aims to characterize the eigenvalues $(\alpha', \beta', \gamma')$ of a triple of $n \times n$ Hermitian matrices (A, B, C) which are subjected to the constraint $A + B = C$, the generalization we consider here aims to characterize the eigenvalues $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ of a *sextuple* of $n \times n$ matrices (A, B, C, D, E, F) subject to four constraints: $A + B = C$, $B + D = F$, $D + C = E$ and $A + F = E$. These constraints arise naturally from building a tetrahedral arrangement of $n \times n$ Hermitian matrices inside the vector space of

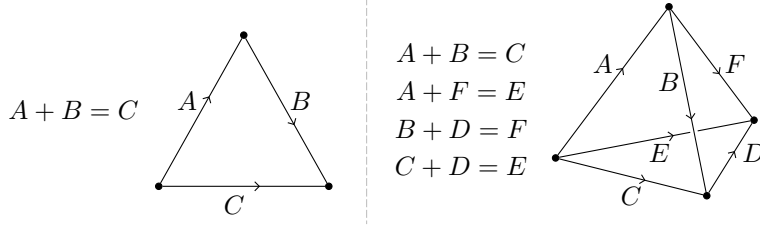


Figure 1: (Left) The *triangular* Horn's problem subjects three Hermitian matrices, (A, B, C) , to the constraint $A + B = C$ forming a triangle in the vector space of Hermitian matrices. (Right) The *tetrahedral* Horn's problem subjects six Hermitian matrices, (A, B, C, D, E, F) , to a quadruple of constraints forming a tetrahedron in the vector space of Hermitian matrices. [fig:tetrahedron_labelling_scheme](#)

such Hermitian matrices with edges labelled according to Fig. 1.

Beyond the superficial connection between the imposed constraints and the resulting shape formed in the vector space of Hermitian matrices (as in Fig. 1), in the special case of 2×2 Hermitian matrices, this connection is much deeper. It is well-known that the space of admissible eigenvalues for a triangular arrangement of 2×2 traceless Hermitian matrices is in correspondence with the moduli space of Euclidean *triangles* in $\mathbb{R}^3$ [HK96; FKL10; Knu00]. This correspondence generalizes to the tetrahedral case in the sense that the admissible eigenvalues for a tetrahedral arrangements of 2×2 traceless Hermitian matrices is in correspondence with the moduli space of Euclidean *tetrahedra* in $\mathbb{R}^3$. While this latter result is straightforward to prove and also well-known, indeed it forms the basis of the semiclassical asymptotics of the $SU(2)$ Wigner $6j$ symbols [Rob99], it leads to a complete solution to the tetrahedral Horn's problem for 2×2 matrices; specifically, the moduli space of Euclidean tetrahedra in $\mathbb{R}^3$ is provided by the positivity of so-called Cayley-Menger determinants which are polynomial inequalities in the side-lengths (a, b, c, d, e, f) [SJS18]. Even for $n = 2$, we remark that there is a strong qualitative difference between the tetrahedral Horn problem and the original Horn problem. In particular, while the Horn's problem is completely characterized by a *finite* number of *linear* inequalities in the eigenvalues (indeed the solution space is an instance of a moment polytope known as *Horn's polytope*), the solution space for the tetrahedral Horn's problem is not convex nor a polytope.

More recently, Alekseev et al. have investigated a few variants of the tetrahedral Horn problem concerning either the singular values of products of invertible matrices or the tropical singular values of concatenations of weighted planar networks [Ale+25]. These variants are termed *multiple* Horn problems in [Ale+25]; the tetrahedral Horn problem we consider here is termed the *additive* multiple Horn problem in the language of [Ale+25].

Quantum marginals & the $6j$ -symbol for the symmetric group S_k : The techniques used in this paper to derive bounds on the $6j$ symbol for the unitary group $U(n)$ are closely related to the techniques used in an earlier paper by Christandl, Şahinoğlu, and Walter which derived analogous bounds on the $6j$ symbol for the symmetric group S_k [CŞW18]. In that paper, the authors considered the problem of determining whether there exists a density matrix ρ_{ABC} with partial traces $\rho_{AB}, \rho_{BC}, \rho_A, \rho_B$, and ρ_C with a given sextuple of eigenvalues [CŞW18]. Such problems arise naturally in the context of quantum information theory and are typically referred to as quantum marginals problems [CŞW18; Fra22]. In particular, they proved the $6j$ symbols for the symmetric group S_k (referred to as the recoupling coefficients) in the limit as $k \rightarrow \infty$ either decay exponentially or can be bounded from below by a polynomial factor depending on whether there exists a density matrix with the prescribed marginal spectra [CŞW18]. Moreover, they also demonstrated how the tetrahedral Horn problem may be viewed as a special case of the quantum marginals problem and, in a complementary manner, furthermore demonstrated how, by restricting to the subgroup $S_k \subset U(k)$ of permutation matrices and applying Nishiyama's restriction theorem [Nis00], the $6j$ symbols for unitary group $U(n)$ may be viewed as a special case of the $6j$ symbols for the symmetric group S_k [CŞW18]. This latter result obtained by [CŞW18] is a generalizes how the Littlewood-Richardson coefficients can be cast as a special case of the Kronecker coefficients [Lit58; Mur55; IP23]. The approach taken by this paper differs from that of [CŞW18] in that we prove the correspondence between solutions to the tetrahedral Horn problem on one hand and the asymptotics of the $6j$ symbol for $U(n)$ on the other hand *directly*, rather than

first pass through the quantum marginals problem or through the $6j$ symbols for the symmetric group. We note that this more direct approach was initiated, in the $n = 2$ case, by Backens [Bac10].

Structure of the paper: TODO: Descide on the structure of the paper? Finally, we prove an upper bound on general $6j$ symbols using the Cauchy-Schwarz inequality which, in the $n \rightarrow \infty$ asymptotic limit, gives rise to a novel proof of a known strongly subadditive inequality involving unnormalized entropies.

2 Tetrahedral eigenvalues

sec:tet_eig

2.1 The tetrahedral Horn problem

prob:tri_problem

Problem 3 (Trigonal Horn problem). A triple $(\alpha', \beta', \gamma') \in (\mathbb{R}_{\text{dec}}^n)^{\times 3}$ of candidate eigenvalues are said to be *trigonal* if there exists a corresponding triple (A, B, C) of $d \times d$ Hermitian matrices satisfying (i) the linear constraint

$$A + B = C, \quad \text{eq:tri_linear_constraints} \quad (4)$$

and (ii) the three eigenvalue constraints

$$\text{eig}(A) = \alpha', \quad \text{eig}(B) = \beta', \quad \text{eig}(C) = \gamma'. \quad \text{eq:tri_eigenvalue_constraints} \quad (5)$$

The problem of determining the set of trigonal eigenvalues $(\alpha', \beta', \gamma')$ is known as *Horn problem*.

prob:tet_problem

Problem 4 (Tetrahedral Horn problem). A sextuple $(\alpha', \beta', \gamma', \delta', \epsilon', \phi') \in (\mathbb{R}_{\text{dec}}^n)^{\times 6}$ of candidate eigenvalues are said to be *tetrahedral* if there exists a corresponding sextuple (A, B, C, D, E, F) of $n \times n$ Hermitian matrices satisfying (i) the four linear constraints

$$A + B = C, \quad B + D = F, \quad C + D = E, \quad A + F = E, \quad \text{eq:tet_linear_constraints} \quad (6)$$

and (ii) the six eigenvalue constraints

$$\text{eig}(A) = \alpha', \quad \text{eig}(B) = \beta', \quad \text{eig}(C) = \gamma', \quad \text{eig}(D) = \delta', \quad \text{eig}(E) = \epsilon', \quad \text{eig}(F) = \phi'. \quad \text{eq:tet_eigenvalue_constraints} \quad (7)$$

The problem of determining the set of tetrahedral eigenvalues $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ will be referred to as the *tetrahedral Horn problem*.

The tetrahedral Horn problem can be viewed as a generalization of the Horn's problem in a variety of ways. For instance, a triplet $(\alpha', \beta', \gamma')$ of eigenvalues is trigonal if and only if the sextuple $(\alpha', \beta', \gamma', 0, \gamma', \beta')$ is tetrahedral. More generally, it is possible to view the tetrahedral Horn problem as the conjunction of four separate instances of Horn's original problem subject to additional matching constraints. Specifically, a sextuple of eigenvalues $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ is tetrahedral only if each of the following four triplets are trigonal:

$$(\alpha', \beta', \gamma'), \quad (\beta', \delta', \phi'), \quad (\gamma', \delta', \epsilon'), \quad (\alpha', \phi', \epsilon'). \quad \text{eq:four_faces} \quad (8)$$

If each of the above triplets are trigonal, then there exists a set of twelve Hermitian matrices with the corresponding eigenvalues satisfying

$$A + B = C, \quad \tilde{B} + D = F, \quad \tilde{C} + \tilde{D} = E, \quad \tilde{A} + \tilde{F} = \tilde{E}. \quad \text{eq:four_faces_matrices} \quad (9)$$

The tetrahedral Horn problem goes beyond four separate instances of Horn's original problem in that it additionally requires the primed and unprimed Hermitian matrices to coincide:²

$$A = \tilde{A}, \quad B = \tilde{B}, \quad C = \tilde{C}, \quad D = \tilde{D}, \quad E = \tilde{E}, \quad F = \tilde{F}. \quad (10)$$

²Note that by conjugating the third and fourth conditions in Eq. (9) by unitaries U and V respectively, it is possible to set $\tilde{C} \mapsto U^{-1}\tilde{C}U = C$ and $\tilde{F} \mapsto V^{-1}\tilde{F}V = F$, and thus one could assume without loss of generality that $C = \tilde{C}$ and $F = \tilde{F}$.

2.2 Low dimensional solutions

Remark 5 (Traces are tetrahedral). In the original version of Horn's problem, the eigenvalues $(\alpha', \beta', \gamma')$ of Hermitian matrices (A, B, C) satisfying $A + B = C$ necessarily must satisfy a basic equality condition known as the *trace condition*:

$$|\alpha'| + |\beta'| = |\gamma'|, \quad \text{\texttt{eq:tri_trace_cond}} \quad (11)$$

where $|\lambda'| := \sum_i \lambda'_i$ is shorthand for the sum of the eigenvalues $\lambda' = (\lambda'_1 \geq \dots \geq \lambda'_n)$. This condition must hold because $A + B = C$ implies $\text{Tr}(C) = \text{Tr}(A + B) = \text{Tr}(A) + \text{Tr}(B)$ and $\text{Tr}(X) = |\lambda'|$ for any Hermitian matrix X with eigenvalues λ' . For the tetrahedral Horn's problem, a set of tetrahedral eigenvalues $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ necessarily satisfy four *trace conditions*:

$$|\alpha'| + |\beta'| = |\gamma'|, \quad |\beta'| + |\delta'| = |\phi'|, \quad |\delta'| + |\gamma'| = |\epsilon'|, \quad |\alpha'| + |\phi'| = |\epsilon'|. \quad \text{\texttt{eq:tet_trace_conds}} \quad (12)$$

Furthermore, for the trivial case where $n = 1$, the above trace constraints are both necessary and sufficient for the tetrahedral Horn problem. exam:d.2

Example 6 (Tetrahedral Horn's problem for $d = 2$). Here we articulate the solution to the tetrahedral Horn's problem for 2×2 Hermitian matrices. We restrict our focus to the case of *traceless* 2×2 Hermitian matrices³. In this context, the sextuple of eigenvalues $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ can be identified with a sextuple of non-negative real numbers $(a, b, c, d, e, f) \in \mathbb{R}_{\geq 0}^6$ such that

$$\alpha' = (a, -a), \quad \beta' = (b, -b), \quad \gamma' = (c, -c), \quad \delta' = (d, -d), \quad \epsilon' = (e, -e), \quad \phi' = (f, -f). \quad \text{\texttt{eq:traceless_two_dim_tetra}} \quad (13)$$

Moreover, each traceless 2×2 Hermitian matrix, X , may be identified with a three-dimensional vector $(x_1, x_2, x_3) \in \mathbb{R}^3$ using the standard Pauli-matrix basis $\{\sigma_1, \sigma_2, \sigma_3\}$ according to the formula:

$$X = x_1 \sigma_1 + x_2 \sigma_2 + x_3 \sigma_3 = \begin{pmatrix} x_3 & x_1 - ix_2 \\ x_1 + ix_2 & -x_3 \end{pmatrix}. \quad (14)$$

Under this identification, the eigenvalues of X are determined by the Euclidean length of the vector (x_1, x_2, x_3) :

$$\text{eig}(X) = \text{eig} \begin{pmatrix} x_3 & x_1 - ix_2 \\ x_1 + ix_2 & -x_3 \end{pmatrix} = (\|x\|, -\|x\|), \quad \text{where,} \quad \|x\| = \sqrt{x_1^2 + x_2^2 + x_3^2}. \quad (15)$$

Furthermore, if A is determined by the coordinates (a_1, a_2, a_3) and B by the coordinates (b_1, b_2, b_3) , then their sum $C = A + B$ is determined by the coordinates $(c_1, c_2, c_3) = (a_1 + b_1, a_2 + b_2, a_3 + b_3)$. Consequently, in this setting, the tetrahedral Horn's problem reduces to problem of describing the space of all possible edge-lengths (a, b, c, d, e, f) of a Euclidean tetrahedron in $\mathbb{R}^3$. This problem admits of a well-studied solution [Ric02; WD09], namely, the conjunction of a quadruplet of triangle-inequalities

$$|a - b| \leq c \leq a + b, \quad |b - d| \leq f \leq b + d, \quad |d - c| \leq e \leq d + c, \quad |a - f| \leq e \leq a + f, \quad \text{\texttt{eq:quadruplet_triangle_constraints}} \quad (16)$$

(which ensure the possibility of forming a Euclidean triangle for each of the four faces), together with a single non-linear inequality of the form

$$D(a, b, c, d, e, f) = \frac{1}{288} \det \begin{pmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & a^2 & c^2 & e^2 \\ 1 & a^2 & 0 & b^2 & f^2 \\ 1 & c^2 & b^2 & 0 & d^2 \\ 1 & e^2 & f^2 & d^2 & 0 \end{pmatrix} \geq 0. \quad \text{\texttt{eq:volume_positive_constraint}} \quad (17)$$

The quantity $D(a, b, c, d, e, f)$ is known as a Cayley-Menger determinant and, when $D(a, b, c, d, e, f)$ is positive and when the triangle inequalities above are also satisfied, $D(a, b, c, d, e, f)$ is always equal to *square* of the volume of the corresponding tetrahedron [SJS18, Equation 5.2]⁴. Figure 2 depicts a three-dimensional projection of the solution space described by Eq. (16) and Eq. (17).

³In Section 2.3, Corollary 11, we will see that this restriction can be made without loss of generality.

⁴The triangle inequalities $|a - b| \leq c \leq a + b$ are also equivalent to the non-negativity of the *area*, $D(a, b, c)$, of the triangle

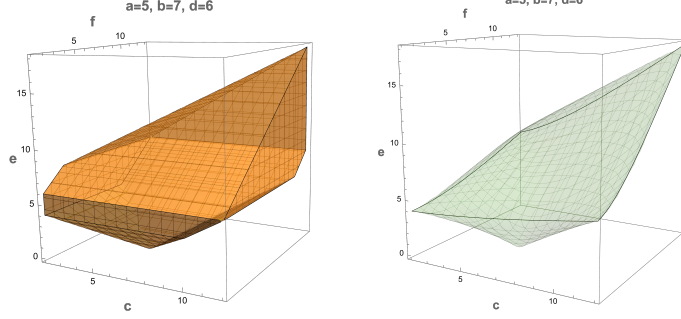


Figure 2: A slice of the set of admissible side-lengths (a, b, c, d, e, f) for Euclidean tetrahedron along the coordinates $(a, b, d) = (5, 7, 6)$ (Left) The polyhedron of values (c, e, f) satisfying the triangle inequalities for each face Eq. (16). (Right) The set of all values (c, e, f) satisfying both Eq. (16) and Eq. (17).

2.3 Symmetries of tetrahedral eigenvalues

`sec:symmetries`

There are numerous types of symmetries present in the statement of Problem 4. The purpose of this section is to describe the scaling, inversion, tetrahedral, and shifting symmetries of the problem. These symmetries can, in many cases, be used to simplify the problem by selecting a canonical representative of the corresponding equivalence class; in other cases, it will be useful to leave Problem 4 as stated.

`prop:scaling`

Proposition 7 (Scaling symmetry). *Let $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ be a sextuple of candidate eigenvalues and let $s \in (0, \infty)$ be a positive real number. Then the scaled eigenvalues, $(s\alpha', s\beta', s\gamma', s\delta', s\epsilon', s\phi')$, are tetrahedral if and only if the original, unscaled eigenvalues $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ are tetrahedral.*

Proof. The statement follows from the fact that $\text{eig}(sX) = s \text{eig}(X)$ holds for all Hermitian matrices X and positive scalars s and that each of the constraints in Eq. (6) are homogeneous in the six Hermitian matrices (A, B, C, D, E, F) ; for example, we have $sA + sB - sC = 0$ if and only if $s(A + B - C) = 0$ if and only if $A + B - C = 0$. $\square$

One may wonder whether it is possible to extend the scaling symmetry to consider scaling by negative $s < 0$. Indeed, this can be accomplished, provided one accounts for the fact that negating a Hermitian matrices both negates and reverses the order of its eigenvalues.

`prop:inversion`

Proposition 8 (Inversion symmetry). *Let $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ be a sextuple of candidate eigenvalues. Given any list of non-increasing eigenvalues $\lambda' = (\lambda'_1 \geq \dots \geq \lambda'_n)$, let λ'^* denote the inverted list of resorted and negated eigenvalues: $\lambda'^* = (-\lambda'_n \geq \dots \geq -\lambda'_1)$. Then $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ are tetrahedral if and only if $(\alpha'^*, \beta'^*, \gamma'^*, \delta'^*, \epsilon'^*, \phi'^*)$ are tetrahedral.*

Proof. The statement holds because (i) (A, B, C, D, E, F) satisfy Eq. (6) if and only if $(A', B', C', D', E', F') := (-A, -B, -C, -D, -E, -F)$ satisfy Eq. (6), and (ii) $\text{eig}(X) = \lambda'$ if and only if $\text{eig}(-X) = \lambda'^*$. $\square$

The inversion symmetry Proposition 8 is just one element belonging to a group of 48 symmetries consisting of inversions of eigenvalues, $\lambda' \rightarrow \lambda'^*$, or permutations of elements in the sextuple $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$.

`prop:tet_symmetry`

Proposition 9 (Oriented Tetrahedral symmetry). *The sextuple $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ of eigenvalues are tetrahedral if and only if any (and thus all) of the following sextuples of eigenvalues are tetrahedral:*

$$(\alpha' \ \phi' \ \epsilon' \ \delta'^* \ \gamma' \ \beta') \quad (\gamma' \ \epsilon'^* \ \delta'^* \ \phi' \ \beta' \ \alpha'^*) \quad (18)$$

`{eq:gens_of_tet_sym}`

with side-lengths a, b and c where $D(a, b, c)^2 = -\frac{1}{16} \det \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & a^2 & c^2 \\ 1 & a^2 & 0 & b^2 \\ 1 & c^2 & b^2 & 0 \end{pmatrix}$ is equivalent to Heron's formula.

Moreover, these two operations generate a finite group of order 48 whose action, grouped by conjugacy classes, produces:

$$\begin{array}{cccc}
(\alpha' & \beta' & \gamma' & \delta' & \epsilon' & \phi') & (\alpha'^* & \gamma' & \beta' & \delta' & \phi' & \epsilon') & (\alpha' & \gamma'^* & \beta'^* & \delta'^* & \phi'^* & \epsilon'^*) & (\alpha'^* & \beta'^* & \gamma'^* & \delta'^* & \epsilon'^* & \phi'^*) \\
(\delta' & \beta' & \phi' & \alpha' & \epsilon' & \gamma') & (\beta'^* & \alpha'^* & \gamma'^* & \epsilon' & \delta' & \phi') & (\beta' & \alpha' & \gamma' & \epsilon'^* & \delta'^* & \phi'^*) & (\delta'^* & \beta'^* & \phi'^* & \alpha'^* & \epsilon'^* & \gamma'^*) \\
(\delta'^* & \epsilon' & \gamma' & \alpha'^* & \beta' & \phi') & (\phi'^* & \beta' & \delta'^* & \gamma'^* & \epsilon'^* & \alpha'^*) & (\phi' & \beta'^* & \delta' & \gamma' & \epsilon' & \alpha') & (\delta' & \epsilon'^* & \gamma'^* & \alpha' & \beta'^* & \phi'^*) \\
(\alpha' & \epsilon'^* & \phi'^* & \delta' & \beta'^* & \gamma'^*) & (\epsilon' & \delta'^* & \gamma' & \beta'^* & \alpha' & \phi'^*) & (\epsilon'^* & \delta' & \gamma'^* & \beta' & \alpha'^* & \phi') & (\delta' & \epsilon'^* & \gamma'^* & \alpha' & \beta'^* & \phi'^*) \\
& & & & & & (\gamma' & \beta'^* & \alpha' & \phi' & \epsilon' & \delta') & (\gamma'^* & \beta' & \alpha'^* & \phi'^* & \epsilon'^* & \delta'^*) & (\alpha'^* & \epsilon' & \phi' & \delta'^* & \beta' & \gamma') \\
& & & & & & (\alpha' & \phi' & \epsilon' & \delta'^* & \gamma' & \beta') & (\alpha^* & \phi'^* & \epsilon'^* & \delta' & \gamma'^* & \beta'^*) & (\alpha'^* & \epsilon' & \phi' & \delta'^* & \beta' & \gamma') \\
(\beta' & \phi'^* & \delta'^* & \epsilon' & \gamma' & \alpha') & & & & & & & & & & & & (\beta'^* & \phi' & \delta' & \epsilon'^* & \gamma'^* & \alpha'^*) \\
(\gamma' & \alpha'^* & \beta' & \phi'^* & \delta'^* & \epsilon'^*) & & & & & & & & & & & & (\gamma'^* & \alpha' & \beta'^* & \phi' & \delta' & \epsilon') \\
(\epsilon' & \gamma'^* & \delta' & \beta' & \phi' & \alpha'^*) & & & & & & & & & & & & (\epsilon'^* & \gamma' & \delta'^* & \beta'^* & \phi'^* & \alpha') \\
(\phi' & \alpha' & \epsilon' & \gamma'^* & \delta' & \beta'^*) & & & & & & & & & & & & (\phi'^* & \alpha'^* & \epsilon'^* & \gamma' & \delta'^* & \beta') \\
(\phi'^* & \delta' & \beta'^* & \gamma' & \alpha' & \epsilon') & & & & & & & & & & & & (\phi' & \delta'^* & \beta' & \gamma'^* & \alpha'^* & \epsilon'^*) \\
(\epsilon'^* & \phi' & \alpha'^* & \beta'^* & \gamma'^* & \delta') & & & & & & & & & & & & (\epsilon' & \phi'^* & \alpha' & \beta' & \gamma' & \delta'^*) \\
(\gamma'^* & \delta'^* & \epsilon'^* & \phi' & \alpha'^* & \beta') & & & & & & & & & & & & (\gamma' & \delta' & \epsilon' & \phi'^* & \alpha' & \beta'^*) \\
(\beta'^* & \gamma' & \alpha' & \epsilon'^* & \phi'^* & \delta'^*) & & & & & & & & & & & & (\beta' & \gamma'^* & \alpha'^* & \epsilon' & \phi' & \delta')
\end{array}
\tag{19}$$

Proof. To begin, notice that each of the operations on $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ appearing in Eq. (18) are invertible (indeed the first operation has order two, while the second operation has order four). The claim that these two operations generate a finite group of order 48 can be checked explicitly using computer software. Now suppose $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ are the eigenvalues of a sextuple of Hermitian matrices (A, B, C, D, E, F) satisfying Eq. (6). Then, for example, the sextuple $(A', B', C', D', E', F') = (A, F, E, -D, C, B)$ (which has the eigenvalues $(\alpha', \phi', \epsilon', \delta'^*, \gamma', \beta')$) is also tetrahedral because

$$\begin{aligned}
A' + B' - C' &= (A + F - E) = 0, \\
B' + D' - F' &= -(B + D - F) = 0, \\
C' + D' - E' &= -(C + D - E) = 0, \\
A' + F' - E' &= (A + B - C) = 0.
\end{aligned}
\tag{20}$$

Moreover, as the operation mapping $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ to $(\alpha'^*, \gamma', \beta', \delta', \phi', \epsilon')$ is invertible, one also has that if $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ is not tetrahedral, then neither is $(\alpha'^*, \gamma', \beta', \delta', \phi', \epsilon')$. An analogous argument holds for the operation mapping $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ to $(\gamma', \epsilon'^*, \delta'^*, \phi', \beta', \alpha'^*)$ with (A', B', C', D', E', F') given by $(C, -E, -D, F, B, -A)$. $\square$

prop:shifting

Proposition 10 (Shifting symmetry). *Let $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ be a sextuple of candidate eigenvalues and let $x, y, z \in \mathbb{R}$ be fixed. Given any list of non-increasing eigenvalues $\lambda' = (\lambda'_1 \geq \dots \geq \lambda'_n)$, let $\lambda' + x := (\lambda'_1 + x \geq \dots \geq \lambda'_n + x)$. Then define shifted eigenvalues by the formula*

$$\begin{aligned}
\tilde{\alpha}' &:= \alpha' + x, & \tilde{\beta}' &:= \beta' + y, & \tilde{\gamma}' &:= \gamma' + x + y, \\
\tilde{\delta}' &:= \delta' + z, & \tilde{\epsilon}' &:= \epsilon' + x + y + z, & \tilde{\phi}' &:= \phi' + y + z.
\end{aligned}
\tag{21}$$

The shifted eigenvalues $(\tilde{\alpha}', \tilde{\beta}', \tilde{\gamma}', \tilde{\delta}', \tilde{\epsilon}', \tilde{\phi}')$ are tetrahedral if and only if the unshifted eigenvalues $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ are tetrahedral.

Proof. Given a Hermitian matrix X with eigenvalues $\text{eig}(X) = \lambda'$ and a constant $x \in \mathbb{R}$, the eigenvalues of $X + xI$ (where I is the identity matrix) are

$$\text{eig}(X + xI) = \lambda' + x = (\lambda'_1 + x \geq \dots \geq \lambda'_n + x).
\tag{22}$$

The statement then follows because

$$(A + xI, B + yI, C + (x + y)I, D + zI, E + (x + y + z)I, F + (y + z)I)
\tag{23}$$

satisfy the tetrahedral constraints in Eq. (6) if and only if (A, B, C, D, E, F) satisfy the tetrahedral constraints in Eq. (6). $\square$

As a corollary of the shifting symmetries (Proposition 10) one concludes that to solve the tetrahedral Horn's problem, in a certain sense, it suffices to solve the tetrahedral Horn's problem merely for traceless matrices (Corollary 11) or, alternatively, for positive definite or positive semidefinite matrices (Corollary 12).

cor:traceless

Corollary 11 (Traceless eigenvalues). *Let $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ be a sextuple of candidate eigenvalues and for any list of n eigenvalues $\lambda' = (\lambda'_1 \geq \dots \geq \lambda'_n)$, let $\lambda'^0 := \lambda' - |\lambda'|/n$ denote the traceless list of eigenvalues obtained by shifting λ' by $-|\lambda'|/n$. Then $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ are tetrahedral if and only if (i) their traces $(|\alpha'|, |\beta'|, |\gamma'|, |\delta'|, |\epsilon'|, |\phi'|)$ are tetrahedral (they satisfy Eq. (12)) and (ii) $(\alpha'^0, \beta'^0, \gamma'^0, \delta'^0, \epsilon'^0, \phi'^0)$ are tetrahedral.*

Proof. This “only if” portion of the statement follows from both the tetrahedral trace conditions (Eq. (12)) and applications of the shifting symmetry Proposition 10 with (x, y, z) values

$$x = -n^{-1} \text{Tr}(A), \quad y = -n^{-1} \text{Tr}(B) \quad z = -n^{-1} \text{Tr}(D). \quad (24)$$

The “if” portion of the statement follows from the shifting symmetry applied to the negated (x, y, z) values. $\square$

The next corollary of the shifting symmetry proves that to solve the tetrahedral Horn's problem it suffices to solve the tetrahedral Horn's problem for positive definite spectra only. Indeed, one can simply take the shifting parameters (x, y, z) appearing in Proposition 10 to be sufficiently large. The following corollary establishes, in a certain sense, the minimum shifting required to ensure strictly positive eigenvalues for all six Hermitian matrices. Note that in subsequent sections we will often explicitly assume the candidate eigenvalues $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ are either non-negative or positive for the sake of technical convenience or sometimes to establish a connection with density matrices from quantum theory (which are positive semidefinite matrices with unit trace).

cor:psd

Corollary 12 (Positive eigenvalues). *If $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ are a tetrahedral sextuple of eigenvalues, then necessarily the smallest eigenvalues satisfy*

$$\gamma'_d \geq \alpha'_d + \beta'_d, \quad \phi'_d \geq \beta'_d + \delta'_d, \quad \epsilon'_d \geq \delta'_d + \gamma'_d, \quad \epsilon'_d \geq \alpha'_d + \phi'_d. \quad (25)$$

{eq:min_eig_ineqs}

Moreover, if $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ are a sextuple of candidate eigenvalues satisfying Eq. (25), then for any $(x, y, z) \in \mathbb{R}^3$ satisfying

$$x > -\alpha'_d, \quad y > -\beta'_d, \quad z > -\delta'_d, \quad (26)$$

{eq:positive_shift}

the resulting shifted eigenvalues $(\tilde{\alpha}', \tilde{\beta}', \tilde{\gamma}', \tilde{\delta}', \tilde{\epsilon}', \tilde{\phi}')$ (given by Eq. (21)) are all positive.

Proof. That tetrahedral eigenvalues satisfy Eq. (25) follows from four applications of the well-known inequality $\gamma'_d \geq \alpha'_d + \beta'_d$ for Horn's problem where $C = A + B$ [Bha01]. As a reminder, the smallest eigenvalue λ'_n of a matrix X is characterized by a minimization: $\lambda'_d = \min_{\|v\|=1} \langle v, Xv \rangle$ which produces

$$\gamma'_d = \min_{\|v\|=1} \langle v, Cv \rangle = \min_{\|v\|=1} (\langle v, Av \rangle + \langle v, Bv \rangle) \geq \min_{\|v\|=1} \langle v, Av \rangle + \min_{\|w\|=1} \langle w, Bw \rangle = \alpha'_d + \beta'_d. \quad (27)$$

Four applications of this inequality produces Eq. (25). Now if (x, y, z) satisfy Eq. (26), we have for all j

$$\alpha'_j + x \geq \alpha'_d + x > 0, \quad (28)$$

meaning $\alpha' + x = (\alpha'_1 + x \geq \dots \geq \alpha'_d + x)$ is strictly positive. Similar statements hold for $\beta', \gamma', \delta', \epsilon'$ and ϕ' ; for instance, for all j ,

$$\epsilon'_j + x + y + z \geq \epsilon'_d + x + y + z > \epsilon'_d - \alpha'_d - \beta'_d - \delta'_d \geq 0, \quad (29)$$

meaning $\epsilon' + x + y + z$ is strictly positive also. $\square$

Remark 13. Although this section listed a handful of symmetries of the tetrahedral Horn's problem, there remains the possibility of additional symmetries which are unknown at this time. For example, in the case 2×2 Hermitian matrices, an additional class of symmetries on the set of edge-lengths (a, b, c, e, d, f)

of tetrahedra, known as the *Regge symmetries*, hold [AI19; Boa07; PR68, Theorem 1.1]. If there exists a tetrahedron with edge lengths (a, b, c, d, e, f) , then there also exists a tetrahedron with edge lengths

$$(s_1 - a, s_1 - b, c, s_1 - d, s_1 - e, f) \quad \text{where} \quad s_1 = \frac{a + b + d + e}{2}. \quad (30)$$

Combining this symmetry with the permutational symmetries given by Proposition 9 (and noting that for the eigenvalues λ' of traceless 2×2 Hermitian matrices, $\lambda' = (x, -x)$, we have $\lambda'^* = \lambda'$) one concludes the co-existence of a family of six tetrahedra with edge lengths given by [VMK88, Chapter 9]:

$$\begin{array}{ll} (a, b, c, d, e, f) & (s_1 - a, s_1 - b, c, s_1 - d, s_1 - e, f) \\ (a, s_3 - b, s_3 - c, d, s_3 - e, s_3 - f) & (s_2 - a, b, s_2 - c, s_2 - d, e, s_2 - f) \\ (s_2 - d, s_1 - e, s_3 - f, s_2 - a, s_1 - b, s_3 - c) & (s_1 - d, s_3 - e, s_2 - f, s_1 - a, s_3 - b, s_2 - c) \end{array} \quad (31)$$

where

$$s_1 = \frac{a + b + d + e}{2}, \quad s_2 = \frac{a + c + d + f}{2}, \quad s_3 = \frac{b + c + e + f}{2}. \quad (32)$$

Remark 14. The Regge symmetry of a tetrahedron can be understood as a consequence of an earlier discovery of a suprising symmetry of Wigner $6j$ symbol for $SU(2)$ originally due to Regge [Reg59]. The paper due to Boalch provides a comprehensive and modern treatment of the Regge symmetry for the $6j$ symbol for $SU(2)$ by viewing it as a consequence of a simpler symmetry of the $6j$ -symbol for $SU(3)$ [Boa07]. That the Regge symmetry of a tetrahedron emerges from the Regge symmetry of the associated $6j$ symbols can be viewed as a consequence of the semiclassical limit of the latter [PR68; Rob99]. It has also be shown that the Regge symmetry of Euclidean tetrahedra is actually a scissors-congruence preserving both the volume and Dhen invariants [AI19].

At the moment, it remains unclear if there exists analogous symmetries for the generalized Wigner $6j$ symbol for $n \geq 3$ (although there are conjectured symmetries for the $6j$ symbols for $U_q(\mathfrak{sl}_n)$ which would follow from the eigenvalue hypothesis [MS18; LS23]). It could also be possible for additional symmetries to be present in the tetrahedral Horn problem for $n \geq 3$ which are *not* present in the $6j$ symbol for $SU(n)$.

3 Representation theory & eigenvalue estimation

sec:rep_theory

3.1 Schur-Weyl duality & eigenvalue estimation

Our main tool for connecting the eigenvalues of an $n \times n$ Hermitian matrix A to the representation theory of the unitary group $U(n)$ will be to examine the Schur-Weyl decomposition of the k -th tensor power $A^{\otimes k}$. To begin, we recall the Schur-Weyl decomposition of $(\mathbb{C}^n)^{\otimes k}$ as a representation of the group $S_k \times U(n)$.

Proposition 15 (Schur-Weyl duality). *Let k and n be positive integers. Then the decomposition of $(\mathbb{C}^n)^{\otimes k}$ into irreducible subrepresentations of $S_k \times U(n)$, known as the Schur-Weyl decomposition, is given by*

$$(\mathbb{C}^n)^{\otimes k} \cong \bigoplus_{\lambda \vdash_n k} W_\lambda \otimes V_\lambda^n \quad \text{(eq:schur_weyl_decomp)} \quad (33)$$

where the direct summands are indexed by Young diagrams λ with k boxes and at most n rows, and where W_λ supports the irreducible representation of S_k and V_λ^n supports the irreducible representation of $U(n)$.

The dimension of the V_λ^n is given by Weyl's dimension formula [Hal08; IN66],

$$\dim(V_\lambda^n) = \frac{\Delta_n(\tilde{\lambda})}{\Delta_n(\rho)}, \quad \text{(eq:weyl_dim)} \quad (34)$$

whereas the dimension of W_λ may be given by [GW00, Corollary 9.1.5],

$$\dim(W_\lambda) = \frac{|\lambda|!}{\tilde{\lambda}_1! \tilde{\lambda}_2! \cdots \tilde{\lambda}_n!} \Delta_n(\tilde{\lambda}) \quad \text{(eq:frob_dim)} \quad (35)$$

where $\tilde{\lambda} = \lambda + \rho$, where $\rho = (n-1, n-2, \dots, 1, 0)$ is known as the Weyl vector and where $\Delta_n(x_1, \dots, x_n) = \prod_{1 \leq i < j \leq n} (x_i - x_j)$ is the Vandermonde determinant. Note that the representation W_λ of S_k is independent of n ; indeed, the right hand side of Eq. (35) does not depend on n and coincides with the Hook-length formula [GW00, Corollary 9.1.6] (any value of n which is at least the depth of λ suffices).

Definition 16. Given a non-zero positive semidefinite $n \times n$ matrix A , a positive integer x , and a Young diagram $\alpha \vdash_n x$, let

$$\Gamma_A(\alpha) := \text{Tr}[\underline{A}^{\otimes x} P_\alpha], \quad (36)$$

where $\underline{A} = \frac{A}{\text{Tr}(A)}$ is the normalization of A and P_α projects onto the isotypic component of $(\mathbb{C}^n)^{\otimes x}$ indexed by α .

As P_α is $U(n)$ -equivariant, one observes that $\Gamma_A(\alpha)$ depends only on the eigenvalues of A . The following result, known as spectrum or eigenvalue estimation, states that the map $\alpha \mapsto \Gamma_A(\alpha)$ for A positive semidefinite is concentrated those Young diagrams α whose shape matches the shape of the eigenvalues $\alpha' = \text{eig}(A)$ [ARS88; KW01; HM02].

prop:eig_est

Proposition 17 (Eigenvalue estimation). *Let A be a positive semidefinite matrix with eigenvalues $\alpha' = \text{eig}(A)$ and let $\alpha \vdash_n x$. Then*

$$\Gamma_A(\alpha) \leq \dim(V_\alpha^n) \exp(-xD(\underline{\alpha} \parallel \underline{\alpha}')), \quad (37)$$

where $D(\underline{\alpha} \parallel \underline{\alpha}')$ denotes the Kullback-Leibler divergence of $\underline{\alpha}$ from $\underline{\alpha}'$.

Proof. See [KW01] for a proof (noting that $\text{Tr}(\underline{A}) = 1$).

□
prop:HS_measure

Proposition 18. *Let $d\mu(A)$ denote the $U(n)$ -invariant Hilbert Schmidt probability measure over the set of $n \times n$ positive semidefinite trace one matrices and let a be a positive integer. Then*

$$\mathbb{E}_A[A^{\otimes a}] := \int d\mu(A) A^{\otimes a} = \sum_{\alpha \vdash_n x} q_\alpha P_\alpha \quad \text{(eq:HS_mean)} \quad (38)$$

where the coefficient q_α equals

$$q_\alpha = \frac{\dim(V_\alpha^n)}{\dim(W_\alpha) \dim(V_{(a)}^{n^2})} \quad \text{(eq:HS_mean_coeff)} \quad (39)$$

Proof. See [ZS01] or [Wal14] for proof details. □

prop:HS_integral_uni

Proposition 19. *Let A be a positive semidefinite $n \times n$ matrix such that $\text{Tr}(A) = 1$. Then*

$$\mathbb{E}_A \Gamma_A(\alpha) := \int d\mu(A) \Gamma_A(\alpha) = \frac{\dim(V_\alpha^n)^2}{\dim(V_{(a)}^{n^2})}. \quad (40)$$

prop:completeness_rel_uni

Proposition 20. *Let A be positive semidefinite $n \times n$ matrix with $\text{Tr}(A) = 1$ with eigenvalues $\alpha' = \text{eig}(A)$ and rank $r(\alpha') = \text{rank}(A)$. Then for any positive integer a ,*

$$\sum_{\alpha \vdash_{r(\alpha')a} a} \Gamma_A(\alpha) = 1, \quad (41)$$

where the sum is over Young diagrams α with size a with depth at most $r(\alpha')$.

TC: Here I'm reformulating everything with two objectives: (i) to invoke eigenvalue estimation as late as possible, and (ii) to *discover* the “T” quantities I had defined in a more organic way. For example, the tri-gamma quantity reemerges as a summand in the binomial expansion theorem.

Definition 21. Let $(\pi_\lambda, V_\lambda^n)$ be an irreducible representation of $U(n)$ labelled by the Young diagram λ . Given an $n \times n$ matrix X , viewed as a linear operator on the defining representation $V_{(1)}^n \simeq \mathbb{C}^n$, we define $\pi_\lambda(X)$ by the formula:

$$\pi_\lambda(X) := \frac{\text{Tr}_{W_\lambda}[\iota_\lambda^\dagger X^{\otimes |\lambda|} \iota_\lambda]}{\dim(W_\lambda)}$$

where $\iota_\lambda : W_\lambda \otimes V_\lambda^n \rightarrow V_{(1)}^{\otimes |\lambda|}$ is the equivariant isometry for the λ -isotypic subspace of $V_{(1)}^{\otimes |\lambda|}$. This definition ensures that $X^{\otimes k} \simeq \bigoplus_{\lambda \vdash k} \pi_\lambda(X)^{\oplus W_\lambda}$ holds for any positive integer k . When X is a unitary, $\pi_\lambda(X)$ is the representation of X on V_λ^n , so this abuse of notation is warranted. If $X \geq 0$ is positive semidefinite, then $\pi_\lambda(X) \geq 0$ is positive semidefinite also. If $\lambda = (1)$, then $\pi_{(1)}(X) = X$ as $\dim(W_{(1)}) = 1$.

Note also that the matrix coefficients of $\pi_\lambda(X)$ are homogeneous degree $|\lambda|$ polynomials in the matrix coefficients of X meaning for any scalar $s \in \mathbb{C}$,

$$\pi_\lambda(sX) = s^{|\lambda|} \pi_\lambda(X). \quad (42)$$

Additionally, if X, Y are two matrices, we have

$$\pi_\lambda(XY) = \pi_\lambda(X)\pi_\lambda(Y). \quad (43)$$

The *character* of X satisfies

$$\chi_\lambda(X) := \text{Tr}(\pi_\lambda(X)) = \frac{\text{Tr}[P_\lambda X^{\otimes |\lambda|}]}{\dim(W_\lambda)} = s_\lambda(\text{eig}(X)),$$

where $s_\lambda(\text{eig}(X))$ is the corresponding Schur-polynomial evaluated on the eigenvalues of X . For fixed positive integer k , we also define the *Plancherel distribution* $\{\mu_\lambda | \lambda \vdash k\}$ over Young diagrams of size k according to

$$\mu_\lambda := \frac{(\dim W_\lambda)^2}{|\lambda|!}. \quad (44)$$

Moreover, we define the rescaling of $\pi_\lambda(X)$ to be

$$\tilde{\pi}_\lambda(X) := \dim(W_\lambda) \pi_\lambda(X) = \text{Tr}_{W_\lambda}[\iota_\lambda^\dagger X^{\otimes |\lambda|} \iota_\lambda],$$

and the corresponding rescaled character is

$$\tilde{\chi}_\lambda(X) := \text{Tr}[\tilde{\pi}_\lambda(X)] = \text{Tr}[P_\lambda X^{\otimes |\lambda|}] = \dim(W_\lambda) s_\lambda(\text{eig}(X)),$$

where $P_\lambda := \iota_\lambda^\dagger \iota_\lambda$ is the orthogonal projection operator onto the λ -isotypic subspace.

These rescaled characters emerge as coefficients in the following decomposition associated to an integral over Haar measure over $U(n)$,

$$\int_{U(n)} dU (UXU^{-1})^{\otimes k} = \sum_{\lambda \vdash k} \tilde{\chi}_\lambda(X) \frac{P_\lambda}{\text{Tr}(P_\lambda)}. \quad (45)$$

This normalization scheme means that $\tilde{\pi}_\lambda(X)$ is *not* a representation, while the character $\tilde{\chi}_\lambda(X)$ satisfies

$$\sum_{\lambda \vdash k} \tilde{\chi}_\lambda(X) = \sum_{\lambda \vdash k} \text{Tr}[X^{\otimes k} P_\lambda] = \text{Tr}[X^{\otimes k}] = \text{Tr}[X]^k. \quad (46)$$

Therefore, whenever X is positive semidefinite and trace one (i.e., when X is a density matrix), we have that the map $\lambda \mapsto \tilde{\chi}_\lambda(X)$ is a probability distribution when restricted to the set of Young diagrams with a fixed number k of boxes:

$$X \geq 0 \implies \tilde{\chi}_\lambda(X) \geq 0, \quad \text{and} \quad \text{Tr}(X) = 1 \implies \sum_{\lambda \vdash k} \tilde{\chi}_\lambda(X) = 1. \quad (47)$$

As the (rescaled) character $\chi_\lambda(X)$ depends only on the eigenvalues $\text{eig}(X)$ of X , we have

$$\chi_\lambda(X) = \chi_\lambda(\text{diag}(\text{eig}(X))). \quad (48)$$

Given a tuple of non-negative sorted eigenvalues $\lambda' = (\lambda'_1 \geq \dots \geq \lambda'_n \geq 0)$ summing to one, $\sum_{i=1}^n \lambda'_i = 1$, the distribution $\lambda \mapsto \tilde{\chi}_\lambda(\text{diag}(\lambda'))$ over $\lambda \vdash k$ is denoted $\text{SW}^k(\lambda')$ and called the *Schur-Weyl* distribution in [OW16].

Proposition 22 (Eigenvalue estimation). *Let $X \geq 0$ be a positive semidefinite $n \times n$ matrix with eigenvalues $\lambda' = \text{eig}(X)$. Then the Schur-Weyl distribution $\tilde{\chi}_\lambda(\underline{X}) = \text{Tr}(P_\lambda \underline{X}^{\otimes k})$ over $\lambda \vdash k$ satisfies*

$$\tilde{\chi}_\lambda(\underline{X}) \leq (k+1)^{\frac{n(n-1)}{2}} \exp(-kD(\lambda \parallel \lambda')) \quad (49)$$

and

$$\tilde{\chi}_\lambda(\underline{X}) \geq (k+n)^{-\frac{(n+2)(n-1)}{2}} \exp(-kD(\lambda \parallel \lambda')) \quad (50)$$

where $D(\lambda \parallel \lambda')$ denotes the Kullback-Leibler divergence of the normalized Young diagram λ from the normalized eigenvalue λ' .

Proposition 23 (Eigenvalue separation). *Let (α', β') be eigenvalues of positive semidefinite $n \times n$ matrices (A, B) and let $s = \|\underline{\alpha}' - \underline{\beta}'\|_1$ be the separation between the eigenvalues of $\underline{A}$ and $\underline{B}$. For each positive integer $k \geq 1$ and positive powers $p, q > 0$, define*

$$\Delta_k^{p,q}(A, B) := \sum_{\lambda \vdash k} \tilde{\chi}_\lambda(A)^p \tilde{\chi}_\lambda(B)^q. \quad (51)$$

By homogeneity, we have $\Delta_k^{p,q}(aA, bB) = a^{pk} b^{qk} \Delta_k^{p,q}(A, B)$. Moreover, we have

$$\Delta_k^{p,q}(\underline{A}, \underline{B}) \leq |\text{Par}(k, n)| m_n(k)^{p+q} \exp\left(-ks^2 \frac{pq}{2(p+q)}\right). \quad (52)$$

Therefore, if $s > 0$, then $\lim_{k \rightarrow \infty} \Delta_k^{p,q}(\underline{A}, \underline{B}) = 0$ for every $p, q > 0$. Otherwise, if $s = 0$, then $\underline{\alpha}' = \underline{\beta}'$ and

$$\Delta_k^{p,q}(\underline{A}, \underline{B}) := \sum_{\lambda \vdash k} \tilde{\chi}_\lambda(\underline{\alpha}')^{p+q} \geq \frac{1}{|\text{Par}(k, n)|^{p+q-1}}. \quad (53)$$

Of special importance is the case where $p = q = \frac{1}{2}$, in which case $\Delta_k^{\frac{1}{2}, \frac{1}{2}}(A, B)$ becomes the fidelity between the probability distributions $\tilde{\chi}_\lambda(\underline{A})$ and $\tilde{\chi}_\lambda(\underline{B})$:

$$F_k(\underline{A}, \underline{B}) := \Delta_k^{\frac{1}{2}, \frac{1}{2}}(\underline{A}, \underline{B}) = \sum_{\lambda \vdash k} \sqrt{\tilde{\chi}_\lambda(\underline{A}) \tilde{\chi}_\lambda(\underline{B})}. \quad (54)$$

Then if $s > 0$ we have

$$F_k(\underline{A}, \underline{B}) \leq |\text{Par}(k, n)| m_n(k) \exp\left(-\frac{ks^2}{8}\right). \quad (55)$$

and otherwise, $s = 0$ and

$$F_k(\underline{A}, \underline{B}) = 1. \quad (56)$$

Proof. The proof follows from eigenvalue estimation for a Young diagram $\lambda \vdash k$ which yields

$$\tilde{\chi}_\lambda(\underline{A}) \leq m_n(k) \exp(-kD(\lambda \parallel \underline{\alpha}')). \quad (57)$$

Furthermore, Pinsker's inequality yields

$$pD(\lambda \parallel \underline{\alpha}') + qD(\lambda \parallel \underline{\beta}') \geq \frac{p}{2} \|\lambda - \underline{\alpha}'\|_1^2 + \frac{q}{2} \|\lambda - \underline{\beta}'\|_1^2 \geq \frac{pq}{2(p+q)} \|\underline{\alpha}' - \underline{\beta}'\|_1^2, \quad (58)$$

where the last inequality arises because the weighted sum of squared norms is minimized when $(p+q)\lambda = p\underline{\alpha}' + q\underline{\beta}'$. For the lower bound in the case where $s = 0$, we use the fact that $\tilde{\chi}_\lambda(\underline{\alpha}')$ is a probability distribution and thus the relevant quantity is minimized for uniform distributions. $\square$

It is well-known that there is a multiplicity-free decomposition of the representation $V_{(k)}^{n^2}$ of $U(n^2)$ under the restriction to the subgroup $U(n) \times U(n)$ known as Howe-duality [How87]:

$$V_{(k)}^{n^2} \cong \bigoplus_{\lambda \vdash k} V_{\lambda}^n \otimes V_{\lambda}^n. \quad (59)$$

Consequently, this provides a probability distribution over Young diagrams $\lambda \vdash k$, which we refer to as the *Howe distribution* with probabilities given by

$$h_n(\lambda) := \frac{\dim(V_{\lambda}^n)^2}{\dim(V_{(k)}^{n^2})}. \quad (60)$$

The Howe distribution also appears when evaluating $\tilde{\pi}_{\lambda}(X)$ for X sampled randomly according to the so-called Hilbert-Schmidt measure [ZS01].

Proposition 24. *Let $X \geq 0$ be a random positive semidefinite $n \times n$ matrix such that $\text{Tr}(X) = 1$ sampled according to the Hilbert-Schmidt measure. Then for $\lambda \vdash k$,*

$$\mathbb{E}_{X \sim 1} \tilde{\pi}_{\lambda}(X) = h_n(\lambda) \frac{\text{id}_{V_{\lambda}^n}}{\dim(V_{\lambda}^n)}. \quad (61)$$

Proof. Given a measure $\mu(X)$ that is invariant under conjugation by $U(n)$, the operator $\mathbb{E}_X[X^{\otimes k}]$ is necessarily $U(n)$ equivariant and thus decomposes as

$$\mathbb{E}_{X \sim 1}[X^{\otimes k}] := \int d\mu(X) X^{\otimes k} = \sum_{\lambda \vdash k} h_{\lambda} \frac{P_{\lambda}}{\text{Tr}(P_{\lambda})}, \quad (62)$$

for some proportionality constants h_{λ} . For the Hilbert Schmidt measure, which is equivalent to the push-forward measure of the Haar measure induced by $U(n^2)$ on $\mathbb{C}^{n^2}$ under the partial trace to operators on $\mathbb{C}^n$ [ZS01], the proportionality constant given by (see also [Wal14]) the Howe distribution parameter $h_n(\lambda)$. Therefore,

$$\mathbb{E}_X[\tilde{\pi}_{\lambda}(X)] = \text{Tr}_{W_{\lambda}}[\iota_{\lambda}^{\dagger} \mathbb{E}_X[X^{\otimes |\lambda|}] \iota_{\lambda}] = h_n(\lambda) \frac{\text{Tr}_{W_{\lambda}}[\iota_{\lambda}^{\dagger} P_{\lambda} \iota_{\lambda}]}{\text{Tr}(P_{\lambda})} = h_n(\lambda) \frac{\text{Tr}_{W_{\lambda}}[\text{id}_{W_{\lambda}} \otimes \text{id}_{V_{\lambda}^n}]}{\text{Tr}(P_{\lambda})} = h_n(\lambda) \frac{\text{id}_{V_{\lambda}^n}}{\dim(V_{\lambda}^n)}. \quad (63)$$

□

Theorem 25 (Binomial theorem). *Let A, B be $n \times n$ matrices. Then*

$$\tilde{\pi}_{\gamma}(A + B) = \sum_{\alpha, \beta} \frac{|\gamma|!}{|\alpha|! |\beta|!} \Phi_{\alpha, \beta}^{\gamma}[\tilde{\pi}_{\alpha}(A) \otimes \tilde{\pi}_{\beta}(B)] \quad (64)$$

where $\Phi_{\alpha, \beta}^{\gamma}$ is the linear superoperator sending operators on $V_{\alpha}^n \otimes V_{\beta}^n$ to operators on V_{γ}^n according to the formula

$$\Phi_{\alpha, \beta}^{\gamma}[\cdot] = \text{Tr}_{C_{\alpha, \beta}^{\gamma}} \left[(\iota_{\alpha, \beta}^{\gamma})^{\dagger} \cdot (\iota_{\alpha, \beta}^{\gamma}) \right], \quad (65)$$

where $\iota_{\alpha, \beta}^{\gamma} : C_{\alpha, \beta}^{\gamma} \otimes V_{\gamma}^n \rightarrow V_{\alpha}^n \otimes V_{\beta}^n$ is the equivariant isometry for the γ -isotypic subspace of $V_{\alpha}^n \otimes V_{\beta}^n$. If $C_{\alpha, \beta}^{\gamma}$ is zero-dimensional, then $\iota_{\alpha, \beta}^{\gamma}$ and $\Phi_{\alpha, \beta}^{\gamma}$ are taken to be zero maps.

Proof. By definition, we have $\tilde{\pi}_{\gamma}(A + B) = \text{Tr}_{W_{\gamma}}[\iota_{\gamma}^{\dagger} (A + B)^{\otimes |\gamma|} \iota_{\gamma}]$. Note that we can expand $(A + B)^{\otimes k}$ using a non-commutative binomial expansion:

$$(A + B)^{\otimes k} = \sum_{\ell=0}^k \binom{k}{\ell} \frac{1}{k!} \sum_{\sigma \in S_k} \sigma^{-1} (A^{\otimes \ell} \otimes B^{\otimes (k-\ell)}) \sigma \quad (66)$$

were $\sigma \in S_k$ acts on $(\mathbb{C}^n)^{\otimes k}$ by permutation of tensor factors. However, as the superoperator $\text{Tr}_{W_\gamma}[\iota_\gamma^\dagger \cdot \iota_\gamma]$ is manifestly $S_{|\gamma|}$ -invariant (due to the partial trace over the representation space W_γ of $S_{|\gamma|}$), the normalized sum over $\sigma \in S_{|\gamma|}$ disappears and one obtains

$$\tilde{\pi}_\gamma(A + B) = \sum_{\ell=0}^k \binom{k}{\ell} \text{Tr}_{W_\gamma}[\iota_\gamma^\dagger (A^{\otimes \ell} \otimes B^{\otimes (k-\ell)}) \iota_\gamma]. \quad (67)$$

Next decompose $A^{\otimes \ell}$ (resp. $B^{\otimes (k-\ell)}$) into isotypic components indexed by $\alpha \vdash \ell$ (resp. $\beta \vdash k - \ell$) to obtain

$$A^{\otimes \ell} \otimes B^{\otimes (k-\ell)} = \sum_{\substack{\alpha \vdash \ell \\ \beta \vdash k-\ell}} (\iota_\alpha \otimes \iota_\beta) [(\text{id}_{W_\alpha} \otimes \pi_\alpha(A)) \otimes (\text{id}_{W_\beta} \otimes \pi_\beta(B))] (\iota_\alpha \otimes \iota_\beta)^\dagger. \quad (68)$$

Combining these results and noting that the $S_{|\gamma|}$ -representation space W_γ recomposes as $W_\gamma \simeq \bigoplus_{\alpha, \beta} C_{\alpha, \beta}^\gamma \otimes W_\alpha \otimes W_\beta$ under the restriction to the subgroup $S_{|\alpha|} \times S_{|\beta|}$ acting on the first ℓ and last $k - \ell$ factors respectively produces

$$\tilde{\pi}_\gamma(A + B) = \sum_{\ell=0}^k \binom{k}{\ell} \sum_{\substack{\alpha \vdash \ell \\ \beta \vdash k-\ell}} \text{Tr}_{W_\gamma}[\iota_\gamma^\dagger (\iota_\alpha \otimes \iota_\beta) [(\text{id}_{W_\alpha} \otimes \pi_\alpha(A)) \otimes (\text{id}_{W_\beta} \otimes \pi_\beta(B))] (\iota_\alpha \otimes \iota_\beta)^\dagger \iota_\gamma] \quad (69)$$

$$= \sum_{\ell=0}^k \binom{k}{\ell} \sum_{\substack{\alpha \vdash \ell \\ \beta \vdash k-\ell}} \dim(W_\alpha) \dim(W_\beta) \text{Tr}_{C_{\alpha, \beta}^\gamma}[(\iota_{\alpha, \beta}^\gamma)^\dagger \pi_\alpha(A) \otimes \pi_\beta(B) (\iota_{\alpha, \beta}^\gamma)], \quad (70)$$

$$= \sum_{\alpha, \beta} \frac{|\gamma|!}{|\alpha|! |\beta|!} \text{Tr}_{C_{\alpha, \beta}^\gamma}[(\iota_{\alpha, \beta}^\gamma)^\dagger \tilde{\pi}_\alpha(A) \otimes \tilde{\pi}_\beta(B) (\iota_{\alpha, \beta}^\gamma)], \quad (71)$$

where the final line follows as the sum in both cases is over all α, β such that $|\gamma| = |\alpha| + |\beta|$ as the multiplicity space $C_{\alpha, \beta}^\gamma$ is zero whenever $|\gamma| \neq |\alpha| + |\beta|$. $\square$

Corollary 26. *For $x > 0$, let $X \sim x$ denote a random positive semidefinite matrix sampled such that its trace is fixed, $\text{Tr}(X) = x$, and its normalization, $\underline{X} := X / \text{Tr}(X)$, is distributed according to the Hilbert-Schmidt measure. Fix $a, b, c \in \mathbb{R}_{\geq 0}$ such that $c = a + b$. Then*

$$\mathbb{E}_{\substack{A \sim a \\ B \sim b}} \tilde{\pi}_\gamma(A + B) = \text{id}_{V_\gamma^n} \sum_{\alpha, \beta} \frac{|\gamma|!}{|\alpha|! |\beta|!} \frac{a^{|\alpha|} b^{|\beta|}}{c^{|\gamma|}} \frac{c_{\alpha, \beta}^\gamma h_\alpha h_\beta}{\dim(V_\alpha^n \otimes V_\beta^n)} \quad (72)$$

Corollary 27.

$$\mathbb{E}_{A, B \sim 1} \Phi_{\alpha, \beta}^\gamma[\tilde{\pi}_\alpha(A) \otimes \tilde{\pi}_\beta(B)] = h_\alpha h_\beta \frac{c_{\alpha, \beta}^\gamma}{\dim(V_\alpha^n \otimes V_\beta^n)} \text{id}_{V_\gamma^n} \quad (73)$$

Remark 28. As a sanity check, in the case where $A = B = X$, the binomial theorem reads

$$\tilde{\pi}_\gamma(2X) = 2^{|\gamma|} \tilde{\pi}_\gamma(X), \quad (74)$$

$$= \sum_{\alpha, \beta} \frac{|\gamma|!}{|\alpha|! |\beta|!} \Phi_{\alpha, \beta}^\gamma[\tilde{\pi}_\alpha(X) \otimes \tilde{\pi}_\beta(X)], \quad (75)$$

$$= \sum_{\alpha, \beta} \frac{|\gamma|!}{|\alpha|! |\beta|!} \frac{\dim(C_{\alpha, \beta}^\gamma \otimes W_\alpha \otimes W_\beta)}{\dim(W_\gamma)} \tilde{\pi}_\gamma(X). \quad (76)$$

Now the coefficients in front of $\tilde{\pi}_\gamma(X)$ match because, for any fixed ℓ , $W_\gamma \cong \bigoplus_{\alpha \vdash \ell, \beta \vdash |\gamma| - \ell} C_{\alpha, \beta}^\gamma \otimes W_\alpha \otimes W_\beta$. Therefore, we have

$$\sum_{\alpha, \beta} \frac{|\gamma|!}{|\alpha|! |\beta|!} \frac{\dim(C_{\alpha, \beta}^\gamma \otimes W_\alpha \otimes W_\beta)}{\dim(W_\gamma)} = \sum_{\ell=0}^{|\gamma|} \binom{|\gamma|}{\ell} \sum_{\alpha \vdash \ell, \beta \vdash |\gamma| - \ell} \frac{\dim(C_{\alpha, \beta}^\gamma \otimes W_\alpha \otimes W_\beta)}{\dim(W_\gamma)} = \sum_{\ell=0}^{|\gamma|} \binom{|\gamma|}{\ell} = 2^{|\gamma|} \quad (77)$$

as expected.

Remark 29. Taking the trace of both sides of the binomial theorem yields

$$\tilde{\chi}_\gamma(A+B) = \sum_{\alpha,\beta} \frac{|\gamma|!}{|\alpha|!|\beta|!} \text{Tr}[P_{\alpha,\beta}^\gamma \tilde{\pi}_\alpha(A) \otimes \tilde{\pi}_\beta(B)] \quad (78)$$

where $P_{\alpha,\beta}^\gamma$ projects onto the γ -isotypic component of $V_\alpha^n \otimes V_\beta^n$. Define $\Gamma_{\alpha,\beta}^\gamma(A, B)$ to be the argument in the sum above:

$$\Gamma_{\alpha,\beta}^\gamma(A, B) := \text{Tr}[P_{\alpha,\beta}^\gamma \tilde{\pi}_\alpha(\underline{A}) \otimes \tilde{\pi}_\beta(\underline{B})]. \quad (79)$$

Alternatively, but equivalently, one has

$$\Gamma_{\alpha,\beta}^\gamma(A, B) = \text{Tr}[P_\gamma(P_\alpha \otimes P_\beta)(A^{\otimes|\alpha|} \otimes B^{\otimes|\beta|})]. \quad (80)$$

3.2 Littlewood-Richardson coefficients & trigonal eigenvalues

sec:lr_tri

The purpose of this section is to reveal how eigenvalue estimation can be used to prove a kind of correspondence between trigonal eigenvalue $(\alpha', \beta', \gamma')$ and Littlewood-Richardson coefficients $c_{\alpha\beta}^\gamma$. Given a triple of Hermitian matrices (A, B, C) with eigenvalues $(\alpha', \beta', \gamma')$ satisfying $A + B = C$, by eigenvalue estimation we see that each of

$$\alpha \mapsto \Gamma_A(\alpha), \quad \beta \mapsto \Gamma_B(\beta), \quad \gamma \mapsto \Gamma_C(\gamma), \quad (81)$$

are respectively concentrated around Young diagrams (α, β, γ) whose normalized shapes $(\underline{\alpha}, \underline{\beta}, \underline{\gamma})$ approximate the normalized eigenvalues $(\underline{\alpha}', \underline{\beta}', \underline{\gamma}')$. As a result of this concentration, one can detect whether there exists Hermitian matrices (A, B, C) with eigenvalues $(\alpha', \beta', \gamma')$ satisfying $A + B = C$ by examining if whether there exists an intersection of subspaces onto which P_α , P_β and P_γ project. To support this claim, we define the following quantity.

Definition 30. Given positive integers a and b , let $c = a + b$. Let (α, β, γ) be a triple of Young diagrams with sizes:

$$\alpha \vdash a, \quad \beta \vdash b, \quad \gamma \vdash c, \quad (82)$$

Define corresponding projection operators acting on $(\mathbb{C}^n)^{\otimes a+b}$ by

$$\Pi_\alpha := P_\alpha \otimes \text{id}^{\otimes b}, \quad \Pi_\beta := \text{id}^{\otimes a} \otimes P_\beta, \quad \Pi_\gamma := P_\gamma. \quad (83)$$

{eq:tri_extended_projs}

Given $n \times n$ positive semidefinite matrices A, B and $C = A + B$, let

$$\Gamma_{A,B}^\Delta(\alpha \beta \gamma) := \text{Tr}[(A^{\otimes a} \otimes B^{\otimes b})(P_\alpha \otimes P_\beta)P_\gamma] \quad (84)$$

defn:lr_coeffs

Definition 31 (Littlewood-Richardson coefficients). Let (α, β, γ) be a triple of Young diagrams with sizes (a, b, c) where $c = a + b$ and let $V_\alpha^n, V_\beta^n, V_\gamma^n$ be a triple of irreducible representations of $U(n)$. Let $C_{\alpha,\beta}^\gamma$ denote the multiplicity space of $U(n)$ -intertwiners from V_γ to $V_\alpha \otimes V_\beta$:

$$C_{\alpha,\beta}^\gamma := \text{Hom}_{U(n)}(V_\gamma, V_\alpha \otimes V_\beta) \simeq \text{Inv}_{U(n)}(V_\gamma^* \otimes V_\alpha \otimes V_\beta). \quad (85)$$

The *Littlewood-Richardson coefficient* is defined as the dimension of this multiplicity space:

$$c_{\alpha,\beta}^\gamma := \dim(C_{\alpha,\beta}^\gamma). \quad (86)$$

The *Clebsch-Gordon isometry* is the $U(n)$ -equivariant isometric linear map

$$Y_{\alpha,\beta}^\gamma : C_{\alpha,\beta}^\gamma \otimes V_\gamma \rightarrow V_\alpha \otimes V_\beta, \quad (87)$$

sending $f \otimes v \in C_{\alpha,\beta}^\gamma \otimes V_\gamma$ to $f(v) \in V_\alpha \otimes V_\beta$ ⁵. The Clebsch-Gordon isometries (and coisometries) will be depicted diagrammatically with multiplicity spaces given a distinguished style as in [AB68]:

$$\begin{array}{ccc} \begin{array}{c} V_\alpha \quad V_\beta \\ \diagdown \quad \diagup \\ \bigcirc \\ \diagup \quad \diagdown \\ C_{\alpha,\beta}^\gamma \quad V_\gamma \end{array} & \equiv & \begin{array}{c} V_\alpha \quad V_\beta \\ \diagdown \quad \diagup \\ \boxed{Y_{\alpha,\beta}^\gamma} \\ \diagup \quad \diagdown \\ C_{\alpha,\beta}^\gamma \quad V_\gamma \end{array} \end{array} \quad \begin{array}{ccc} \begin{array}{c} C_{\alpha,\beta}^\gamma \quad V_\gamma \\ \diagdown \quad \diagup \\ \bigcirc \\ \diagup \quad \diagdown \\ V_\alpha \quad V_\beta \end{array} & \equiv & \begin{array}{c} C_{\alpha,\beta}^\gamma \quad V_\gamma \\ \diagdown \quad \diagup \\ \boxed{Y_{\alpha,\beta}^{\gamma,*}} \\ \diagup \quad \diagdown \\ V_\alpha \quad V_\beta \end{array} \end{array} \quad (88)$$

prop:tri_HS_mean

Proposition 32. Let (α, β, γ) be Young diagrams with sizes (a, b, c) where $c = a + b$. Then

$$\mathbb{E}_{A,B} \Gamma_{A,B}^\Delta(\alpha \beta \gamma) = h_\alpha h_\beta \frac{\dim V_\gamma^n}{\dim V_\alpha^n \otimes V_\beta^n} c_{\alpha,\beta}^\gamma = R_{\alpha,\beta;\gamma} c_{\alpha,\beta}^\gamma \quad (89)$$

where $c_{\alpha,\beta}^\gamma$ is the Littlewood-Richardson coefficient and $R_{\alpha,\beta;\gamma}$ is the ratio of dimensions:

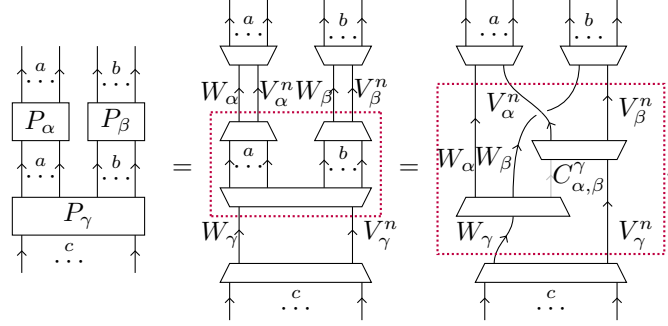
$$R_{\alpha,\beta;\gamma} = \frac{\dim(V_\alpha^n) \dim(V_\beta^n) \dim(V_\gamma^n)}{\dim(V_{(a)}^{n^2}) \dim(V_{(b)}^{n^2})}. \quad (90)$$

⁵If $C_{\alpha,\beta}^\gamma = 0$, then $Y_{\alpha,\beta}^\gamma$ is identified with the zero map.

Proof. Using Proposition 18 we obtain

$$\mathbb{E}_{A,B} \Gamma_{A,B}^\Delta(\alpha \beta \gamma) = q_\alpha q_\beta \text{Tr}[(P_\alpha \otimes P_\beta) P_\gamma], \quad \text{\textcolor{blue}{\{eq:mean_tri_gamma\}}} \quad (91)$$

where q_α and q_β are given by Eq. (39). Moreover, the operator $(P_\alpha \otimes P_\beta) P_\gamma$ decomposes as



$$\text{\textcolor{blue}{\{eq:lr_triple_prod\}}} \quad (92)$$

and therefore taking the trace yields

$$\text{Tr}[(P_\alpha \otimes P_\beta) P_\gamma] = \dim(W_\alpha) \dim(W_\beta) \dim(C_{\alpha,\beta}^\gamma) \dim(V_\gamma^n). \quad \text{\textcolor{blue}{\{eq:trace_tri_formula\}}} \quad (93)$$

Combining Eq. (91) and Eq. (93) produces the claimed formula as $c_{\alpha,\beta}^\gamma = \dim(C_{\alpha,\beta}^\gamma)$. $\square$

Our first result establishes a connection between $\Gamma_{A,B}^\Delta(\alpha \beta \gamma)$ and the corresponding Littlewood-Richardson coefficient.

cor:tri_indep_upper_bound

Corollary 33. *Let (α, β, γ) be Young diagrams with sizes (a, b, c) satisfying $c = a + b$. Then for any positive semidefinite matrices (A, B, C) satisfying $C = A + B$, we have*

$$c_{\alpha,\beta}^\gamma = 0 \implies \Gamma_{A,B}^\Delta(\alpha \beta \gamma) = 0. \quad \text{\textcolor{blue}{\{eq:tri_gamma_lr_upper_bound\}}} \quad (94)$$

Proof. From the definition of $\Gamma_{A,B}^\Delta(\alpha \beta \gamma)$, and the fact that $\underline{A}^{\otimes a} \otimes \underline{B}^{\otimes b}$ is trace one positive semidefinite, we have the bound

$$\Gamma_{A,B}^\Delta(\alpha \beta \gamma) \leq \|(P_\alpha \otimes P_\beta) P_\gamma\|_\infty \quad (95)$$

Using the decomposition of $(P_\alpha \otimes P_\beta) P_\gamma$ into isometries given by Eq. (92), we also have

$$\|(P_\alpha \otimes P_\beta) P_\gamma\|_\infty = \|\text{id}_{C_{\alpha,\beta}^\gamma}\|_\infty = \begin{cases} 1 & c_{\alpha,\beta}^\gamma > 0 \\ 0 & c_{\alpha,\beta}^\gamma = 0 \end{cases}. \quad (96)$$

Therefore if $c_{\alpha,\beta}^\gamma = 0$, then $(P_\alpha \otimes P_\beta) P_\gamma = 0$ and thus $\Gamma_{A,B}^\Delta(\alpha \beta \gamma) = 0$ also. $\square$

The following proposition is the analogue of Proposition 20.

prop:tri_completeness

Proposition 34. *Let (A, B, C) be positive semidefinite $n \times n$ matrices satisfying $C = A + B$ with eigenvalues $(\alpha', \beta', \gamma')$. Then*

$$\sum_{\alpha,\beta,\gamma} \Gamma_{A,B}^\Delta(\alpha \beta \gamma) = 1. \quad \text{\textcolor{blue}{\{eq:tri_completeness\}}} \quad (97)$$

where the sum is taken over Young diagrams with the following sizes:

$$\alpha \vdash_{r(\alpha')} a, \quad \beta \vdash_{r(\beta')} b, \quad \gamma \vdash_{r(\gamma')} c. \quad \text{\textcolor{blue}{\{eq:three_yd_sizes_rank\}}} \quad (98)$$

Proof. To begin, note that Eq. (97) holds when the sum is taken over Young diagrams with unrestricted rank, meaning (α, β, γ) have sizes (a, b, c) with $c = a + b$ but arbitrary rank/depth. That one can restrict to Young diagrams with depths bounded according to Eq. (98) follows because $\Gamma_{A,B}^\Delta(\alpha \beta \gamma) = 0$ whenever the depth of *any* of (α, β, γ) exceeds the corresponding ranks $(r(\alpha'), r(\beta'), r(\gamma'))$. $\square$

Theorem 35. Let $(\alpha', \beta', \gamma')$ be a triple of eigenvalues of non-zero positive semidefinite (A, B, C) satisfying $C = A + B$. Let (α, β, γ) be Young diagrams with sizes given by

$$\alpha \vdash a = \lceil k |\alpha'| \rceil, \quad \beta \vdash b = \lceil k |\beta'| \rceil, \quad \gamma \vdash c = \lceil k |\alpha'| \rceil + \lceil k |\beta'| \rceil, \quad \text{\textcolor{blue}{\eq:ceil_tri_yd_sizes}} \quad (99)$$

for any positive integer $k \geq \tau_{\min}^{-1}$ where $\tau_{\min} = \min\{|\alpha'|, |\beta'|, |\gamma'|\}$. Then

$$\Gamma_{A,B}^{\Delta}(\alpha \beta \gamma) \leq p_{\Delta}(k) \exp(-k I_{\alpha'\beta'\gamma'}^{\alpha\beta\gamma}/2). \quad \text{\textcolor{blue}{\eq:tri_gamma_upper_bound}} \quad (100)$$

where $p_{\Delta}(k)$ grows polynomially in k and where $I_{\alpha'\beta'\gamma'}^{\alpha\beta\gamma}$ is a quantity which measures the deviation of the primed quantities from the unprimed quantities:

$$I_{\alpha'\beta'\gamma'}^{\alpha\beta\gamma} = |\alpha'| D(\underline{\alpha} \parallel \underline{\alpha}') + |\beta'| D(\underline{\beta} \parallel \underline{\beta}') + |\gamma'| D(\underline{\gamma} \parallel \underline{\gamma}'). \quad (101)$$

Proof. The first portion of the proof does not depend on the value of k . For now, we suppose the Young diagrams (α, β, γ) have sizes (a, b, c) with $c = a + b$. Then

$$\Gamma_{A,B}^{\Delta}(\alpha \beta \gamma)^2 = \text{Tr}[(\underline{A}^{\otimes a} \otimes \underline{B}^{\otimes b})(P_{\alpha} \otimes P_{\beta})P_{\gamma}]^2, \quad (102)$$

$$\leq \text{Tr}[\underline{A}^{\otimes a} P_{\alpha}] \text{Tr}[\underline{B}^{\otimes b} P_{\beta}] \text{Tr}[(\underline{A}^{\otimes a} \otimes \underline{B}^{\otimes b})P_{\gamma}] \quad (103)$$

The third term can be relaxed further to an upper bound in terms of $C = A + B$ because $C^{\otimes c} = (A + B)^{\otimes a+b}$ and therefore

$$\text{Tr}(P_{\gamma} C^{\otimes c}) = \sum_{j=0}^c \binom{c}{j} \text{Tr}(P_{\gamma} (A^{\otimes j} \otimes B^{\otimes c-j})). \quad (104)$$

Here we have used the S_c -equivariance of P_{γ} . Since each of the summands above are non-negative, we conclude for all $0 \leq j \leq c$ that

$$\text{Tr}(P_{\gamma} C^{\otimes c}) \geq \binom{c}{j} \text{Tr}(P_{\gamma} (A^{\otimes j} \otimes B^{\otimes c-j})). \quad (105)$$

Letting $j = a$ and $c - j = c - a = b$ and rearranging yields

$$\text{Tr}(P_{\gamma} (A^{\otimes a} \otimes B^{\otimes b})) \leq \frac{a!b!}{c!} \text{Tr}(P_{\gamma} C^{\otimes c}). \quad (106)$$

Altogether we obtain

$$\Gamma_{A,B}^{\Delta}(\alpha \beta \gamma)^2 \leq \frac{a!b!}{c!} \frac{|\gamma'|^c}{|\alpha'|^a |\beta'|^b} \text{Tr}[\underline{A}^{\otimes a} P_{\alpha}] \text{Tr}[\underline{B}^{\otimes b} P_{\beta}] \text{Tr}[\underline{C}^{\otimes c} P_{\gamma}] \quad \text{\textcolor{blue}{\eq:tri_gamma_abc_upper_bound}} \quad (107)$$

Our objective now will be to use the k -dependent values for (a, b, c) to relax this upper bound further. From Eq. (99), we obtain the following estimates:

$$k |\alpha'| \leq a < k |\alpha'| + 1, \quad k |\beta'| \leq b < k |\beta'| + 1, \quad k |\gamma'| \leq c < k |\gamma'| + 2. \quad (108)$$

An application of Lemma 54 for binomial distributions ($m = 2$) implies the following bound

$$\frac{c!}{a!b!} \frac{|\alpha'|^a |\beta'|^b}{|\gamma'|^c} \geq 12^2 (k |\gamma'| + 2)^{\frac{1}{2}}. \quad (109)$$

Moreover, from Weyl's dimension formula, Eq. (34), we obtain [CM06, Equation 7]:

$$\dim(V_{\gamma}^n) \leq (|\gamma| + 1)^{n(n-1)/2} \leq (k |\gamma'| + 3)^{n(n-1)/2}, \quad (110)$$

because $(\gamma_i - i) - (\gamma_j - j) \leq |\gamma| + 1$ as well as $j - i \geq 1$. Using eigenvalue estimation, Proposition 17, we can relax the bound on $\text{Tr}[\underline{C}^{\otimes c} P_{\gamma}]$ to obtain a bound in terms of k :

$$\text{Tr}[\underline{C}^{\otimes c} P_{\gamma}] \leq \dim(V_{\gamma}^n) \exp(-c D(\underline{\gamma} \parallel \underline{\gamma}')) \leq (k |\gamma'| + 3)^{n(n-1)/2} \exp(-k |\gamma'| D(\underline{\gamma} \parallel \underline{\gamma}')) \quad (111)$$

with similar bounds on $\text{Tr}[A^{\otimes a} P_\alpha]$ and $\text{Tr}[B^{\otimes b} P_\beta]$. Combining these bounds together yields

$$\Gamma_{A,B}^\Delta(\alpha_k \beta_k \gamma_k)^2 \leq 12^2 (k |\gamma'| + 2)^{\frac{1}{2}} [(k |\alpha'| + 2)(k |\beta'| + 2)(k |\gamma'| + 3)]^{n(n-1)/2} \exp(-k I_{\alpha' \beta' \gamma'}^{\alpha \beta \gamma}). \quad (112)$$

Therefore, the claim holds if $p_\Delta(k)$ is taken to be

$$p_\Delta(k) = 12(k\tau_{\max} + 3)^{\frac{3n(n-1)+1}{4}}. \quad \text{\texttt{eq:p_of_k_tri}} \quad (113)$$

where $\tau_{\max} = \max\{|\alpha'|, |\beta'|, |\gamma'|\}$. $\square$

While the previous theorem states that $\Gamma_{A,B}^\Delta(\alpha \beta \gamma)$ concentrates in magnitude around particular Young diagrams, it is not directly applicable to Horn problem simply because its value depends on more than just the eigenvalues of the matrices (A, B, C) . Nevertheless, combining the previous theorem the completeness relation (Eq. (97)) yields a necessary condition for $(\alpha', \beta', \gamma')$ to be tetrahedral.

Corollary 36. *Let $(\alpha', \beta', \gamma')$ be the eigenvalues of non-zero positive semidefinite matrices (A, B, C) satisfying $C = A + B$ with minimum trace $\tau_{\min}$. Then there exists a sextuple of sequences of Young diagrams $(\alpha_k, \beta_k, \gamma_k)$ indexed by positive integers $k > \tau_{\min}^{-1}$ with sizes given by Eq. (99) satisfying (i) the convergence condition*

$$I_{\alpha' \beta' \gamma'}^{\alpha_k \beta_k \gamma_k} \leq \frac{2}{k} \log p_\Delta(k) q_\Delta(k) \xrightarrow{k \rightarrow \infty} 0, \quad \text{\texttt{eq:tri_convergence_rate}} \quad (114)$$

where $q_\Delta(k) = (k\tau_{\max} + 2)^{3(r_{\max}-1)}$, and (ii) the non-vanishing of the Littlewood-Richardson coefficient

$$c_{\alpha_k, \beta_k}^{\gamma_k} \neq 0. \quad (115)$$

Proof. To proof the claim, we construct the sequence implicitly. For each k , it suffices to take $(\alpha_k, \beta_k, \gamma_k)$ to be any sextuple of Young diagrams which maximizes the value of $\Gamma_{A,B}^\Delta(\alpha \beta \gamma)$ with sizes determined by Eq. (99):

$$(\alpha_k, \beta_k, \gamma_k) := \underset{\alpha \beta \gamma}{\operatorname{argmax}} \Gamma_{A,B}^\Delta(\alpha \beta \gamma). \quad (116)$$

While this maximal choice may not be unique, it will necessarily satisfy a lower bound due to the completeness relation satisfied by $\Gamma_{A,B}^\Delta(\alpha \beta \gamma)$ Eq. (97). Specifically, this maximal choice must satisfy

$$\Gamma_{A,B}^\Delta(\alpha_k \beta_k \gamma_k) \geq \frac{1}{q_\Delta(k)} \quad \text{\texttt{eq:tri_gamma_lower_bound}} \quad (117)$$

where $q_\Delta(k)$ is any upper bound on the number of possible summands in the completeness relation Eq. (97) viewed as a function of k . As there less than or equal to $(s+1)^{r-1}$ Young diagrams with r rows with size at most s , letting $q_\Delta(k)$ be as stated in the claim ensures Eq. (117) holds because the size (resp. rank) of any of $(\alpha_k, \beta_k, \gamma_k)$ is at most $k\tau_{\max} + 2$ (resp. $r_{\max}$). Moreover, applying Eq. (100) yields

$$I_{\alpha' \beta' \gamma'}^{\alpha_k \beta_k \gamma_k} \leq \frac{2}{k} \log(q_\Delta(k) p_\Delta(k)) \quad (118)$$

and thus Eq. (114) holds because $p_\Delta(k)$ and $q_\Delta(k)$ are polynomials in k with coefficients depending on the maximum trace $\tau_{\max}$ and maximum rank $r_{\max}$.

The non-vanishing of the Littlewood-Richardson coefficient is implied by Corollary 33. $\square$

Remark 37. The convergence condition above (Eq. (114)) in conjunction with the size conditions given by Eq. (99) together ensure that the Young diagrams $(\alpha_k, \beta_k, \gamma_k)$ converge to the eigenvalues $(\alpha', \beta', \gamma')$ in the following sense that

$$\frac{\alpha_k}{k} \rightarrow \alpha', \quad \frac{\beta_k}{k} \rightarrow \beta', \quad \frac{\gamma_k}{k} \rightarrow \gamma', \quad \text{\texttt{eq:tri_convergence}} \quad (119)$$

where $\frac{\lambda_k}{k} \rightarrow \lambda'$ is shorthand for $\lim_{k \rightarrow \infty} (\lambda_k)_i / k = \lambda'_i$ for all $i \in \{1, \dots, n\}$.

While the existence of a convergent sequence $(\alpha_k, \beta_k, \gamma_k)$ of primed Young diagrams is necessary for $(\alpha', \beta', \gamma')$ to be tetrahedral by Corollary 47, our next result proves it is also sufficient. In other words, if $(\alpha', \beta', \gamma')$ are not tetrahedral, then no such sequence can exist.

Corollary 38. Let $(\alpha', \beta', \gamma')$ be a non-trigonal triple of positive semidefinite and non-zero eigenvalues. Then for all sequences of Young diagrams $(\alpha_k, \beta_k, \gamma_k)$ with sizes given by Eq. (99) and converging to $(\alpha', \beta', \gamma')$ in the sense that Eq. (114) holds, one necessarily has

$$\forall k \geq K : \quad c_{\alpha_k, \beta_k}^{\gamma_k} = 0. \quad (120)$$

for some sufficiently large K .

Proof. **TODO:** write the proof using Proposition 32. □

TODO: Summarize [Chr08]. **TODO:** Briefly mention the direct and pedagogical Kempf-Ness style proofs of stronger results in [BVW18, Proposition 2.3] also using the strong duality result of [FW20, Theorem 1 & Proposition 3.1] and the quantitative Kempf-Ness theorem [Bür+19, Theorem 1.17].

3.3 Generalized 6j symbols & tetrahedral eigenvalues

sec:sixj-tet

The purpose of this section is to introduce a representation theoretic object, namely the generalized Wigner 6j symbol, and then show how its asymptotic properties encode whether a given sextuple $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ is tetrahedral in a manner which analogous to how the Littlewood-Richardson coefficient encodes whether a given triple $(\alpha', \beta', \gamma')$ is trigonal. For the case where $n = 2$, this correspondence is well-known [Rob99; PR68; Bac10], while for the case where $n \geq 3$, this correspondence was already established indirectly by Christandl, Şahinoğlu, and Walter [CŞW18]. Here we establish this correspondence directly.

defn:tet_gamma

Definition 39. Given positive integers a, b , and d , let $c = a + b$, $f = b + d$, and $e = a + b + d$. Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be a sextuple of Young diagrams with sizes:

$$\alpha \vdash a, \quad \beta \vdash b, \quad \gamma \vdash c, \quad \delta \vdash d, \quad \epsilon \vdash e, \quad \phi \vdash f. \quad (121)$$

{eq:six_yd_sizes}

Define corresponding projection operators acting on $(\mathbb{C}^n)^{\otimes a+b+d}$ by

$$\begin{aligned} \Pi_\alpha &:= P_\alpha \otimes \text{id}^{\otimes b} \otimes \text{id}^{\otimes d}, & \Pi_\beta &:= \text{id}^{\otimes a} \otimes P_\beta \otimes \text{id}^{\otimes d}, & \Pi_\gamma &:= P_\gamma \otimes \text{id}^{\otimes d}, \\ \Pi_\delta &:= \text{id}^{\otimes a} \otimes \text{id}^{\otimes b} \otimes P_\delta, & \Pi_\epsilon &:= P_\epsilon, & \Pi_\phi &:= \text{id}^{\otimes a} \otimes P_\phi. \end{aligned} \quad (122)$$

{eq:tet_extended_projs}

Given $n \times n$ positive semidefinite matrices A, B, D with $C = A + B$, $F = B + D$, and $E = A + B + D$, let

$$\Gamma_{A,B,D}^{\otimes} \begin{pmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{pmatrix} = \text{Tr}[(\underline{A}^{\otimes a} \otimes \underline{B}^{\otimes b} \otimes \underline{D}^{\otimes d}) \Pi_\alpha \Pi_\beta \Pi_\gamma \Pi_\delta \Pi_\epsilon \Pi_\phi]. \quad (123)$$

Remark 40. In the analysis that follows, it will be convenient to avoid certain edge cases pertaining to the candidate sextuple of positive semidefinite eigenvalues $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$. Specifically, let the extremal traces be denoted by

$$\tau_{\min} := \min\{|\alpha'|, |\beta'|, |\gamma'|, |\delta'|, |\epsilon'|, |\phi'|\}, \quad \tau_{\max} := \max\{|\alpha'|, |\beta'|, |\gamma'|, |\delta'|, |\epsilon'|, |\phi'|\}, \quad (124)$$

and the extremal ranks be denoted by

$$r_{\min} := \min\{r(\alpha'), r(\beta'), r(\gamma'), r(\delta'), r(\epsilon'), r(\phi')\}, \quad r_{\max} := \max\{r(\alpha'), r(\beta'), r(\gamma'), r(\delta'), r(\epsilon'), r(\phi')\}. \quad (125)$$

In order to avoid the edge case where one of the eigenvalues $\lambda' \in \{\alpha', \beta', \gamma', \delta', \epsilon', \phi'\}$ is totally zero, we sometimes demand that the minimum trace is strictly positive or equivalently that the minimum rank is at least one:

$$\tau_{\min} > 0, \quad r_{\min} \geq 1. \quad (126)$$

Another technical difficulty which complicates subsequent analysis is that the set of projection operators $\{\Pi_\alpha, \Pi_\beta, \Pi_\gamma, \Pi_\delta, \Pi_\epsilon, \Pi_\phi\}$ defined by Eq. (122) is *almost* pairwise mutually commuting.

lem:tet_com_rels

Lemma 41. Every pair of projection operators in $\{\Pi_\alpha, \Pi_\beta, \Pi_\gamma, \Pi_\delta, \Pi_\epsilon, \Pi_\phi\}$ is commuting with only possible exception being Π_γ and Π_ϕ . Therefore, the operators $\Pi_{\alpha\beta\delta\epsilon}^\gamma$ and $\Pi_{\alpha\beta\delta\epsilon}^\phi$ defined by

$$\Pi_{\alpha\beta\delta\epsilon}^\gamma = \Pi_\gamma \Pi_\alpha \Pi_\beta \Pi_\delta \Pi_\epsilon, \quad \text{and} \quad \Pi_{\alpha\beta\delta\epsilon}^\phi = \Pi_\phi \Pi_\alpha \Pi_\beta \Pi_\delta \Pi_\epsilon, \quad (127)$$

{eq:commuting_extended_projs}

are projection operators.

Proof. That all projection operators except the pair $\{\Pi_\gamma, \Pi_\phi\}$ commute follows from elementary applications of Schur's lemma. The obstruction for the pair $\{\Pi_\gamma, \Pi_\phi\}$ is simply that Π_γ is not necessarily $\text{SU}(n)$ -equivariant with respect to the diagonal action on $\mathbb{C}^{\otimes b+d}$ and likewise Π_ϕ is not necessarily $\text{SU}(n)$ -equivariant with respect to the diagonal action on $\mathbb{C}^{\otimes a+b}$. $\square$

[defn:sixj](#)

Definition 42 (Generalized 6j operators). Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be a sextuple of Young diagrams with sizes (a, b, c, d, e, f) given by Eq. (121). Then we define the $\text{U}(n)$ -6j operator to be the operator⁶

$$\left\{ \begin{matrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{matrix} \right\} : C_{\alpha, \beta}^\gamma \otimes C_{\delta, \gamma}^\epsilon \rightarrow C_{\alpha, \phi}^\epsilon \otimes C_{\beta, \delta}^\phi, \quad (128)$$

defined by

$$\left\{ \begin{matrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{matrix} \right\} := \frac{1}{\dim(V_\epsilon)} \text{Tr}_{V_\epsilon} [Y_{\alpha, \phi}^{\epsilon \dagger} (\text{id}_{V_\alpha} \otimes Y_{\beta, \delta}^{\phi \dagger}) (Y_{\alpha, \beta}^\gamma \otimes \text{id}_{V_\delta}) Y_{\gamma, \delta}^\epsilon], \quad (129)$$

[eq:sixj](#)

and depicted diagrammatically by

$$\left\{ \begin{matrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{matrix} \right\} := \frac{1}{\dim(V_\epsilon)} V_\alpha V_\beta V_\gamma V_\delta V_\epsilon = \frac{1}{\dim(V_\epsilon)} \text{tetrahedron diagram} \quad (130)$$

The following proposition is the analogue of Proposition 19.

[prop:HS_integral_tet](#)

Proposition 43. Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be Young diagrams with sizes given by Eq. (121). Then

$$\begin{aligned} \mathbb{E}_{A, B, D} \Gamma_{A, B, D}^{\Delta} \left(\begin{matrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{matrix} \right) &:= \iiint d\mu(A) d\mu(B) d\mu(D) \Gamma_{A, B, D}^{\Delta} \left(\begin{matrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{matrix} \right) \\ &= h_n(\alpha) h_n(\beta) h_n(\delta) \frac{\dim V_\epsilon^n}{\dim(V_\alpha^n \otimes V_\beta^n \otimes V_\delta^n)} \left\| \left\{ \begin{matrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{matrix} \right\} \right\|_2^2. \end{aligned} \quad (131)$$

[eq:tet_mean_value](#)

Proof. To prove Eq. (131), we use Proposition 18 and Lemma 41 to see that

$$\mathbb{E}_{A, B, D} \Gamma_{A, B, D}^{\Delta} \left(\begin{matrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{matrix} \right) = q_\alpha q_\beta q_\delta \text{Tr}(\Pi_\alpha \Pi_\beta \Pi_\gamma \Pi_\delta \Pi_\epsilon \Pi_\phi). \quad (132)$$

[eq:tet_mean_step_1](#)

We claim that the product $\Pi_\alpha \Pi_\beta \Pi_\gamma \Pi_\delta \Pi_\epsilon \Pi_\phi$ is related to the generalized 6j operator $\left\{ \begin{matrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{matrix} \right\}$ in that there exists appropriately-typed isometries ι_I and ι_{II} such that

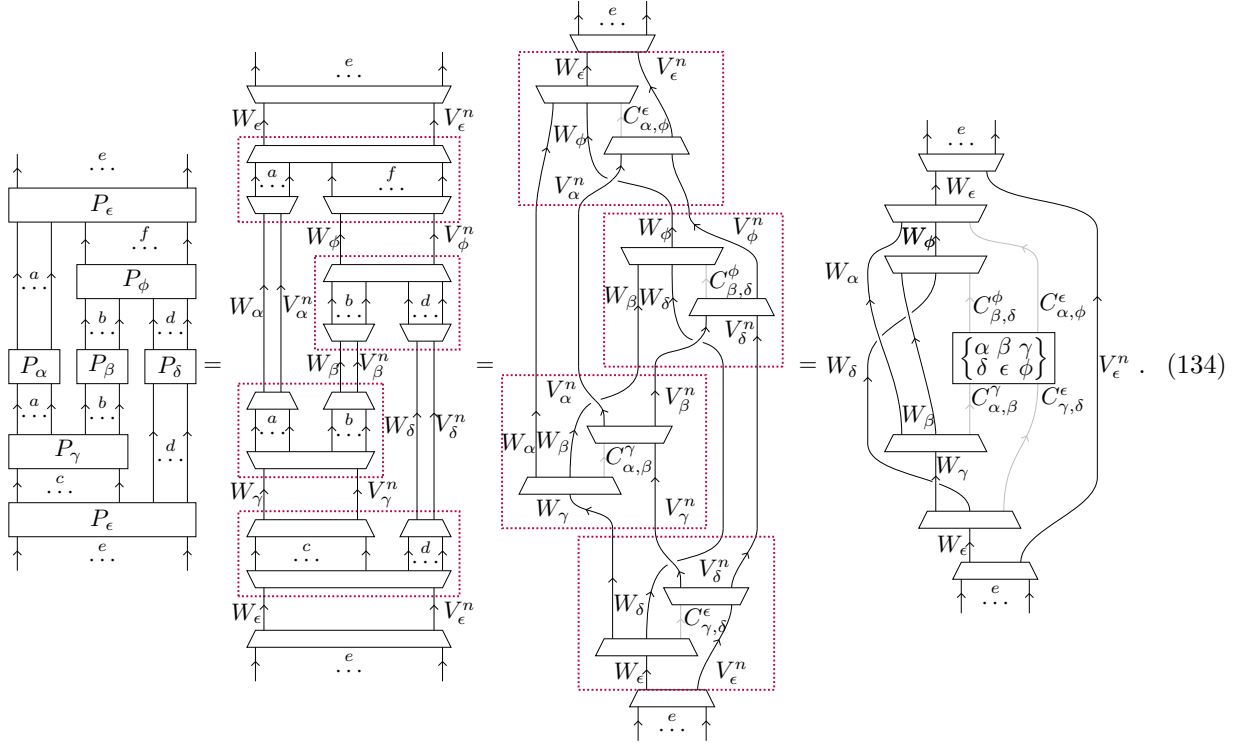
$$\Pi_\alpha \Pi_\beta \Pi_\gamma \Pi_\delta \Pi_\epsilon \Pi_\phi = \iota_{II} \left(\text{id}_{V_{\epsilon'}^n} \otimes \text{id}_{W_{\alpha'}} \otimes \text{id}_{W_{\beta'}} \otimes \text{id}_{W_{\delta'}} \otimes \left\{ \begin{matrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{matrix} \right\} \right) \iota_I^{\dagger} \quad (133)$$

[eq:ud_sixj_prod_proj_iso](#)

The form of these isometries can be deduced by applying Eq. (92) four times in the following diagrammatic

⁶It is important to note that there does not seem to exist a universal convention concerning how 6j symbols are normalized; Coquereaux discusses this problem in [Coq23].

calculation:



From Eq. (133) we conclude

$$\text{Tr}(\Pi_\alpha \Pi_\beta \Pi_\gamma \Pi_\delta \Pi_\epsilon \Pi_\phi) = \dim(V_\epsilon^n) \dim(W_{\alpha'}) \dim(W_{\beta'}) \dim(W_{\delta'}) \left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_2^2, \quad (135)$$

Combining Eq. (135) with Eq. (132) and Eq. (39) then yields Eq. (131). $\square$

The following proposition is the analogue of Proposition 20.

Proposition 44. *Let (A, B, C, D, E, F) be tetrahedral positive semidefinite matrices with eigenvalues $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$. Then*

$$\sum_{\substack{\alpha\beta\gamma \\ \delta\epsilon\phi}} \Gamma_{A,B,D}^{\Diamond} \left(\begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right) = 1. \quad (136)$$

where the sum is taken over Young diagrams with the following sizes:

$$\alpha \vdash_{r(\alpha')} a, \quad \beta \vdash_{r(\beta')} b, \quad \gamma \vdash_{r(\gamma')} c, \quad \delta \vdash_{r(\delta')} d, \quad \epsilon \vdash_{r(\epsilon')} e, \quad \phi \vdash_{r(\phi')} f. \quad (137)$$

Proof. To begin, note that Eq. (136) holds when the sum is taken over Young diagrams with unrestricted rank according to Eq. (121) because $\sum_{\lambda \vdash_n \ell} P_\lambda = \text{id}_n^{\otimes \ell}$ for all positive integers ℓ and $\lambda \in \{\alpha, \beta, \gamma, \delta, \epsilon, \phi\}$. That one can restrict to Young diagrams with depths bounded according to Eq. (137) follows because $\Gamma_{A,B,D}^{\Diamond} \left(\begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right) = 0$ whenever the depth of *any* of $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ exceeds the corresponding ranks $(r(\alpha'), r(\beta'), r(\gamma'), r(\delta'), r(\epsilon'), r(\phi'))$.⁷ $\square$

Our next goal will be to derive two upper bounds on the absolute value of $\Gamma_{A,B,D}^{\Diamond} \left(\begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right)$. Our first upper bound is *independent* of the given matrices and follows as a corollary of Lemma 41.

⁷This claim actually follows from Theorem 46 and will be proven shortly. Importantly, Theorem 46 does not rely on Proposition 44.

cor:abd_indep_upper_bound

Corollary 45. Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be Young diagrams of sizes given by Eq. (121). Then for all tetrahedral, non-zero positive semidefinite (A, B, C, D, E, F) , we have

$$\left| \Gamma_{A,B,D}^{\boxtimes} \left(\begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right) \right| \leq \left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_{\infty}. \quad \text{feq:tet_gamma_sizej_upper_bound} \quad (138)$$

Proof. From Eq. (127) we have

$$\Gamma_{A,B,D}^{\boxtimes} \left(\begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right) = \text{Tr}(\Pi_{\alpha\beta\delta\epsilon}^{\phi} (\underline{A}^{\otimes a} \otimes \underline{B}^{\otimes b} \otimes \underline{D}^{\otimes d}) \Pi_{\alpha\beta\delta\epsilon}^{\gamma}). \quad (139)$$

Moreover, as $\Omega := \underline{A}^{\otimes a} \otimes \underline{B}^{\otimes b} \otimes \underline{D}^{\otimes d}$ is positive semidefinite trace one, and $\Pi_1 := \Pi_{\alpha\beta\delta\epsilon}^{\phi}$ and $\Pi_2 := \Pi_{\alpha\beta\delta\epsilon}^{\gamma}$ are projection operators by Lemma 41, we have

$$\left| \Gamma_{A,B,D}^{\boxtimes} \left(\begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right) \right| = |\text{Tr}(\Pi_1 \Omega \Pi_2)| \leq |\langle v, \Pi_1 \Pi_2 v \rangle| \leq \|\Pi_1 \Pi_2\|_{\infty} = \left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_{\infty}. \quad (140)$$

where the last equality follows from the isometry decomposition provided by Eq. (133). $\square$

Our second upper bound on the magnitude of $\Gamma_{A,B,D}^{\boxtimes} \left(\begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right)$ depends on the *eigenvalues* $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ of the given tetrahedral matrices (A, B, C, D, E, F) where $C = A + B$, $F = B + D$ and $E = A + B + D$. This result proves that in the limit of large Young diagrams, the magnitude of $\Gamma_{A,B,D}^{\boxtimes} \left(\begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right)$ is sharply concentrated around those Young diagrams $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ whose shapes resemble the sextuple of eigenvalues $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ of (A, B, C, D, E, F) .

thm:abd_dep_upper_bound

Theorem 46. Let everything be as in Definition 39 and let the Young diagrams $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ have sizes given by

$$\begin{aligned} \alpha \vdash a = \lceil k |\alpha'| \rceil, \quad \beta \vdash b = \lceil k |\beta'| \rceil, \quad \gamma \vdash c = \lceil k |\alpha'| \rceil + \lceil k |\beta'| \rceil, \\ \delta \vdash d = \lceil k |\delta'| \rceil, \quad \epsilon \vdash e = \lceil k |\alpha'| \rceil + \lceil k |\beta'| \rceil + \lceil k |\delta'| \rceil, \quad \phi \vdash f = \lceil k |\beta'| \rceil + \lceil k |\delta'| \rceil, \end{aligned} \quad \text{feq:ceil_yd_sizes} \quad (141)$$

for some positive integer $k > \tau_{\min}^{-1}$ where $\tau_{\min}$ is the minimum trace. Then

$$\left| \Gamma_{A,B,D}^{\boxtimes} \left(\begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right) \right| \leq p(k) \exp(-k I_{\alpha'\beta'\gamma'\delta'\epsilon'\phi'}^{\alpha\beta\gamma\delta\epsilon\phi} / 6). \quad \text{feq:tet_gamma_upper_bound} \quad (142)$$

where $p(k)$ grows polynomially in k and where $I_{\alpha'\beta'\gamma'\delta'\epsilon'\phi'}^{\alpha\beta\gamma\delta\epsilon\phi}$ is a quantity which measures the deviation of the primed quantities from the unprimed quantities:

$$I_{\alpha'\beta'\gamma'\delta'\epsilon'\phi'}^{\alpha\beta\gamma\delta\epsilon\phi} = |\alpha'| D(\underline{\alpha} \parallel \underline{\alpha}') + |\beta'| D(\underline{\beta} \parallel \underline{\beta}') + |\gamma'| D(\underline{\gamma} \parallel \underline{\gamma}') + |\delta'| D(\underline{\delta} \parallel \underline{\delta}') + |\epsilon'| D(\underline{\epsilon} \parallel \underline{\epsilon}') + |\phi'| D(\underline{\phi} \parallel \underline{\phi}'). \quad (143)$$

Proof. The first portion of the proof does not depend on the value of k . For now, we suppose the Young diagrams $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ have sizes determined by Eq. (121). As in the proof of Corollary 45, we let $\Omega := \underline{A}^{\otimes a} \otimes \underline{B}^{\otimes b} \otimes \underline{D}^{\otimes d}$. Using the commutation relations provided by Lemma 41 we apply the Cauchy-Schwarz inequality to $\langle X, Y \rangle = \text{Tr}(X^* Y) \leq \text{Tr}(X^* X)^{\frac{1}{2}} \text{Tr}(Y^* Y)^{\frac{1}{2}}$ for $X = \Omega^{\frac{1}{2}} \Pi_{\phi} \Pi_{\alpha} \Pi_{\beta}$ and $Y = \Omega^{\frac{1}{2}} \Pi_{\gamma} \Pi_{\delta} \Pi_{\epsilon}$ to obtain

$$\begin{aligned} \left| \Gamma_{A,B,D}^{\boxtimes} \left(\begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right) \right|^6 &= |\text{Tr}[(\Pi_{\phi} \Pi_{\alpha} \Pi_{\beta}) \Omega (\Pi_{\gamma} \Pi_{\delta} \Pi_{\epsilon})]|^6 \\ &\leq \text{Tr}(\Pi_{\phi} \Pi_{\alpha} \Pi_{\beta} \Omega)^3 \text{Tr}(\Pi_{\gamma} \Pi_{\delta} \Pi_{\epsilon} \Omega)^3 \\ &\leq \text{Tr}(\Pi_{\phi} \Omega) \text{Tr}(\Pi_{\alpha} \Omega) \text{Tr}(\Pi_{\beta} \Omega) \text{Tr}(\Pi_{\gamma} \Omega) \text{Tr}(\Pi_{\delta} \Omega) \text{Tr}(\Pi_{\epsilon} \Omega), \end{aligned} \quad (144)$$

where the last inequality follows because $\Pi_{\lambda} \leq \text{id}_n^{\otimes |\lambda|}$ for each λ in $\{\alpha, \beta, \gamma, \delta, \epsilon, \phi\}$.

Now, for each $\lambda \in \{\alpha, \beta, \gamma, \delta, \epsilon, \phi\}$, it is possible to upper-bound $\text{Tr}(\Pi_{\lambda} \Omega)$ using the eigenvalue estimation technique (Proposition 17). For a detailed example, we will consider value of $\text{Tr}(\Pi_{\gamma} \Omega)$:

$$\text{Tr}(\Pi_{\gamma} \Omega) = \frac{\text{Tr}[(P_{\gamma} \otimes \text{id}^{\otimes d})(A^{\otimes a} \otimes B^{\otimes b} \otimes D^{\otimes d})]}{|\alpha'|^a |\beta'|^b |\delta'|^d} = \frac{\text{Tr}[P_{\gamma}(A^{\otimes a} \otimes B^{\otimes b})]}{|\alpha'|^a |\beta'|^b}. \quad (145)$$

Under the assumption that $C = A + B$, we have $C^{\otimes c} = (A + B)^{\otimes a+b}$ which implies

$$\mathrm{Tr}(P_\gamma C^{\otimes c}) = \sum_{k=0}^c \binom{c}{k} \mathrm{Tr}(P_\gamma (A^{\otimes k} \otimes B^{\otimes c-k})). \quad (146)$$

Here we have used the S_c -equivariance of P_γ . Since each of the summands above are non-negative, we conclude for all $0 \leq j \leq c$ that

$$\mathrm{Tr}(P_\gamma C^{\otimes c}) \geq \binom{c}{j} \mathrm{Tr}(P_\gamma (A^{\otimes j} \otimes B^{\otimes c-j})). \quad (147)$$

Letting $j = a$ and rearranging yields

$$\mathrm{Tr}(P_\gamma (A^{\otimes a} \otimes B^{\otimes b})) \leq \frac{a!b!}{c!} \mathrm{Tr}(P_\gamma C^{\otimes c}). \quad (148)$$

Combining this result with the spectrum estimation bound (Proposition 17) yields

$$\mathrm{Tr}(\Pi_\gamma \Omega) \leq \frac{|\gamma'|^c}{|\alpha'|^a |\beta'|^b} \frac{a!b!}{c!} \dim(V_\gamma^n) \exp(-cD(\underline{\gamma} \parallel \underline{\gamma}')). \quad (149)$$

Analogous calculations for $\lambda \in \{\alpha, \beta, \gamma, \delta, \epsilon, \phi\}$ produce the following set of upper bounds:

$$\begin{aligned} \mathrm{Tr}(\Pi_\alpha \Omega) &\leq \dim(V_\alpha^n) \exp(-aD(\underline{\alpha} \parallel \underline{\alpha}')), \\ \mathrm{Tr}(\Pi_\beta \Omega) &\leq \dim(V_\beta^n) \exp(-bD(\underline{\beta} \parallel \underline{\beta}')), \\ \mathrm{Tr}(\Pi_\gamma \Omega) &\leq \dim(V_\gamma^n) \frac{|\gamma'|^c}{|\alpha'|^a |\beta'|^b} \frac{a!b!}{c!} \exp(-cD(\underline{\gamma} \parallel \underline{\gamma}')), \\ \mathrm{Tr}(\Pi_\delta \Omega) &\leq \dim(V_\delta^n) \exp(-dD(\underline{\delta} \parallel \underline{\delta}')), \\ \mathrm{Tr}(\Pi_\epsilon \Omega) &\leq \dim(V_\epsilon^n) \frac{|\epsilon'|^e}{|\alpha'|^a |\beta'|^b |\delta'|^d} \frac{a!b!d!}{e!} \exp(-eD(\underline{\epsilon} \parallel \underline{\epsilon}')), \\ \mathrm{Tr}(\Pi_\phi \Omega) &\leq \dim(V_\phi^n) \frac{|\phi'|^f}{|\beta'|^b |\delta'|^d} \frac{b!d!}{f!} \exp(-fD(\underline{\phi} \parallel \underline{\phi}')). \end{aligned} \quad (150)$$

Our objective now will be to use the k -dependent values for (a, b, c, d, e, f) to upper bound the multinomial prefactors appearing in the bounds for γ , ϵ , and ϕ above in terms of k . From Eq. (141), we obtain the following estimates:

$$\begin{aligned} k|\alpha'| &\leq a < k|\alpha'| + 1, & k|\beta'| &\leq b < k|\beta'| + 1, & k|\gamma'| &\leq c < k|\gamma'| + 2, \\ k|\delta'| &\leq d < k|\delta'| + 1, & k|\epsilon'| &\leq e < k|\epsilon'| + 3, & k|\phi'| &\leq f < k|\phi'| + 2. \end{aligned} \quad (151)$$

An application of Lemma 54 for binomial distributions ($m = 2$) implies the following bound

$$\frac{|\gamma'|^c}{|\alpha'|^a |\beta'|^b} \frac{a!b!}{c!} \leq 12^2 (k|\gamma'| + 2)^{\frac{1}{2}}. \quad (152)$$

Moreover, from Weyl's dimension formula, Eq. (34), we obtain [CM06, Equation 7]:

$$\dim(V_\gamma^n) \leq (|\gamma| + 1)^{n(n-1)/2} \leq (k|\gamma'| + 3)^{n(n-1)/2}, \quad (153)$$

because $(\gamma_i - i) - (\gamma_j - j) \leq |\gamma| + 1$ as well as $j - i \geq 1$. Altogether, we can relax the bound on $\mathrm{Tr}(\Pi_\gamma \Omega)$ to obtain

$$\mathrm{Tr}(\Pi_\gamma \Omega) \leq 12^2 (k|\gamma'| + 3)^{\frac{n(n-1)}{2} + \frac{1}{2}} \exp(-k|\gamma'| D(\underline{\gamma} \parallel \underline{\gamma}')). \quad (154)$$

A similar calculations for $\mathrm{Tr}(\Pi_\epsilon \Omega)$ yields (noting to apply Lemma 54 for trinomial distributions when $m = 3$) yields:

$$\mathrm{Tr}(\Pi_\epsilon \Omega) \leq 12^3 (k|\epsilon'| + 4)^{\frac{n(n-1)}{2} + 1} \exp(-k|\epsilon'| D(\underline{\epsilon} \parallel \underline{\epsilon}')). \quad (155)$$

It is also not too difficult to see that the polynomial in k prefactor in the upper bound on $\text{Tr}(\Pi_\epsilon \Omega)$ above is the largest such prefactor and thus Eq. (142) indeed holds if we take $p(k)$ to be

$$p(k) = 12^3(k\tau_{\max} + 4)^{\frac{n(n-1)}{2}+1}. \quad \text{\texttt{eq:p_of_k}} \quad (156)$$

□

While Theorem 46 states that $\Gamma_{A,B,D}^{\Diamond} \left(\begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right)$ concentrates in magnitude around particular Young diagrams, it is not directly applicable to the tetrahedral Horn problem simply because its value depends on more than just the eigenvalues of the matrices (A, B, C, D, E, F) . Nevertheless, combining Theorem 46 with the completeness relation (Eq. (136)) yields a necessary condition for $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ to be tetrahedral. \texttt{cor:tet_size_lower_bound}

Corollary 47. *Let $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ be the eigenvalues of non-zero positive semidefinite tetrahedral Hermitian matrices (A, B, C, D, E, F) with minimum trace $\tau_{\min}$. Then there exists a sextuple of sequences of Young diagrams $(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k)$ indexed by positive integers $k > \tau_{\min}^{-1}$ with sizes given by Eq. (141) satisfying both (i) the convergence condition*

$$I_{\alpha' \beta' \gamma' \delta' \epsilon' \phi'}^{\alpha_k \beta_k \gamma_k \delta_k \epsilon_k \phi_k} \leq \frac{6}{k} \log p(k) q(k) \xrightarrow{k \rightarrow \infty} 0, \quad \text{\texttt{eq:sextuple_convergence_rate}} \quad (157)$$

and (ii) the lower bound

$$\left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\|_{\infty} \geq \frac{1}{q(k)}, \quad \text{\texttt{eq:ud_size_lower_bound}} \quad (158)$$

where $p(k)$ is given by Eq. (156) whereas $q(k)$ is given by

$$q(k) = (k\tau_{\max} + 4)^{6(r_{\max}-1)}. \quad \text{\texttt{eq:q_of_k}} \quad (159)$$

Proof. To proof the claim, we construct the sequence implicitly. For each k , it suffices to take $(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k)$ to be any sextuple of Young diagrams which maximizes the value of $\left| \Gamma_{A,B,D}^{\Diamond} \left(\begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right) \right|$ with sizes determined by Eq. (141):

$$(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k) := \operatorname{argmax}_{\alpha \beta \gamma \delta \epsilon \phi} \left| \Gamma_{A,B,D}^{\Diamond} \left(\begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right) \right|. \quad (160)$$

While this maximal choice may not be unique, it will necessarily satisfy a lower bound due to the completeness relation satisfied by $\Gamma_{A,B,D}^{\Diamond} \left(\begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right)$ Eq. (136). Specifically, this maximal choice must satisfy

$$\left| \Gamma_{A,B,D}^{\Diamond} \left(\begin{smallmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{smallmatrix} \right) \right| \geq \frac{1}{q(k)} \quad \text{\texttt{eq:tet_gamma_lower_bound}} \quad (161)$$

where $q(k)$ is any upper bound on the number of possible summands in the completeness relation Eq. (136) viewed as a function of k . As there less than or equal to $(s+1)^{r-1}$ Young diagrams with r rows with size at most s , letting $q(k)$ be given by Eq. (159) ensures Eq. (161) holds because the size (resp. rank) of any of $(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k)$ is at most $k\tau_{\max} + 3$ (resp. $r_{\max}$). Moreover, applying Eq. (142) yields

$$I_{\alpha' \beta' \gamma' \delta' \epsilon' \phi'}^{\alpha_k \beta_k \gamma_k \delta_k \epsilon_k \phi_k} \leq \frac{6}{k} \log(q(k)p(k)) \quad (162)$$

and thus Eq. (157) holds because $p(k)$ and $q(k)$ are polynomials in k with coefficients depending on the maximum trace $\tau_{\max}$ and maximum rank $r_{\max}$. Also Eq. (158) follows from Eq. (161) combined with the upper bound provided by Eq. (138). □

Remark 48. The convergence condition above (Eq. (157)) in conjunction with the size conditions given by Eq. (141) together ensure that the Young diagrams $(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k)$ converge to the eigenvalues $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ in the following sense that

$$\begin{aligned} \frac{\alpha_k}{k} &\rightarrow \alpha', & \frac{\beta_k}{k} &\rightarrow \beta', & \frac{\gamma_k}{k} &\rightarrow \gamma', \\ \frac{\delta_k}{k} &\rightarrow \delta', & \frac{\epsilon_k}{k} &\rightarrow \epsilon', & \frac{\phi_k}{k} &\rightarrow \phi', \end{aligned} \quad \text{\texttt{eq:sextuple_convergence}} \quad (163)$$

where $\frac{\lambda_k}{k} \rightarrow \lambda'$ is shorthand for $\lim_{k \rightarrow \infty} (\lambda_k)_i / k = \lambda'_i$ for all $i \in \{1, \dots, n\}$.

While the existence of a convergent sequence $(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k)$ of primed Young diagrams is necessary for $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ to be tetrahedral by Corollary 47, our next result proves it is also sufficient. In other words, if $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ are not tetrahedral, then no such sequence can exist.

Corollary 49. *Suppose $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ is a non-tetrahedral sextuple of positive semidefinite and non-zero eigenvalues with minimum trace $\tau_{\min}$. Then for all sequences of Young diagrams $(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k)$ indexed by $k > \tau_{\min}^{-1}$ with sizes given by Eq. (141) and converging to $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ in the sense that Eq. (157) holds, we have*

$$\left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\|_2^2 \leq \exp \left(-\frac{k\tau_{\min}}{12} D_*^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'} + O(\log k) \right), \quad \text{\textcolor{blue}{\small{eq:seq_ud_size_upper_bound}}} \quad (164)$$

where $D_*^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'} > 0$ is a positive constant which measures the distance from $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ to the closed set of tetrahedral eigenvalues $(\tilde{\alpha}', \tilde{\beta}', \tilde{\gamma}', \tilde{\delta}', \tilde{\epsilon}', \tilde{\phi}')$ with matching traces (i.e., $|\alpha'| = |\tilde{\alpha}'|$, $|\beta'| = |\tilde{\beta}'|$, etc.):

$$D_*^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'} = \min_{\substack{\tilde{\alpha}', \tilde{\beta}', \tilde{\gamma}', \tilde{\delta}', \tilde{\epsilon}', \tilde{\phi}' \\ \text{tetrahedral} \\ \forall \lambda': |\tilde{\lambda}'| = |\lambda'|}} \left(\|\underline{\alpha}' - \underline{\tilde{\alpha}'}\|_1^2 + \|\underline{\beta}' - \underline{\tilde{\beta}'}\|_1^2 + \|\underline{\gamma}' - \underline{\tilde{\gamma}'}\|_1^2 + \|\underline{\delta}' - \underline{\tilde{\delta}'}\|_1^2 + \|\underline{\epsilon}' - \underline{\tilde{\epsilon}'}\|_1^2 + \|\underline{\phi}' - \underline{\tilde{\phi}'}\|_1^2 \right). \quad (165)$$

Proof. Using Eq. (131), we have

$$R_{\alpha\beta\delta;\epsilon} \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_2^2 = \iiint d\mu(A) d\mu(B) d\mu(D) \Gamma_{A,B,D}^{\boxtimes} \left(\begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right) \quad (166)$$

where $R_{\alpha\beta\delta;\epsilon}$ is a ratio of dimensions,

$$R_{\alpha\beta\delta;\epsilon} = h_n(\alpha) h_n(\beta) h_n(\delta) \frac{\dim V_\epsilon^n}{\dim(V_\alpha^n \otimes V_\beta^n \otimes V_\delta^n)} = \frac{\dim(V_\alpha^n) \dim(V_\beta^n) \dim(V_\delta^n) \dim(V_\epsilon^n)}{\dim(V_{(a)}^{n^2}) \dim(V_{(b)}^{n^2}) \dim(V_{(d)}^{n^2})}, \quad (167)$$

and where the integral is taken over a triple of $n \times n$ trace one positive semidefinite matrices A, B, D . For each integrand, define a tetrahedral sextuple $(\tilde{A}, \tilde{B}, \tilde{C}, \tilde{D}, \tilde{E}, \tilde{F})$ as follows

$$\begin{aligned} \tilde{A} &:= |\alpha'| A, & \tilde{B} &:= |\beta'| B, & \tilde{C} &:= |\alpha'| A + |\beta'| B, \\ \tilde{D} &:= |\delta'| D, & \tilde{E} &:= |\alpha'| A + |\beta'| B + |\delta'| D, & \tilde{F} &:= |\beta'| B + |\delta'| D. \end{aligned} \quad \text{\textcolor{blue}{\small{eq:tilde_defs}}} \quad (168)$$

Furthermore, let $(\tilde{\alpha}', \tilde{\beta}', \tilde{\gamma}', \tilde{\delta}', \tilde{\epsilon}', \tilde{\phi}')$ be the eigenvalues of $(\tilde{A}, \tilde{B}, \tilde{C}, \tilde{D}, \tilde{E}, \tilde{F})$. In this notation, the integrand above is also equal to $\Gamma_{\tilde{A}, \tilde{B}, \tilde{D}}^{\boxtimes} \left(\begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right)$. Moreover, by Eq. (142) we have a bound on the integrand,

$$\left| \Gamma_{\tilde{A}, \tilde{B}, \tilde{D}}^{\boxtimes} \left(\begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right) \right| \leq p(k) \exp(-k I_{\alpha'\beta'\gamma'\delta'\epsilon'\phi'}^{\alpha\beta\gamma\delta\epsilon\phi} / 6). \quad (169)$$

Now by construction, the eigenvalues $(\tilde{\alpha}', \tilde{\beta}', \tilde{\gamma}', \tilde{\delta}', \tilde{\epsilon}', \tilde{\phi}')$ are tetrahedral while by assumption, the eigenvalues $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ are not tetrahedral (although they have matching traces due to Eq. (168)). Therefore we conclude that

$$(\alpha', \beta', \gamma', \delta', \epsilon', \phi') \neq (\tilde{\alpha}', \tilde{\beta}', \tilde{\gamma}', \tilde{\delta}', \tilde{\epsilon}', \tilde{\phi}') \quad (170)$$

and therefore $I_{\alpha'\beta'\gamma'\delta'\epsilon'\phi'}^{\alpha\beta\gamma\delta\epsilon\phi} > 0$.

Our goal now will be to show that for any sequence of Young diagrams $(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k)$ converging to the non-tetrahedral eigenvalues $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$, one can also guarantee that for sufficiently large k , the exponent $I_{\alpha'\beta'\gamma'\delta'\epsilon'\phi'}^{\alpha_k\beta_k\gamma_k\delta_k\epsilon_k\phi_k}$ above is also bounded from below by a positive constant. Let λ' be a placeholder variable indexing $\lambda' \in \{\alpha', \beta', \gamma', \delta', \epsilon', \phi'\}$. By Pinsker's inequality for the relative entropy, we obtain a sequence of lower bounds on $I_{\alpha'\beta'\gamma'\delta'\epsilon'\phi'}^{\alpha\beta\gamma\delta\epsilon\phi}$:

$$I_{\alpha'\beta'\gamma'\delta'\epsilon'\phi'}^{\alpha\beta\gamma\delta\epsilon\phi} = \sum_{\lambda'} |\lambda'| D(\underline{\lambda} \| \underline{\lambda}') \geq \frac{1}{2} \sum_{\lambda'} |\lambda'| \|\underline{\lambda} - \underline{\lambda}'\|_1^2 \geq \frac{\tau_{\min}}{2} \sum_{\lambda'} \|\underline{\lambda} - \underline{\lambda}'\|_1^2 \geq \frac{\tau_{\min}}{2} \max_{\lambda'} \|\underline{\lambda} - \underline{\lambda}'\|_1^2, \quad (171)$$

where $\tau_{\min}$ is the minimum trace over the eigenvalues (the lower indices). Applying the last bound to $I_{\alpha'\beta'\gamma'\delta'\epsilon'\phi'}^{\alpha_k\beta_k\gamma_k\delta_k\epsilon_k\phi_k}$ yields

$$\max_{\lambda'} \|\underline{\lambda}_k - \underline{\lambda}'\|_1 \leq \sqrt{2\tau_{\min}^{-1} I_{\alpha'\beta'\gamma'\delta'\epsilon'\phi'}^{\alpha_k\beta_k\gamma_k\delta_k\epsilon_k\phi_k}} \quad (172)$$

whereas applying the second to last bound to $I_{\alpha'\beta'\tilde{\gamma}'\tilde{\delta}'\tilde{\epsilon}'\tilde{\phi}'}^{\alpha_k\beta_k\gamma_k\delta_k\epsilon_k\phi_k}$ (noting that $\tau_{\min}$ is also the minimum trace of $(\tilde{\alpha}', \tilde{\beta}', \tilde{\gamma}', \tilde{\delta}', \tilde{\epsilon}', \tilde{\phi}')$) yields

$$\sum_{\lambda'} \left\| \underline{\lambda}_k - \tilde{\underline{\lambda}}' \right\|_1^2 \leq 2\tau_{\min}^{-1} I_{\alpha'\beta'\tilde{\gamma}'\tilde{\delta}'\tilde{\epsilon}'\tilde{\phi}'}^{\alpha_k\beta_k\gamma_k\delta_k\epsilon_k\phi_k}. \quad (173)$$

Furthermore, by the reverse triangle inequality we have

$$\sum_{\lambda'} \left\| \underline{\lambda}_k - \tilde{\underline{\lambda}}' \right\|_1^2 \geq \sum_{\lambda'} \left(\|\underline{\lambda}_k - \underline{\lambda}'\|_1 - \|\underline{\lambda}' - \tilde{\underline{\lambda}}'\|_1 \right)^2 \geq \sum_{\lambda'} \left\| \underline{\lambda}' - \tilde{\underline{\lambda}}' \right\|_1^2 - 2 \|\underline{\lambda}_k - \underline{\lambda}'\|_1 \left\| \underline{\lambda}' - \tilde{\underline{\lambda}}' \right\|_1. \quad (174)$$

Also by the definition of $D_*^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'}$ we have

$$D_*^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'} \leq \sum_{\lambda'} \left\| \underline{\lambda}' - \tilde{\underline{\lambda}}' \right\|_1^2. \quad (175)$$

Since $\underline{\lambda}'$ and $\tilde{\underline{\lambda}}'$ are both probability distributions we also have $\left\| \underline{\lambda}' - \tilde{\underline{\lambda}}' \right\|_1 \leq 2$. Also, $6 \max_{\lambda'} \|\underline{\lambda}_k - \underline{\lambda}'\|_1 \geq \sum_{\lambda'} \|\underline{\lambda}_k - \underline{\lambda}'\|_1$ as there are six summands. Combining all these bounds together yields

$$\left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_2^2 \leq \exp \left(-k \left\{ \frac{\tau_{\min} D_*^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'}}{12} - 2\sqrt{2\tau_{\min} I_{\alpha'\beta'\gamma'\delta'\epsilon'\phi'}^{\alpha_k\beta_k\gamma_k\delta_k\epsilon_k\phi_k}} - \frac{1}{k} \log(R_{\alpha_k, \beta_k, \delta_k; \epsilon_k}^{-1} p(k)) \right\} \right). \quad (176)$$

By assumption, the Young diagrams $(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k)$ converge in normalization to the eigenvalues $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ in the sense that $I_{\alpha'\beta'\gamma'\delta'\epsilon'\phi'}^{\alpha_k\beta_k\gamma_k\delta_k\epsilon_k\phi_k}$ tends to zero as $k \rightarrow \infty$ at a rate given by Eq. (157). Moreover, $R_{\alpha_k, \beta_k, \delta_k; \epsilon_k}^{-1}$ can be upper bounded by a polynomial in k and $p(k)$ is a polynomial in k so Eq. (164) holds as claimed. $\square$

4 The large n limit & an entropic inequality

sec:large_n

The purpose of this section is to show how to use the necessary condition for $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ to be tetrahedral provided by Corollary 47 to derive inequalities satisfied by tetrahedral eigenvalues. Our strategy will be to prove an upper bound the generalized $6j$ symbol which depends only on the dimensions of the underlying irreducible representations and the multiplicity spaces between them. By combining this upper bound with the lower bound provided by Corollary 47, we obtain a new constraint on the convergent sequence of Young diagrams $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$.

thm:sixj_ssa_bound

Theorem 50. *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be a sextuple of Young diagrams. Then*

$$\left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_2^2 \leq \left(c_{\alpha, \phi}^{\epsilon} c_{\beta, \delta}^{\phi} c_{\alpha, \beta}^{\gamma} c_{\gamma, \delta}^{\epsilon} \right)^{\frac{1}{2}} \frac{\dim(V_{\gamma}^n) \dim(V_{\phi}^n)}{\dim(V_{\beta}^n) \dim(V_{\epsilon}^n)}. \quad (177)$$

Proof. Here we provide a diagrammatic proof of the claimed inequality. For the sake of clarity, we abbreviate the irreducible representation labels, V_{λ}^n , and omit the multiplicity space labels on the intermediate steps.

$$\left(\begin{array}{c} \text{Diagram 1} \\ \text{Diagram 2} \end{array} \right) = \left(\begin{array}{c} \text{Diagram 3} \\ \text{Diagram 4} \end{array} \right) \leq \left(\begin{array}{c} \text{Diagram 5} \\ \text{Diagram 6} \end{array} \right)^{\frac{1}{2}} = \left(\begin{array}{c} \text{Diagram 7} \\ \text{Diagram 8} \end{array} \right)^{\frac{1}{2}} \quad (178)$$

The first equality follows from Definition 42. The inequality follows from an application of the Cauchy-Schwarz inequality with respect to the annotated contraction. The final equality follows from applying Schur's lemma to components of the diagram mapping V_{β}^n to itself on the left and V_{ϵ}^n to itself on the right. Note that the factors of $\dim(V_{\epsilon}^n)^2$ appearing on left-most and right-most sides of the above inequality cancel producing the claimed inequality. $\square$

An immediate application of Theorem 50 is to combine it with Corollary 47. If $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ are tetrahedral eigenvalues, then by Corollary 47 there exists a converging sequence of Young diagrams $(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k)$ such that⁸

$$\frac{1}{q(k)^2} \leq \left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\|_2^2 \leq \left(c_{\alpha_k, \phi_k}^{\epsilon_k} c_{\beta_k, \delta_k}^{\phi_k} c_{\alpha_k, \beta_k}^{\gamma_k} c_{\gamma_k, \delta_k}^{\epsilon_k} \right)^{\frac{1}{2}} \frac{\dim(V_{\gamma_k}^n) \dim(V_{\phi_k}^n)}{\dim(V_{\beta_k}^n) \dim(V_{\epsilon_k}^n)}. \quad \text{\textcolor{blue}{eq:combined_bound}} \quad (179)$$

Remarkably, both $q(k)$ (given by Eq. (159)) and the Littlewood-Richardson coefficients $c_{\alpha_k, \phi_k}^{\epsilon_k}$, $c_{\beta_k, \delta_k}^{\phi_k}$, $c_{\alpha_k, \beta_k}^{\gamma_k}$, and $c_{\gamma_k, \delta_k}^{\epsilon_k}$ are *independent* of the dimension parameter n defining the group $U(n)$ (assuming n is at least as large as the maximum depth of any of the involved Young diagrams, i.e., $n \geq r_{\max}$ where $r_{\max}$ is the maximum rank of any of the tetrahedral eigenvalues). This observation provides one with an opportunity to optimize the above bound by varying the value of n .

In particular, it will useful to better understand how the dimension, $\dim(V_{\lambda}^n)$, of the irreducible representation V_{λ}^n of $U(n)$ grows as n becomes much larger than the number of rows of λ . In general, for fixed λ , the function $n \mapsto V_{\lambda}^n$ can grow at an exponential rate with increasing n . One can find estimates for $\dim(V_{\lambda}^n)$ in the regime where n is large compared to $|\lambda|$ and also the last row $\lambda_{\min}$ is large compared to the depth of λ in paper by Itzykson and Nauenberg [IN66, Section III]. Here we use an elegant presentation of this scaling, which relates the asymptotics of $\dim(V_{\lambda}^n)$ to the value of $\dim(W_{\lambda'})$ (see [Har23, Lemma 2] or [CHW07, Equations (25, 26)]).

⁸Here we used the well-known inequality $\|X\|_{\infty} \leq \|X\|_2$.

prop:harrow_formula

Proposition 51. *Let λ be a Young diagram with depth at most n . Then*

$$\frac{\dim(V_\lambda^n)}{n^{|\lambda|}} = \frac{\dim(W_\lambda)}{|\lambda|!} \prod_{(i,j) \in \lambda} \left(1 + \frac{j-i}{n}\right). \quad (180)$$

Proof. The statement follows from the dimension formulas for $\dim(V_\lambda^n)$ and $\dim(W_\lambda)$ in Eq. (34) and Eq. (35). See also [Har23, Appendix A]. $\square$

From Proposition 51 we obtain the following corollary.

cor:harrow_ssa_lim

Corollary 52. *Let $\gamma, \phi, \beta, \epsilon$ be a quadruple of Young diagrams such that*

$$|\gamma| + |\phi| = |\beta| + |\epsilon|. \quad (181)$$

{eq:balanced_sizes}

Then

$$\lim_{n \rightarrow \infty} \frac{\dim(V_\gamma^n) \dim(V_\phi^n)}{\dim(V_\beta^n) \dim(V_\epsilon^n)} = \frac{|\beta|! |\epsilon|! \dim(W_\gamma) \dim(W_\phi)}{|\gamma|! |\phi|! \dim(W_\beta) \dim(W_\epsilon)} \quad (182)$$

Proof. From Proposition 51 one evidently has for any Young diagram λ the limit

$$\lim_{n \rightarrow \infty} \frac{\dim(V_\lambda^n)}{n^{|\lambda|}} = \frac{\dim(W_\lambda)}{|\lambda|!}. \quad (183)$$

The claim then follows because the missing prefactor, $\frac{n^{|\beta|} n^{|\epsilon|}}{n^{|\gamma|} n^{|\phi|}}$, is equal to one by Eq. (181). $\square$

We are now in a position to prove the main result of this section.

Theorem 53. *Let $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ be tetrahedral eigenvalues. Then*

$$I(\gamma', \phi'; \beta', \epsilon') = s(\gamma') + s(\phi') - s(\beta') - s(\epsilon') \geq 0, \quad (184)$$

where

$$s(\lambda') := - \sum_{i=1}^n \lambda'_i \log \lambda'_i. \quad (185)$$

{eq:little_s_defn}

Proof. As $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ are tetrahedral, the trace conditions (Eq. (12)) necessarily imply that Eq. (181) holds. Therefore, by Corollary 52 and the discussion surrounding the bound in Eq. (179) we conclude that along the convergent sequence of Young diagrams satisfies

$$I' := \lim_{k \rightarrow \infty} \frac{1}{k} \log \left[\frac{|\beta_k|! |\epsilon_k|! \dim(W_{\gamma_k}) \dim(W_{\phi_k})}{|\gamma_k|! |\phi_k|! \dim(W_{\beta_k}) \dim(W_{\epsilon_k})} \right] \geq 0 \quad (186)$$

provided the limit I' exists. An application of Stirling's approximation [Rob55] to the dimension formula for $\dim(W_{\lambda_k})$ together with the approximation $\lambda_k \approx k\lambda'$ yields

$$\frac{1}{k} \log \left(\frac{\dim(W_{\lambda_k})}{|\lambda|!} \right) = s(\lambda') + |\lambda'| (1 - \log k) + O(k^{-1} \log k) \quad (187)$$

where $s(\lambda')$ is given in Eq. (185). Moreover by Eq. (181), the $(1 - \log k)$ terms cancel and therefore $I' = I(\gamma', \phi'; \beta', \epsilon') \geq 0$ as claimed above. $\square$

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A On the oriented tetrahedral symmetry

The group G of order 48 appearing in Proposition 9 corresponds to the subgroup of all possible signed permutations of 6 elements generated by permutations and negations of the elements of the sextuple (A, B, C, D, E, F) which preserves the set of linear constraints in Eq. (6). The group G is isomorphic to both the direct product $S_2 \times S_4$ and to the signed permutation group of three elements, i.e., $(S_2 \times S_2 \times S_2) \rtimes S_3$.

The first isomorphism, $G \simeq S_2 \times S_4$, can be understood as S_2 acting by the inversion of all eigenvalues,

$$(\alpha', \beta', \gamma', \delta', \epsilon', \phi') \leftrightarrow (\alpha'^*, \beta'^*, \gamma'^*, \delta'^*, \epsilon'^*, \phi'^*), \quad (188)$$

while S_4 acts by permutation of the *vertices* of the tetrahedron in Fig. 1: each vertex permutation induces a permutation of the six directed edges with the possibility of inverting the orientation of a subset of the edges (such edge orientation reversals can be accounted for a corresponding inversion of eigenvalues).

The second isomorphism, $G \simeq (S_2 \times S_2 \times S_2) \rtimes S_3$, can be understood by considering the triplet $(A, D), (B, E), (C, F)$ of non-adjacent pairs of edges in Fig. 1. Accordingly, it is helpful to arrange the sextuple $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ into a 2×3 grid using the following identification:

$$(\alpha' \ \beta' \ \gamma' \ \delta' \ \epsilon' \ \phi') \cong \begin{pmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{pmatrix}. \quad (189)$$

Then one can see that each S_2 factor in the direct product $S_2 \times S_2 \times S_2$ acts faithfully on the sextuple by both negating a certain subset of the columns and permuting the upper and lower rows leaving just one column unchanged.

$$\begin{pmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{pmatrix} \leftrightarrow \begin{pmatrix} \alpha' & \epsilon'^* & \phi'^* \\ \delta' & \beta'^* & \gamma'^* \end{pmatrix}, \quad \begin{pmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{pmatrix} \leftrightarrow \begin{pmatrix} \delta' & \beta' & \phi' \\ \alpha' & \epsilon' & \gamma' \end{pmatrix}, \quad \begin{pmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{pmatrix} \leftrightarrow \begin{pmatrix} \delta'^* & \epsilon' & \gamma' \\ \alpha'^* & \beta' & \phi' \end{pmatrix}. \quad (190)$$

The symmetric group S_3 of order $3! = 6$ acts instead by permuting the columns and negating a subset of the entries in the top row, producing a subgroup of order 6 of the form:

$$\begin{pmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{pmatrix} \quad \begin{pmatrix} \beta' & \gamma'^* & \alpha'^* \\ \epsilon' & \phi' & \delta' \end{pmatrix} \quad \begin{pmatrix} \gamma'^* & \alpha' & \beta'^* \\ \phi' & \delta' & \epsilon' \end{pmatrix} \\ \begin{pmatrix} \alpha'^* & \gamma' & \beta' \\ \delta' & \phi' & \epsilon' \end{pmatrix} \quad \begin{pmatrix} \beta'^* & \alpha'^* & \gamma'^* \\ \epsilon' & \delta' & \phi' \end{pmatrix} \quad \begin{pmatrix} \gamma' & \beta'^* & \alpha' \\ \phi' & \epsilon' & \delta' \end{pmatrix} \quad (191)$$

We further remark that these symmetries also corresponding to symmetries of the Wigner $6j$ operator $\left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\}$ in Definition 42 with the involution $\lambda \mapsto \lambda^*$ corresponding taking the dual representation $V_\lambda^n \mapsto (V_\lambda^n)^*$. In the case where $n = 2$, the involution map becomes the identity map (as every irreducible representation of $SU(2)$ is self-dual, or the reversal of a pair $(x, -x)$ of traceless eigenvalues is trivial), and thus the symmetry group collapses to a group of order $|S_4| = 4! = 24$.

B Multinomial distribution inequalities

lem:multinomial_dist_lower_bound

Lemma 54. Fix $m \geq 2$ and let $(w_1, \dots, w_m) \in \mathbb{R}_{>0}^m$ have minimum $w_{\min} := \min\{w_1, \dots, w_m\} > 0$. Let k be a positive integer satisfying $k \geq w_{\min}^{-1}$ and define positive integers $(z_1, \dots, z_m) \in \mathbb{N}_{>0}^m$ by the equation

$$z_i := \lceil kw_i \rceil. \quad (192)$$

Let $Z = z_1 + \dots + z_m$ and $W = w_1 + \dots + w_m$. Then

$$\frac{Z!}{z_1! \dots z_m!} \frac{w_1^{z_1} \dots w_m^{z_m}}{W^Z} \geq \left(\frac{1}{12} \right)^m (kW + m)^{\frac{m-1}{2}} \quad \text{\textit{eq:multinomial_dist_lower_bound}} \quad (193)$$

Proof. The proof follows from a version of Stirling's approximation due to Robbins [Rob55] which states

$$k! = \sqrt{2\pi k} k^{k+\frac{1}{2}} e^{-k} e^{r_k}, \quad \text{where} \quad \frac{1}{12k+1} \leq r_k \leq \frac{1}{12k}. \quad (194)$$

Therefore,

$$\frac{Z!}{z_1! \cdots z_m!} \frac{w_1^{z_1} \cdots w_m^{z_m}}{W^Z} = \left(\frac{1}{2\pi Z} \right)^{\frac{m-1}{2}} \prod_{i=1}^m \left(\frac{Z}{z_i} \right)^{\frac{1}{2}} \left(\frac{Z w_i}{z_i W} \right)^{z_i} \exp \left(r_Z - \sum_{i=1}^m r_{z_i} \right). \quad (195)$$

In this setting, we have let $z_i = \lceil k w_i \rceil$ which implies

$$k w_i \leq z_i < k w_i + 1, \quad \text{and} \quad k W \leq Z < k W + m. \quad (196)$$

Consequently⁹,

$$\left(\frac{Z w_i}{z_i W} \right)^{z_i} \geq \left(\frac{k w_i}{k w_i + 1} \right)^{k w_i + 1} \geq \frac{1}{4}. \quad (197)$$

The remainder term satisfies

$$r_Z - \sum_i r_{z_i} \geq \frac{1}{12Z + 1} - \sum_i \frac{1}{12z_i} \geq -\frac{m}{12}. \quad (198)$$

Also $Z \geq z_i$ so altogether, we have

$$\frac{Z!}{z_1! \cdots z_m!} \frac{w_1^{z_1} \cdots w_m^{z_m}}{W^Z} \geq \left(\frac{1}{2\pi} \right)^{\frac{m-1}{2}} \left(\frac{1}{4} \right)^m \exp \left(-\frac{m}{12} \right) \left(\frac{1}{k W + m} \right)^{\frac{m-1}{2}} \geq \left(\frac{1}{12} \right)^m (k W + m)^{\frac{m-1}{2}}, \quad (199)$$

as claimed because $\sqrt{2\pi} > 1$ and $\frac{1}{12} < \frac{e^{-\frac{1}{12}}}{4\sqrt{2\pi}} = 0.09176 \dots$. $\square$

C A character formula for $6j$ symbols

Throughout Section 3.3 we proved a number of results concerning the square of the Hilbert-Schmidt norm of the $6j$ operator defined by Definition 42. The following result provides a formula for the square of the Hilbert-Schmidt norm as an integral over four copies of $U(n)$.

lem:character_formula

Lemma 55 (Character formula). *Let $\left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} : C_{\alpha,\beta}^\gamma \otimes C_{\delta,\gamma}^\epsilon \rightarrow C_{\alpha,\phi}^\epsilon \otimes C_{\beta,\delta}^\phi$ be the $6j$ operator defined in Definition 42. Then*

$$\dim(V_\epsilon)^2 \left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_2^2 = \int_{U(n)^4} du_1 du_2 du_3 du_4 X_{\delta,\epsilon,\phi}^{\alpha,\beta,\gamma}(u_1, u_2, u_3, u_4). \quad (200)$$

eq:character_formula

where

$$X_{\delta,\epsilon,\phi}^{\alpha,\beta,\gamma}(u_1, u_2, u_3, u_4) := \chi_\alpha(u_1^{-1}u_2) \chi_\beta(u_4^{-1}u_2) \chi_\gamma(u_2^{-1}u_3) \chi_\delta(u_4^{-1}u_3) \chi_\epsilon(u_3^{-1}u_1) \chi_\phi(u_1^{-1}u_4). \quad (201)$$

Note that when $n = 2$, Eq. (200) recovers the well-known formula for the square of the $6j$ -symbol for $SU(2)$ (see [FL03, Equation (1)] and also [Bar98; FK00]).

Proof of Lemma 55. The proof follows from (i) the following identities concerning the character $\chi_\lambda(u) = \text{Tr}(\pi_\lambda(u))$ and duality $\pi_\lambda^*(u) = (\pi_\lambda(u^{-1}))^T$ relations

$$\chi_\lambda(u) = \boxed{\pi_\lambda(u)} \quad \pi_\lambda^*(u) = \boxed{\pi_\lambda(u^{-1})}, \quad (202)$$

⁹Here we use the identity that $x \geq 1$ implies $\left(\frac{x}{x+1} \right)^{x+1} \geq \frac{1}{4}$.

Proof. That C' is positive semidefinite and has rank at most n follows because C' can be decomposed as $C' = T^*T$ where T is the $n \times 2n$ block matrix

$$T = \begin{bmatrix} A^{\frac{1}{2}} & B^{\frac{1}{2}} \end{bmatrix}. \quad (208)$$

That C' and $A+B$ share the same eigenvalues (up to padding by n zeros) follows because for any rectangular matrix T , the eigenvalues of T^*T and TT^* are equal (up to padding by zeros) and in this instance

$$T^*T = \begin{bmatrix} A^{\frac{1}{2}} \\ B^{\frac{1}{2}} \end{bmatrix} \begin{bmatrix} A^{\frac{1}{2}} & B^{\frac{1}{2}} \end{bmatrix} = \begin{bmatrix} A & A^{\frac{1}{2}}B^{\frac{1}{2}} \\ B^{\frac{1}{2}}A^{\frac{1}{2}} & B \end{bmatrix} = C' \quad TT^* = \begin{bmatrix} A^{\frac{1}{2}} & B^{\frac{1}{2}} \end{bmatrix} \begin{bmatrix} A^{\frac{1}{2}} \\ B^{\frac{1}{2}} \end{bmatrix} = A + B. \quad (209)$$

□

From the perspective of Lemma 56, one can view the Horn's problem involving the eigenvalues $(\alpha', \beta', \gamma')$ of $n \times n$ Hermitian matrices (A, B, C) satisfying $A + B = C$ as a special case of a more general eigenvalue problem concerned with the relations between the eigenvalues of $2n \times 2n$ Hermitian matrices Σ of the form

$$C' = \begin{bmatrix} A' & Z \\ Z^* & B' \end{bmatrix}, \quad \text{\texttt{eq:block_ABX}} \quad (210)$$

and the eigenvalues of its two $n \times n$ diagonal blocks Z and Z . It is known, by [HJ12, Theorem 7.7.9] or more generally [MKX19, Theorem 5.11] that if C' is a 2×2 block matrix such that $A', B' \geq 0$, then $C' \geq 0$ if and only if the off-diagonal block Z is of the form

$$Z = A'^{\frac{1}{2}} K B'^{\frac{1}{2}} \quad (211)$$

for some contraction matrix K (largest eigenvalue of $(K^*K)^{\frac{1}{2}}$ is at most unity). Additionally, when $Z = A'^{\frac{1}{2}} K B'^{\frac{1}{2}}$, one has the decomposition

$$C' = \begin{bmatrix} A' & Z \\ Z^* & B' \end{bmatrix} = \begin{bmatrix} A'^{\frac{1}{2}} & 0 \\ 0 & B'^{\frac{1}{2}} \end{bmatrix} \begin{bmatrix} I & K \\ K^* & I \end{bmatrix} \begin{bmatrix} A'^{\frac{1}{2}} & 0 \\ 0 & B'^{\frac{1}{2}} \end{bmatrix}, \quad (212)$$

noting that K is a contraction if and only if the middle block matrix above is positive semidefinite [MKX19, Lemma 5.10].

Therefore, the special case covered by Lemma 56 corresponds to the case where the corresponding contraction matrix K is the $n \times n$ identity matrix $K = I$ (or more generally any contractive matrix satisfying $A'^{\frac{1}{2}} K B'^{\frac{1}{2}} = A'^{\frac{1}{2}} B'^{\frac{1}{2}}$). A simple observation which distinguishes between eigenvalue problem concerning diagonal blocks and the eigenvalue problem concerning sums is simply that the rank of C' can in general be up to $2n$ whereas the rank of the sum $C = A + B$ is at most n . Despite this additional freedom, there remains non-trivial relations constraining the eigenvalues of C' and of the blocks A' and B' in terms of properties of Z (see [Gum+18; BL22]). Nevertheless, when the rank of C' is constrained to be at most n , we obtain the following theorem which provides an equivalent formulation of Horn's problem.

\texttt{thm:trigonal_block}

Theorem 57. *Let $(\alpha', \beta', \gamma')$ be a triple of positive semidefinite eigenvalues of length at most n . The following are equivalent:*

- (i) *there exists a triple of Hermitian matrices (A, B, C) with eigenvalues $(\alpha', \beta', \gamma')$ satisfying $A + B = C$,*
- (ii) *there exists an $2n \times 2n$ Hermitian matrix,*

$$C' = \begin{bmatrix} A' & Z \\ Z^* & B' \end{bmatrix}, \quad (213)$$

such that $\text{eig}(A') = \alpha'$, $\text{eig}(B') = \beta'$, and $\text{eig}(C') = \gamma'$.

Proof. Since we explicitly assume all of the eigenvalues in the triple $(\alpha', \beta', \gamma')$ are non-negative, the Hermitian matrices A, B, C , and A', B', C' involved in the statements above are necessarily positive semidefinite. The claim (i) $\implies$ (ii) follows from Lemma 56. That (ii) $\implies$ (i) follows by noticing that if C' has

eigenvalues γ' , then it has rank at most n . Therefore, as $C' \geq 0$, there exists a $n \times 2n$ block matrix T of the form

$$T = \begin{bmatrix} T_A & T_B \end{bmatrix} \quad (214)$$

such that $C' = T^*T$. The diagonal blocks of C' are therefore $A' = T_A^*T_A$, and $B' = T_B^*T_B$. As in the proof of Lemma 56, we note that T^*T and TT^* have the same eigenvalues up to padding by zeros. Therefore, by letting $A = T_A T_A^*$ and $B = T_B T_B^*$ we see that

$$\begin{aligned} \text{eig}(A) &= \text{eig}(T_A T_A^*) \simeq \text{eig}(T_A^* T_A) = \text{eig}(A') = \alpha', \\ \text{eig}(B) &= \text{eig}(T_B T_B^*) \simeq \text{eig}(T_B^* T_B) = \text{eig}(B') = \beta', \end{aligned} \quad (215)$$

while the eigenvalues of $C = A + B = T_A T_A^* + T_B T_B^*$ match the eigenvalues of C' because

$$\text{eig}(C) = \text{eig}(A + B) = \text{eig}(T_A T_A^* + T_B T_B^*) = \text{eig}(TT^*) \simeq \text{eig}(T^*T) = \text{eig}(C') = \gamma'. \quad (216)$$

□

Here we established a correspondence between (i) the existence of $n \times n$ positive semidefinite matrices (A, B, C) with eigenvalues $(\alpha', \beta', \gamma')$ satisfying $A + B = C$ and (ii) the positivity of corresponding Littlewood-Richardson coefficients $c_{\alpha'\beta'}^{\gamma'}$ which govern the branching problem for the diagonal subgroup $\text{GL}(n) \hookrightarrow \text{GL}(n) \times \text{GL}(n)$ sending X to (X, X) . Recalling Theorem 57, there is a further correspondence to (iii) the existence of a $2n \times 2n$ positive semidefinite matrix Σ with eigenvalues γ' and $n \times n$ blocks A', B' with eigenvalues α', β' . One may wonder how to show the correspondence between (ii) and (iii) *directly* without passing through (i). This can be accomplished by considering the branching multiplicities $m_{\gamma'}^{\alpha'\beta'}$ for the blockwise subgroup $\text{GL}(a) \times \text{GL}(b) \hookrightarrow \text{GL}(a+b)$ sending (A, B) to $A \oplus B$. In fact there exists a representation-theoretic analogue of Theorem 57, specifically [GW00, Theorem 9.2.3]. In particular, the values of $m_{\gamma'}^{\alpha'\beta'}$ and $c_{\alpha'\beta'}^{\gamma'}$ coincide provided the Young diagrams $(\alpha', \beta', \gamma')$ involved have depths at most $(\min\{a, n\}, \min\{b, n\}, \min\{a+b, n\})$; this ensures that $V_{\alpha'}^a, V_{\beta'}^b, V_{\gamma'}^{a+b}$ and $V_{\alpha'}^n, V_{\beta'}^n, V_{\gamma'}^n$ are all non-zero representation spaces and therefore $m_{\gamma'}^{\alpha'\beta'}$ and $c_{\alpha'\beta'}^{\gamma'}$ are both meaningfully defined. The distinction between $m_{\gamma'}^{\alpha'\beta'}$ and $c_{\alpha'\beta'}^{\gamma'}$ lies with the fact that it is possible for $c_{\alpha'\beta'}^{\gamma'}$ to be zero while $m_{\gamma'}^{\alpha'\beta'}$ is non-zero (this can happen when the depth of γ' is greater than n but at most $a+b$ so that $V_{\gamma'}^n$ is zero but $V_{\gamma'}^{a+b}$ is non-zero). This parallels the fact that, in general, the rank of the $2n \times 2n$ matrix Σ in Theorem 57 could be greater than n while the rank of the $n \times n$ matrix C is at most n .

As a corollary of this correspondence between the Horn's problem and the eigenvalues of blocks of 2×2 matrices, we obtain an equivalent formulation of the tetrahedral Horn problem.

thm:tetrahedral_block

Theorem 58. *Let $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ be a sextuple of positive semidefinite eigenvalues of length at most n . Then $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ are tetrahedral if and only if there exists a $3n \times 3n$ matrix E' of the form*

$$E' = \begin{bmatrix} A' & Z & Y \\ Z^* & B' & X \\ Y^* & X^* & D' \end{bmatrix}, \quad \text{with blocks} \quad C' = \begin{bmatrix} A' & Z \\ Z^* & B' \end{bmatrix}, \quad F' = \begin{bmatrix} B' & X \\ X^* & D' \end{bmatrix} \quad \text{\textit{eq:block_3by3}} \quad (217)$$

such that

$$\text{eig}(A') = \alpha', \quad \text{eig}(B') = \beta', \quad \text{eig}(C') = \gamma', \quad \text{eig}(D') = \delta', \quad \text{eig}(E') = \epsilon', \quad \text{eig}(F') = \phi'. \quad \text{\textit{eq:block_eigenvalue_tet}} \quad (218)$$

Proof. The proof is analogous to the proofs of Lemma 56 and Theorem 57. First we prove the “only if” portion of the claim which is a corollary of Lemma 56. Suppose (A, B, C, D, E, F) is a tetrahedral sextuple of Hermitian matrices with eigenvalues $(\alpha', \beta', \gamma', \delta', \epsilon', \gamma')$. Let E' be given by

$$E' = \begin{bmatrix} A^{\frac{1}{2}} & B^{\frac{1}{2}} & D^{\frac{1}{2}} \end{bmatrix}^* \begin{bmatrix} A^{\frac{1}{2}} & B^{\frac{1}{2}} & D^{\frac{1}{2}} \end{bmatrix} = \begin{bmatrix} A & A^{\frac{1}{2}} B^{\frac{1}{2}} & A^{\frac{1}{2}} D^{\frac{1}{2}} \\ B^{\frac{1}{2}} A^{\frac{1}{2}} & B & B^{\frac{1}{2}} D^{\frac{1}{2}} \\ D^{\frac{1}{2}} A^{\frac{1}{2}} & D^{\frac{1}{2}} B^{\frac{1}{2}} & D \end{bmatrix} \quad (219)$$

Then the eigenvalues of the corresponding principal submatrices of E' satisfy Eq. (218) by Lemma 56. Now we prove the “if” portion of the claim which is a corollary of Theorem 57. If E' is a matrix of form given by Eq. (217) with eigenvalues ϵ' with length at most n , then E' is positive semidefinite with rank at most n . Therefore, there exists an $n \times 3n$ block matrix

$$T = \begin{bmatrix} T_A & T_B & T_D \end{bmatrix} \quad (220)$$

such that $E' = T^*T$ and thus

$$\begin{aligned} A' &= T_A^*T_A, & B' &= T_B^*T_B, & D' &= T_D^*T_D, \\ X &= T_B^*T_D, & Y &= T_A^*T_D, & Z &= T_A^*T_B. \end{aligned} \quad (221)$$

Now let (A, B, C, D, E, F) be given by

$$\begin{aligned} A &= T_A T_A^*, & B &= T_B T_B^*, & C &= T_A T_A^* + T_B T_B^*, \\ D &= T_D T_D^*, & E &= T_A T_A^* + T_B T_B^* + T_D T_D^*, & F &= T_B T_B^* + T_D T_D^*. \end{aligned} \quad (222)$$

From here, one may directly verify that these $n \times n$ Hermitian matrices satisfy Eq. (6) and have eigenvalues $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ (using a similar argument presented in the proof of Theorem 57); i.e., one concludes that $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ are tetrahedral. $\square$

TODO: Describe the same correspondence for the blockwise $6j$ symbols.

E Inequalities for tetrahedral eigenvalues

[sec:ineqs](#)

The purpose of this section is to describe various inequalities which are necessarily satisfied by tetrahedral eigenvalues $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$. In general, however, the task of finding a complete description of the space of tetrahedral eigenvalues for $n \geq 3$ remains an open problem.

E.1 Horn’s inequalities

Recall that $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ are tetrahedral only if each of $(\alpha', \beta', \gamma')$, (β', γ', ϕ') , $(\gamma', \delta', \epsilon')$, and $(\alpha', \phi', \epsilon')$ are trigonal as in Eq. (8). Consequently, every inequality constraining trigonal eigenvalues can be lifted to produce a quadruplet of inequalities constraining the set of tetrahedral eigenvalues. These inequalities can be understood as generalizations of the triangle inequalities, Eq. (16), which hold for case where $n = 2$. Therefore, before proceeding, we briefly summarize the inequalities satisfied by trigonal eigenvalues $(\alpha', \beta', \gamma')$. Of particular interest are Horn’s inequalities [Hor62] which are known to be necessary and sufficient [Kly98; KT99]. For a more complete historical account, the reader is directed to the survey articles [Bha01; Ful98; Knu00] or [NP06, Appendix D]. In addition to the aforementioned trace condition, $\sum_{i=1}^n \alpha'_i + \sum_{j=1}^n \beta'_j = \sum_{k=1}^n \gamma'_k$, one always has the following pair of elementary inequalities,

$$\alpha'_1 + \beta'_1 \geq \gamma'_1, \quad \text{and} \quad \alpha'_n + \beta'_n \leq \gamma'_n, \quad (223)$$

which constrain the largest and smallest eigenvalues respectively (note that under inversion symmetry, these inequalities are equivalent). The first somewhat non-trivial inequality, derived by Weyl in 1912, is now known as *Weyl’s inequality* and states for all i, j such that $i + j \leq n + 1$ [Wey12]¹⁰

$$\alpha'_i + \beta'_j \geq \gamma'_{i+j-1}. \quad (224)$$

In 1949, Fan derived a majorization inequality constraining the first r largest eigenvalues [Fan49], and shortly thereafter Lidskii [Lid53] & Wielandt [Wie55] found proofs of another family of inequalities. Finally, in 1962, Horn determined a complete collection of inequalities for $(\alpha', \beta', \gamma')$ are trigonal for the case where $n = 3$

¹⁰For 2×2 matrices one can see that these necessary inequalities are also sufficient to ensure that $(\alpha', \beta', \gamma')$ are trigonal; upon shifting to traceless eigenvalues using Corollary 11, $\alpha' = (a, -a)$, $\beta' = (b, -b)$ and $\gamma' = (c, -c)$, and these inequalities become equivalent to the triangle inequalities $c \leq a + b$ and $c \geq \max\{b - a, a - b\}$.

[Hor62]. Moreover, also in [Hor62], Horn provided an inductively defined set of subsets $I, J, K \subset \{1, \dots, n\}$ of equal cardinality and conjectured that inequalities of the form

$$\sum_{i \in I} \alpha'_i + \sum_{j \in J} \beta'_j \geq \sum_{k \in K} \gamma'_k, \quad \text{\textcolor{blue}{\{eq:IJK_ineq\}}} \quad (225)$$

were both necessary and sufficient to ensure that $(\alpha', \beta', \gamma')$ are trigonal. While proofs for some of these inequalities were found (see e.g., [TF71; TF70]), it took until the late 1990s for Horn's problem to be to be solved by Klyachko [Kly98] who proved that $(\alpha', \beta', \gamma')$ if and only if in addition to the trace equality, $|\alpha'| + |\beta'| = |\gamma'|$, Eq. (225) holds for all (I, J, K) such that (i) $|I| = |J| = |K| = p$ for some p , and (ii) the Littlewood-Richardson coefficient $c_{\alpha, \beta}^{\gamma}$ is non-zero (here $\alpha = (i_p - p, \dots, i_1 - 1)$, $\beta = (j_p - p, \dots, j_1 - 1)$, and $\gamma = (k_p - p, \dots, k_1 - 1)$). Shortly thereafter, Knutson and Tao proved the saturation conjecture for Littlewood-Richardson coefficients [KT99], which when combined with Klyachko's solution, proved Horn's conjecture.

For the purposes of characterizing the set of tetrahedral eigenvalues further, we will henceforth be exclusively concerned with inequalities constraining the set of tetrahedral eigenvalues which are not already implied by Horn's inequalities.

E.2 A simple quadratic inequality

In Section 2.3 we saw that the shifting symmetry implies, in a certain sense, one may assume without loss of generality that the candidate eigenvalues $(\alpha', \beta', \gamma', \delta', \epsilon')$ are all non-negative or even strictly positive. While almost all of the following inequalities require this positivity assumption, we note that they can be extended to eigenvalues with negative entries in all cases.

To begin, we provide an example of a non-linear inequality constraining tetrahedral eigenvalues which appears to be new.

\text{\textcolor{blue}{prop:simple_quadratic}}

Proposition 59. *Let $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ be non-negative tetrahedral eigenvalues of $n \times n$ Hermitian matrices. Then*

$$2 \sum_{j=1}^n \alpha'_j \delta'_{n-j+1} \leq \|\beta'\|_2^2 + \|\epsilon'\|_2^2 - \|\gamma'\|_2^2 - \|\phi'\|_2^2 \leq 2 \sum_{j=1}^n \alpha'_j \delta'_j. \quad (226)$$

Proof. Let (A, B, C, D, E, F) be the tetrahedral sextuple of positive semidefinite matrices with eigenvalues $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ satisfying $A + B = C$, $B + D = F$, and $A + F = C + D = E$. Then the following identity holds:

$$\begin{aligned} & \text{Tr}[B^2] + \text{Tr}[E^2] - \text{Tr}[C^2] - \text{Tr}[F^2] \\ &= \text{Tr}[(B)^2] + \text{Tr}[(A + B + D)^2] - \text{Tr}[(A + B)^2] - \text{Tr}[(B + D)^2] \\ &= (\text{Tr}[B^2]) + (\text{Tr}[A^2] + \text{Tr}[B^2] + \text{Tr}[D^2] + 2 \text{Tr}[AB] + 2 \text{Tr}[BD] + 2 \text{Tr}[AD]) \\ &\quad - (\text{Tr}[A^2] + \text{Tr}[B^2] + 2 \text{Tr}[AB]) - (\text{Tr}[B^2] + \text{Tr}[D^2] + 2 \text{Tr}[BD]) \\ &= 2 \text{Tr}[AD]. \end{aligned} \quad \text{\textcolor{blue}{\{eq:quad_identity\}}} \quad (227)$$

At this stage, one may note that for all $A, D \geq 0$ positive semidefinite, $\text{Tr}(AD) \geq 0$ and therefore,

$$\text{Tr}[B^2] + \text{Tr}[E^2] - \text{Tr}[C^2] - \text{Tr}[F^2] \geq 0. \quad \text{\textcolor{blue}{\{eq:quadratic_difference_pos\}}} \quad (228)$$

This inequality is well-known and indeed follows as a special of [Sag13, Proposition 2.3]. The improvement made here is to note that while the value of $\text{Tr}(AD)$ cannot, in general, be determined from the eigenvalues $\alpha' = \text{eig}(A)$ and $\delta' = \text{eig}(D)$, one can apply the von Neumann trace inequalities ([MOA10, H.1.g & H.1.h] or [Bau+01, Proposition 4.4]):

$$\sum_j \alpha'_j \delta'_{n-j+1} \leq \text{Tr}[AD] \leq \sum_j \alpha'_j \delta'_j, \quad (229)$$

where $\delta'_{n-j+1} = -\delta_j^*$. The claim then follows because $\text{Tr}[X^2] = \sum_j \lambda_j^2$ for any Hermitian operator X with eigenvalues $\text{eig}(X) = \lambda'$. $\square$

E.3 Ptolemy & Cayley-Menger inequalities

In this section, we prove inequalities constraining the 2-norms $\|\cdot\|_2$ of a tetrahedral sextuple of eigenvalues $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ associated to the Hermitian matrices (A, B, C, D, E, F) . These inequalities correspond to Ptolemy's inequality and the Cayley-Menger determinant inequality for the edge-lengths of a Euclidean tetrahedron in $\mathbb{R}^3$. The core idea is to consider the Hilbert-Schmidt inner product $\langle X, Y \rangle = \text{Tr}(X^\dagger Y)$ on $n \times n$ Hermitian matrices and the induced Hilbert-Schmidt norm $\|X\| = \sqrt{\langle X, X \rangle}$. For any Hermitian matrix X with eigenvalues λ' , we have $\|X\| = \|\lambda'\|_2 = \left(\sum_{j=1}^n \lambda_j'^2\right)^{\frac{1}{2}}$. Recall that the four vertices of the tetrahedron of Hermitian matrices are

$$V_0 = 0, \quad V_1 = A, \quad V_2 = A + B = C, \quad V_3 = A + B + D = E. \quad (230)$$

E.3.1 Ptolemy's Inequality

Ptolemy's inequality applied any four points V_0, V_1, V_2, V_3 in a normed space reads

$$\|V_3 - V_0\| \|V_1 - V_2\| \leq \|V_2 - V_0\| \|V_1 - V_3\| + \|V_1 - V_0\| \|V_2 - V_3\|. \quad \text{\texttt{\{eq:vvvv_ptolemy\}}} \quad (231)$$

Applied to the vertices of the tetrahedron of Hermitian matrices, we obtain

$$\|E\| \|B\| \leq \|C\| \|F\| + \|A\| \|D\|. \quad \text{\texttt{\{eq:abcdef_ptolemy\}}} \quad (232)$$

It is known that every real inner product space with norm $\|X\| = \sqrt{\langle X, X \rangle}$ is Ptolemaic in the sense that Ptolemy's inequality holds. A converse of this result also holds due to Schoenberg; namely, every real semi-normed space satisfying Ptolemy's inequality comes from a real inner product space [Sch52]. Here we will prove the former result, that a real inner product space satisfies Ptolemy's inequality.

Proof. Note that to prove the inequality, one can take $V_0 = 0$ (by shifting) and assume the remaining vectors are non-zero (since otherwise the inequality holds trivially). Under these assumptions, we aim to prove for all V_1, V_2, V_3 non-zero in a real inner product space,

$$\|V_3\| \|V_1 - V_2\| \leq \|V_2\| \|V_1 - V_3\| + \|V_1\| \|V_2 - V_3\|. \quad \text{\texttt{\{eq:shifted_ptolemy\}}} \quad (233)$$

Let $\tilde{V}_1, \tilde{V}_2, \tilde{V}_3$ be rescaled versions of V_1, V_2, V_3 :

$$\tilde{V}_1 = \frac{V_1}{\|V_1\|^2}, \quad \tilde{V}_2 = \frac{V_2}{\|V_2\|^2}, \quad \tilde{V}_3 = \frac{V_3}{\|V_3\|^2}. \quad (234)$$

Under this rescaling, the pairwise distances are rescaled as

$$\|\tilde{V}_i - \tilde{V}_j\| = \sqrt{\frac{1}{\|V_i\|^2} + \frac{1}{\|V_j\|^2} - \frac{2|\langle V_i, V_j \rangle|^2}{\|V_i\|^2 \|V_j\|^2}} = \frac{\|V_i - V_j\|}{\|V_i\| \|V_j\|}. \quad (235)$$

Then, the triangle inequality for $\|\cdot\|$ applied to $\tilde{V}_1, \tilde{V}_2, \tilde{V}_3$,

$$\|\tilde{V}_1 - \tilde{V}_2\| \leq \|\tilde{V}_1 - \tilde{V}_3\| + \|\tilde{V}_2 - \tilde{V}_3\|, \quad (236)$$

is then equivalent to Eq. (233). $\square$

E.3.2 Cayley-Menger Determinant

First we note that the four Hermitian matrices $(V_0, V_1, V_2, V_3) = (0, A, C, E)$ necessarily lie inside a three-dimensional subspace of the set of $n \times n$ Hermitian matrices and thus there exists an orthonormal set of Hermitian matrices L_1, L_2, L_3 such that for each $i \in \{0, 1, 2, 3\}$

$$V_i = \sum_{j=1}^3 (v_i)_j L_j \quad (237)$$

In this basis, the vectors of coefficients $\{v_0, v_1, v_2, v_3\}$ constitute the vertices of a tetrahedra in $\mathbb{R}^3$ and moreover $\langle V_i, V_j \rangle = \text{Tr}(V_i^* V_j) = \langle v_i, v_j \rangle$. Following the wikipedia derivation, define a 4×4 matrix by

$$P = \begin{pmatrix} v_0 & v_1 & v_2 & v_3 \\ 1 & 1 & 1 & 1 \end{pmatrix} \quad (238)$$

such that $|\det(P)| = 3! \text{Vol}(v_0, v_1, v_2, v_3)$ where $\text{Vol}(v_0, v_1, v_2, v_3)$ is the volumes of the tetrahedron with vertices $v_0, v_1, v_2, v_3 \in \mathbb{R}^3$. Then one augments P to

$$\tilde{P} = \begin{pmatrix} v_0 & v_1 & v_2 & v_3 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ \|v_0\|^2 & \|v_1\|^2 & \|v_2\|^2 & \|v_3\|^2 & 1 \end{pmatrix} \quad (239)$$

such that $\det(P) = \det(\tilde{P})$ and also defines a 5×5 matrix Q by

$$Q = \begin{pmatrix} -2 & 0 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 & 0 \\ 0 & 0 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix} \quad (240)$$

Then $\tilde{P}^\dagger Q \tilde{P}$ is a 5×5 matrix

$$\tilde{\Sigma} = \tilde{P}^\dagger Q \tilde{P} = \begin{pmatrix} 0 & d_{01}^2 & d_{02}^2 & d_{03}^2 & 1 \\ d_{10}^2 & 0 & d_{12}^2 & d_{13}^2 & 1 \\ d_{20}^2 & d_{21}^2 & 0 & d_{23}^2 & 1 \\ d_{30}^2 & d_{31}^2 & d_{32}^2 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{pmatrix} \quad (241)$$

where

$$d_{ij}^2 := \|v_i - v_j\|^2 = \|v_i\|^2 + \|v_j\|^2 - 2 \langle v_i, v_j \rangle. \quad (242)$$

The determinant $\det(\tilde{\Sigma})$ is equal to

$$\det(\tilde{\Sigma}) = \det(\tilde{P}^\dagger Q \tilde{P}) = \det(Q) \det(P)^2 = 8 \cdot (3!)^2 \cdot \text{Vol}^2 \geq 0. \quad (243)$$

Using the isometric embedding of $(V_0, V_1, V_2, V_3) = (0, A, C, E)$ into $\mathbb{R}^3$, the Cayley-Menger determinant inequality takes the form

$$\det \begin{pmatrix} 0 & \|A\|^2 & \|C\|^2 & \|E\|^2 & 1 \\ \|A\|^2 & 0 & \|B\|^2 & \|F\|^2 & 1 \\ \|C\|^2 & \|B\|^2 & 0 & \|D\|^2 & 1 \\ \|E\|^2 & \|F\|^2 & \|D\|^2 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{pmatrix} \geq 0 \quad (244)$$

E.4 Strongly subadditive/Submodular inequalities

DISCLAIMER FROM TC: The remainder of this section is written with respect to the A, B, C convention instead of the A, B, D convention. When discussing the tetrahedral Horn problem and its relation to the ordinary Horn problem or when matching the problem to the properties of the $6j$ symbols, the A, B, D convention is preferred. However when talking about submodular/strong subadditivity inequalities and comparing with the literature, the A, B, C convention is preferred. Before finalizing the paper, it will be desirable to establish a globally consistent convention. An equivalent formulation is to consider eigenvalues of four PSD matrices W, X, Y, Z with eigenvalues $(\omega, \chi, \nu, \zeta)$ such that $X + Y = W + Z$ and $W \geq X \geq Z$ and $W \geq Y \geq Z$.

Another strategy for understanding eigenvalues of positive semidefinite matrices and their sums is to consider the expansion of tensors powers of sums of matrices in relation to the tensor powers of their

summands.¹¹ For instance, given a positive integer k and positive semidefinite A and B , using the non-commutative binomial expansion of $(A + B)^{\otimes k}$ and dropping all summands involving both A and B , one obtains [BS15, Proposition 1.1]:

$$(A + B)^{\otimes k} \geq A^{\otimes k} + B^{\otimes k}. \quad (245)$$

This technique, was generalized by [TCL11] [TCL11, Lemma 2.2] to obtain the following result.¹² prop:operator_ineq

Proposition 60. *Let $A, B, C \geq 0$ be positive semidefinite matrices and let k be a positive integer. Then*

$$(A + B + C)^{\otimes k} + B^{\otimes k} \geq (A + B)^{\otimes k} + (B + C)^{\otimes k}. \quad (246)$$

From Proposition 60 one can obtain an inequalities pertaining only to the eigenvalues of $B, A + B, B + C$, and $A + B + C$ by pairing the operator inequality with any $U(n)$ -equivariant positive semidefinite operator acting on $(\mathbb{C}^n)^{\otimes k}$.¹³ cor:schur_ineq

Corollary 61. *Let $A, B, C \geq 0$ be positive semidefinite matrices and let s_λ be the Schur polynomial indexed by the Young diagram $\lambda = (\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n)$ with size $|\lambda| := \lambda_1 + \dots + \lambda_n = k$. Then*

$$s_{\lambda'}(\text{eig}(A + B + C)) + s_{\lambda'}(\text{eig}(B)) \geq s_{\lambda'}(\text{eig}(A + B)) + s_{\lambda'}(\text{eig}(B + C)). \quad (247) \quad \text{eq:schur_ineq}$$

Proof. Given a Young diagram λ , let P_λ be the Hermitian projection operator onto the isotypic subspace $V_\lambda^n \otimes W_\lambda \subseteq (\mathbb{C}^d)^{\otimes k}$ (see Section 3 for details). Given positive semidefinite matrix $X \geq 0$, one can show that

$$\text{Tr}(P_\lambda X^{\otimes k}) = \text{Tr}(P_\lambda \text{diag}(\text{eig}(X))^{\otimes k}) = \dim(W_\lambda) s_{\lambda'}(\text{eig}(X)) \geq 0. \quad (248)$$

Now we apply the positivity preserving map $\text{Tr}(P_\lambda \cdot)$ to Proposition 60 to obtain

$$\text{Tr}[P_{\lambda'}(A + B + C)^{\otimes k}] + \text{Tr}[P_{\lambda'}B^{\otimes k}] \geq \text{Tr}[P_{\lambda'}(A + B)^{\otimes k}] + \text{Tr}[P_{\lambda'}(B + C)^{\otimes k}] \quad (249)$$

and since λ has at most n rows, we conclude $\dim(W_\lambda) \neq 0$ and thus Eq. (247) holds. $\square$

In the special where where $\lambda = (\underbrace{1, \dots, 1}_n)$, where $\text{Tr}(P_{(\underbrace{1, \dots, 1}_n)} X^{\otimes n}) = \det(X)$ one obtains the well-known determinant inequality [TCL11, Lemma 2.2]:

$$\det(A + B + C) + \det(B) \geq \det(A + B) + \det(B + C). \quad (250) \quad \text{eq:lin_det}$$

Another well-known determinant inequality is the multiplicative variant of Eq. (250) which states that for all $A, B, C \geq 0$ positive semidefinite,

$$\det(A + B + C) \det(B) \leq \det(A + B) \det(B + C). \quad (251) \quad \text{eq:mul_det}$$

This inequality follows as a special case of a general result due to Sagnol [Sag13, Proposition 2.3]. thm:sagnol_ineq

Theorem 62. *Let $A, B, C \geq 0$ be positive semidefinite matrices. Let f be a real valued and differentiable function over the non-negative real numbers $\mathbb{R}_{\geq 0}$ whose derivative f' is an operator monotone function over the positive real numbers $\mathbb{R}_{> 0}$. Then the following inequality holds*

$$\text{Tr } f(A + B + C) + \text{Tr } f(B) \geq \text{Tr } f(A + B) + \text{Tr } f(B + C). \quad (252) \quad \text{eq:sagnol_ineq}$$

¹¹We will exploit a variation of this strategy repeatedly in Section 3.

¹²Further extensions of this technique include inequalities involving the sum $A + C$ which are outside the scope of the tetrahedral Horn problem (see [Lin14; BS15; NS23])

¹³Paksoy, Turkmen, and Zhang has used a similar approach to prove analogous inequalities concerning so-called generalized matrix functions d_χ^H associated to irreducible characters χ of subgroups H of the symmetric group S_k [MM65; HM21]. Unfortunately, these generalized matrix functions and generally not invariant under unitary conjugation and thus depend on more just eigenvalues; the only exception is the determinant which is already a special case of Corollary 61.

Proof. Here we sketch the proof in [Sag13, Section 2.1]. Let $F = \text{Tr} \circ f$. The claim then amounts to proving for all $A, B, C \geq 0$,

$$\Delta_A \Delta_C F(B) = F(A + B + C) - F(A + B) - F(B + C) + F(B) \geq 0, \quad \text{\texttt{\{eq:sagnol_diff\}}} \quad (253)$$

where Δ_X is the difference operator $[\Delta_X F](Y) := F(X + Y) - F(Y)$. By standard continuity arguments (and the assumption that f is continuous), we may assume everything is positive definite in which case

$$\partial_A \Delta_C F(B) = \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} \Delta_{\varepsilon A} \Delta_C F(B) \quad (254)$$

$$= \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} (\Delta_C F(B + \varepsilon A) - \Delta_C F(B)) \quad (255)$$

$$= \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} (F(\varepsilon A + B + C) - F(\varepsilon A + B) - F(B + C) + F(B)) \quad (256)$$

$$= \text{Tr}[(f'(B + C) - f'(B))A]. \quad \text{\texttt{\{eq:sagnol_final_step\}}} \quad (257)$$

Using the assumption that f' is an operator monotone (and the assumption of positive definiteness) together with $B + C \geq B$ (because $C \geq 0$), we have $f'(B + C) \geq f'(B)$ and thus $f'(B + C) - f'(B) \geq 0$ is a positive semidefinite operator. Therefore, the function $\Delta_C F(B)$ is non-decreasing under $B \mapsto A + B$ and thus Eq. (253) holds. $\square$

As immediate corollaries to Theorem 62, letting $f(x) = \log x$ and noting $\text{Tr}(\log X) = \log \det(X)$, one recovers Eq. (251). Moreover, taking $f(x) = -x^p$ for $p \in [0, 1]$, one obtains [BGS08, Lemma 5.3] which states for all $A, B, C \geq 0$,

$$\forall p \in [0, 1]: \quad \sum_j \lambda'_j(A + B)^p + \sum_j \lambda'_j(B + C)^p \geq \sum_j \lambda'_j(B)^p + \sum_j \lambda'_j(A + B + C)^p. \quad \text{\texttt{\{eq:bouhtou_ineq\}}} \quad (258)$$

In the limit where $p \rightarrow 0^+$, Sagnol also obtains at first order the well-known rank inequality [Sag13, Proposition A.1]

$$\text{rank}(A + B) + \text{rank}(B + C) \geq \text{rank}(B) + \text{rank}(A + B + C). \quad (259)$$

On the other hand, letting $f(x) = x^p$ for $p \in (1, 2]$ yields for all $A, B, C \geq 0$

$$\forall p \in (1, 2]: \quad \sum_j \lambda'_j(A + B)^p + \sum_j \lambda'_j(B + C)^p \leq \sum_j \lambda'_j(B)^p + \sum_j \lambda'_j(A + B + C)^p. \quad \text{\texttt{\{eq:bouhtou_ineq_reverse\}}} \quad (260)$$

Of particular interest to us in Section 4 is the function $f(x) = x \log x$ which implies for all $A, B, C \geq 0$ the inequality

$$S(A + B) + S(B + C) \geq S(B) + S(A + B + C) \quad \text{\texttt{\{eq:ssa_like_ineq\}}} \quad (261)$$

where $S(X) = s(\text{eig}(X)) := -\sum_{i=1}^n \lambda'_i(X) \log \lambda'_i(X)$ is the *unnormalized* von Neumann entropy of X . Note that a similar inequality involving entropies of sums of positive semidefinite matrices (subject to additional eigenvalue constraints) has appeared in [CŞW18]. Moreover, while it can be shown that Eq. (261) is strictly tighter than the inequality in [CŞW18] for fixed matrices, it is also possible to obtain Eq. (261) from the result in [CŞW18] by downscaling the involved matrices in the sense of Proposition 59.

Remark 63. One consequence of Theorem 58, which relates the eigenvalues of sums of positive semidefinite matrices to the eigenvalues of diagonal blocks of larger positive semidefinite matrices with bounded rank, is that many of the aforementioned inequalities can now be seen as consequences of more general inequalities which constrain the eigenvalues of blocks.

For instance, the Koteljanskii determinant inequality [JB93] (in the notation of Theorem 58) implies for any 3×3 block matrix, that

$$\det(C') \det(F') \geq \det(E') \det(B'). \quad (262)$$

Using Theorem 58, the Koteljanskii determinant inequality becomes Eq. (251). More generally, Audenaert, Hiai, and Petz were interested in functions f which are strongly superadditive in the sense that for any 3×3 block matrices E' (again in the notation of Theorem 58) the following inequality holds [AHP10]:

$$\text{Tr } f(E') + \text{Tr } f(B') \geq \text{Tr } f(C') + \text{Tr } f(F'). \quad \text{\textcolor{blue}{eq:audenaert}} \quad (263)$$

Their conclusion, also discovered by Friedland and Gaubert [FG13] was that it is *sufficient* for f' to be an operator monotone to conclude Eq. (263) holds. In light of Theorem 58, the results of Audenaert, Hiai, and Petz and [FG13] give an alternative proof of Theorem 62. In [LS14], Lewin and Sabin further proved that f' being an operator monotone is also *necessary* for Eq. (263) to hold for all 3×3 block matrices E' . Note that from the proof of Theorem 62 above, it is not too difficult to see that f' being an operator monotone is also necessary for Eq. (252) to hold (otherwise it becomes possible for Eq. (257) to become negative).

Remark 64. Much of the interest in characterizing the eigenvalues of principal submatrices of positive semidefinite matrices is closely related to the correspondence between positive semidefinite matrices and covariance matrices of Gaussian multivariate random variables [Lam+17]. From this perspective, principal submatrices become covariance matrices of Gaussian marginal distributions, and thus one may view the principal submatrix eigenvalue problem as a kind of “spectral” classical Gaussian marginal problem. Note that from this perspective, the differential entropies of the underlying Gaussian distributions are equal to logarithms of determinants up to a additive constants [Lam+17]. The Gaussian entropy region has been partially characterized by Shadbakht and Hassibi [SH11; HS08].

E.5 The Courtade-Sra determinant inequality

Many of the previously discussed inequalities constrain the eigenvalues of positive semidefinite matrices $B, A + B, B + C$ and $A + B + C$. Here we show how to improve the determinant inequality $\det(A + B) \det(B + C) \geq \det(A + B + C) \det(B)$ in a manner which also accounts for the determinants $\det(A)$ and $\det(C)$. This inequality exploits Minkowski’s determinant inequality for positive semidefinite $n \times n$ matrices A and B which reads (see [HJ85, Theorem 7.8.21] or [Bha97, Equation II.44]):

$$\det(A + B)^{\frac{1}{n}} \geq \det(A)^{\frac{1}{n}} + \det(B)^{\frac{1}{n}}. \quad (264)$$

The following inequality can be attributed to an information-theoretic result obtained by Courtade [Cou16]¹⁴, but the elementary proof we present here is do actually due to Sra [Sra]. This inequality also appears in the recent review paper by Niculescu and Sra [NS23, Section 6.2].

prop:cs_ineq

Proposition 65 (Courtade-Sra determinant inequality). *Let $A, B, C \geq 0$ be positive semidefinite $n \times n$ matrices. Then*

$$\det(A + B)^{\frac{1}{n}} \det(B + C)^{\frac{1}{n}} \geq \det(A + B + C)^{\frac{1}{n}} \det(B)^{\frac{1}{n}} + \det(A)^{\frac{1}{n}} \det(C)^{\frac{1}{n}}. \quad \text{\textcolor{blue}{eq:cs_ineq}} \quad (265)$$

Proof. By standard continuity arguments, it suffices to prove the inequality for $B > 0$ positive definite (if it holds for positive definite $B + \varepsilon I$, then continuity permits taking the limit $\varepsilon \rightarrow 0$). As B is positive definite, it is also invertible and we may define

$$X = B^{-\frac{1}{2}} A B^{-\frac{1}{2}}, \quad Y = B^{-\frac{1}{2}} C B^{-\frac{1}{2}}. \quad (266)$$

By dividing Eq. (265) by $\det(B)^{\frac{2}{n}} \neq 0$, it then suffices to prove

eq:cs_ineq_xy

$$\det(I + X)^{\frac{1}{n}} \det(I + Y)^{\frac{1}{n}} \geq \det(I + X + Y)^{\frac{1}{n}} + \det(X)^{\frac{1}{n}} \det(Y)^{\frac{1}{n}}. \quad (267)$$

Let $\lambda'(X) = \text{eig}(X)$ and $\lambda'(Y) = \text{eig}(Y)$. Then for any permutation $\pi \in S_n$ let

$$L_\pi = \text{diag}\{1 + \lambda'_i(X) + \lambda'_{\pi(i)}(Y)\}_{i=1}^n, \quad R_\pi = \text{diag}\{\lambda'_i(X) \lambda'_{\pi(i)}(Y)\}_{i=1}^n. \quad (268)$$

¹⁴Note in [Cou17], Courtade conjectures a generalization of the Brunn-Minkowski inequality for bounded convex sets A, B, C in $\mathbb{R}^n$ of the form $V(A + B)^{\frac{1}{n}} V(B + C)^{\frac{1}{n}} \geq V(A + B + C)^{\frac{1}{n}} V(B)^{\frac{1}{n}} + V(A)^{\frac{1}{n}} V(C)^{\frac{1}{n}}$. See also [CC84].

Then we have the identity

$$\det(I + X) \det(I + Y) = \prod_{i=1}^n (1 + \lambda'_i(X)) \prod_{i=1}^n (1 + \lambda'_i(Y)), \quad (269)$$

$$= \prod_{i=1}^n (1 + \lambda'_i(X) + \lambda'_{\pi(i)}(Y) + \lambda'_i(X) \lambda'_{\pi(i)}(Y)), \quad (270)$$

$$= \det(L_\pi + R_\pi) \quad (271)$$

Here we can apply Minkowski's theorem to obtain

$$\det(L_\pi + R_\pi)^{\frac{1}{n}} \geq \det(L_\pi)^{\frac{1}{n}} + \det(R_\pi)^{\frac{1}{n}}. \quad (272)$$

By definition, we have $\det(R_\pi) = \det(X) \det(Y)$ for any permutation π , while a theorem due to Fiedler [Fie71] states (see also [Bha97, Exercise VI.7.2, Theorem VI.7.1])

$$\det(I + X + Y) \leq \max_{\pi} \prod_{i=1}^n (\lambda'_i(I + X) + \lambda'_{\pi(i)}(Y)) = \max_{\pi} \det(L_\pi). \quad (273)$$

Combining these bounds together yields Eq. (267) and thus the claim holds. $\square$

TC: On [mathoverflow](https://mathoverflow.net/questions/257562/a-determinant-inequality) <https://mathoverflow.net/questions/257562/a-determinant-inequality> Sra also proves

$$E_k(B)E_k(A + B + C) \leq E_k(A + B)E_k(B + C) \quad (274)$$

where $E_k(X) = e_k(\text{eig}(X))$ where e_k is the k th elementary symmetric polynomial. TC: I conjecture that Eq. (265) also holds for more general hyperbolic polynomials, although I do not have a proof yet:

Conjecture 66. *Let $A, B, C \geq 0$ be positive semidefinite $n \times n$ matrices. Then*

$$E_k^{\frac{1}{k}}(A + B)E_k^{\frac{1}{k}}(B + C) \geq E_k^{\frac{1}{k}}(A + B + C)E_k^{\frac{1}{k}}(B) + E_k^{\frac{1}{k}}(A)E_k^{\frac{1}{k}}(C), \quad \text{\textit{\{eq:cs_ineq_conjecture\}}} \quad (275)$$

where $E_k(X) = e_k(\text{eig}(X))$ and where e_k is the k th elementary symmetric polynomial.

Proving this conjecture has proven quite challenging.

What I believe I'm managed to prove is the following.

Let $\phi : \text{PSD}_n \rightarrow \mathbb{R}$ be a function on the cone of positive semidefinite matrices. If ϕ is such that $\log \phi$ is 2-monotone, i.e.,

$$-(\partial_A \log \phi)(B) \geq 0, \quad +(\partial_C \partial_A \log \phi)(B) \geq 0, \quad (276)$$

then it is fairly clear that

$$\log \phi(A + B) + \log \phi(B + C) \geq \log \phi(B) + \log \phi(A + B + C), \quad (277)$$

or equivalently,

$$\phi(A + B)\phi(B + C) \geq \phi(B)\phi(A + B + C). \quad (278)$$

If in addition to $\log \phi$ being 2-monotone, ϕ satisfies a kind of 3-montonicity, namely

$$-\partial_W(\phi^2 \partial_C \partial_A \log \phi)(B) \geq 0, \quad (279)$$

then,

$$[\phi(A + B + W)\phi(B + C + W) - \phi(A + B + C + W)\phi(B + W)] - [\phi(A + B)\phi(B + C) - \phi(A + B + C)\phi(B)] \geq 0 \quad (280)$$

which means, assuming $\phi(0) = 0$ and for $B \mapsto 0$ and $W \mapsto B$, we have desired inequality:

$$\phi(A + B)\phi(B + C) - \phi(A + B + C)\phi(B) - \phi(A)\phi(C) \geq 0. \quad (281)$$

Then what remains is to prove that $\phi = E_k^{\frac{1}{k}}$ has this property.

TC: References that might be useful: The strongly subadditive inequality relies on the log concavity result for hyperbolic polynomials due to Bauschke et al. [Bau+01]. An excellent crash course into Gårding's theory of hyperbolic polynomials (as well as some important results) is also [Bau+01]. For the particular case of the real vector space of $n \times n$ Hermitian matrices of dimension $n(n+1)/2$, the polynomial $X \mapsto \det(X)$ is hyperbolic in the direction of the identity matrix id_n (i.e., Hermitian matrices only have real eigenvalues). Moreover, the homogeneous polynomial $p_m(X) = \text{Tr}(P_{(1,m,1)} X^{\otimes m})$ is the sum of the principal minors of X of size m and also equals the elementary symmetric polynomial of the eigenvalues of X , i.e.,

$$p_m(X) = E_m(\text{eig}(X)). \quad (282)$$

The homogeneous polynomial $p_m(X)$, like the determinant $\det(X) = p_n(X)$ itself, is also hyperbolic in the direction of the identity matrix id_n . In this context, together with the positivity $p_m(\text{id}_n) > 0$, Gårding's inequality generalizes the Minkowski determinant inequality and yields superlinearity of $p_m(X)^{\frac{1}{m}}$:

$$p_m(X + Y)^{\frac{1}{m}} \geq p_m(X)^{\frac{1}{m}} + p_m(Y)^{\frac{1}{m}} \quad (283)$$

for all X, Y positive semidefinite (the hyperbolicity cone for p_m). This inequality is due to Marcus and Lopes [ML57, Theorem 2], but it can also be proven using joint concavity of $(A_1, \dots, A_m) \mapsto A_1^{\frac{1}{m}} \otimes \dots \otimes A_m^{\frac{1}{m}}$ and thus concavity of $(A^{\frac{1}{m}})^{\otimes m}$ and thus by homogeneity the operator inequality $((A+B)^{\frac{1}{m}})^{\otimes m} \geq (A^{\frac{1}{m}})^{\otimes m} + (B^{\frac{1}{m}})^{\otimes m}$ ([Bha09, Exercise 4.2.9]).

See [ML57; McL59; AMT84] for using inequalities and proof techniques which might be relevant. See also [AMT84] or [Bha09, Chapter 4] for many properties of matrix means and their ability to prove minkowski-like inequalities. See also Khovanskii [Kho84] for inequalities mirroring the Aleksandrov-Fenchel type inequalities for hyperbolic polynomials (i.e., the mixed discriminant inequalities of Aleksandrov... see Shenfeld and Van Handel for proofs [SV19].) See [KMS19] for representation theorems and other results concerning complete monotonicity.

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